DCU3 Jtagging of an ST5105 device based STB

MSSFWP

04-21-2008, 12:11 AM

From what i have been able to find out through google and some document searching. DCU3 is a newer control interface (double data rate) with a different register than standard DCU control. By default in most applications, both DCU and DCU3 control are enabled. However most standard jtag programs will not communicate with the mb400 board design through normal DCU control because of it's use of the ST5105 processor, and the unusual way that the 5105 maps blocks of memory.

The ST5105 has been superceeded by the ST5119, but it probably also maps memory in

the same manner as the 5105 and it a safe bet that it also uses DCU3 control for Jtagging.

The first place to start would probably be the register locations, and any pertanant values that may be located there.

ST_5105_InitFMI

```
##  ------------------------------------------------------
## Default setup for configuration registers
## poke -d 0x20302000 0x00000000 ## CONFIG_CONTROL_A
## poke -d 0x20302004 0x00000000 ## CONFIG_CONTROL_B
## poke -d 0x20402000 0x00000000 ## CONFIG_CONTROL_C
x = (peek 0x20402004 -d -q)
poke -d 0x20402004 ((x) & 0xFFFFE1FF) ## CONFIG_CONTROL_D
## poke -d 0x20402008 0x00000000 ## CONFIG_CONTROL_E
## poke -d 0x2040200C 0x00000000 ## CONFIG_CONTROL_F
## poke -d 0x20402010 0x00000000 ## CONFIG_CONTROL_G
x = (peek 0x20402014 -d -q)
## poke -d 0x20402014 ((x) | 0x02000000) ## CONFIG_CONTROL_H
poke -d 0x20402014 ((x) | 0x0E000000) ## CONFIG_CONTROL_H
##  ------------------------------------------------------


RAM Locations and Configurations:

## 32MB RAM
## memory EXTERNAL 0xC0000000 (8*_MB_ - 65*_KB_) RAM
## memory mytracebuffer 0xC07EFC00 (64*_KB_) RAM
## memory TRAPHANDLER 0xC07FFC00 (1*_KB_) DEBUG
## memory AVMEM 0xC0800000 (16*_MB_) RAM
## memory NONCACHED 0xC1800000 (8*_MB_) RAM


Flash Base and register setup:

FLASH_BASE = (0x40000000)
}

memory FLASH (FLASH_BASE) (FLASH_P_SIZE) ROM
## memory STEM0 0x41000000 (32*_MB_) DEVICE
## memory STEM1 0x43000000 (32*_MB_) DEVICE
memory ST_FMI_CI 0x7F400000 (4*_MB_) DEVICE
memory ST_FMI_C 0x7F800000 (2*_MB_) DEVICE
memory ST_FMI_D 0x7FC00000 (2*_MB_) DEVICE
bootiptr (addressof -q FLASH)


And finally some stack init info, and some basic DCU control:

## initstack 0x80000100
initstack 0x80000400
```

```
## divide the stack and heap
## stack EXTERNAL (25 * _KB_)
## YK_SONG(M2.5)
if (SDRAM_SIZE_CONFIG=="BUILT_FOR_64MBIT") {
stack EXTERNAL (40 * _KB_)
heap EXTERNAL (10 * _KB_)
} else {
stack EXTERNAL
heap EXTERNAL
}

PlaceDebugTrapHandler EXTERNAL
}

##########################################################
##############################
## procedures called during link time ##
##########################################################
##############################
proc board_init {
Mem5105Space (0)
}


proc board_init_hex {
## Mem5105Space (1)
Mem5105Space_Flash
}
proc board_boot_init_hex {
Mem5105Space_BootFlash
}

##########################################################
##############################
## procedure called to run from DCU ##
##########################################################
##############################
proc board_runtime_init {
write
write  ------------------------------------------------
write MB400/STi5105 BOARD CONFIGURATION
write

## start initialising the board
## YK_SONG
## reset
reset -full

board_init

ST20C1MemoryInit ## clear C1 channel memory 0x80000000 to 0x80000040
mb400_setup

## Setup Complete, but display some basic version info
## 5105 Device ID, and Cut
## write
## temp = peek (STI5105_AF_DEVICE_ID) -q

## write "STi5105 DeviceId = " -d ((temp)>>12 & 0x000003ff)
## write "STi5105 CutNumber = " -d ((temp)>>28)

## EPLD FW version
## temp = peek (0x41000000) -q
## write MB400 EPLD Firmware Ver = -d (temp & 0x000000ff)
```

write
write Setup Complete
write --
write
}

BTW: DCU3 traphandler base is at &h30000000

That should be enough to get everyone going.
And last but not least the DCU3 control bin from ST.
The cpp r/w control file from SST is in there too for the 39VF3201 (4MB)

I'm not much for C/C++ so if jvvh or one of the other guys want to plug this info into the programs they are working on maybe we can get to further convertion of these maggie/skyview bins without worrying about puking the darn thing by having a viable jtag solution.

---

jvvh5897

04-21-2008, 01:56 PM

I get pulled in so many directions because I try to help where I can that jtagging these boxes was left to others to work out after I posted some of what I was able to figure out. What you have posted is much more than I had.

---

slugworth

04-21-2008, 03:49 PM

I relayed the info to the usbjtag guy,he was delighted.

MSSFWP

04-21-2008, 09:35 PM

I get pulled in so many directions because I try to help where I can that jtagging these boxes was left to others to work out after I posted some of what I was able to figure out. What you have posted is much more than I had. I know that slugworth and myself along with a few others have been working on these units since day one trying to figure out how to jtag these things. Our group has collected info from ST micro/SST/the OEM manufacturers, and also TR coders.

Our belief is that one day this unit may take the place of the 2500/2700 "conversion" units that you and others are working on now bro. You know as well as i do that even with the great coding/code crunching your doing on those 1MB units. Their days are seriously numbered.

With the Maggies/skyview/and at least three other "dual boot euro/american" models with the same generic board design. It would appear as if they are almost as common as the FLU/Pan design.

Pull the standard tuner and replace it with a now popular QPSK tuner that has superceeded the one in the maggie, and you'll have another new popular unit running the updated 5119 - the processor. Most of the coding is backwards compatible though at first glance. Not that you would really need to, but you could if you wanted. ;)

EDIT:
I just noticed in the first post the the DCU3 traphandler base is stated as 0x30000000. That is actually the DCU3 control register address. If you look at the references above, the DCU3 trap handler/base is loaded at 0xC07FFC00.

---

thebroken

10-05-2008, 03:14 AM

Trapper data...
Need Jtag :)
Treats your JTAG as a DCU
Magnum Base Image file...
STBURNER...

Since JVVH wants to help
st20run -t myuctap burner.lku

YLG80

11-10-2008, 04:34 AM

Hello,
The attached mak file found in an obscure location somewhere on the web could be of some help for you guys.
Regards.
YLG

YLG80

11-14-2008, 12:38 PM

Hello,
I'm stucked in the corner with the STx5105 Power on Reset vector.
I've dumped the flash of a new STB (rebadged from a Jiuzhou dts1601 STB STI5105) and I cannot find any Reset vector/jump @ 0x7FFFFFFEh.
From the MSSFWP post here above (thanks to him for that key information), I see that the FLASH BASE is @ 0x40000000h which was the address that I've used for the JTAG dump.
A few lines below in his post, I see : bootiptr (addressof -q FLASH)
Does that mean that this processor starts from 40000000h i.o 7FFFFFFEh ?
My binary dump should be OK as it looks very similar to a Goldvision Magic STB dump with the same processor and the same amount of flash mem (2 Mb).

Any help ... perhaps from Slugworth or JVVH5897 who were working on that processor ?
Thanks for your help

YLG80

11-14-2008, 12:47 PM

Here attached another schematic of a STx5105 set top box.
(CTV 2991 DVB-T)
Rather difficult to read (image size and resolution), but still could be interesting.

Regards

---

jvvh5897

11-14-2008, 01:49 PM

The 5105 does not use the same vectors as 5518/5500 type processors. Look for the entry point at the first byte in flash. The C1 code is very similar to the C2 code in many ways though, you have a two byte jump to within 254 bytes of the entry point, the second word is optional Wptr addr, the third word is sector table pointer and fourth word is start routines data table pointer (reversed order but the same words as you find in c2 compiled code).

I found that IDA Pro will do a fair job on dissassembly of the code, I've never seen that anyone built anything like Dave2's dasm, nor have I seen something like st20emu fo c1--it does look like you could set up the st20emu for c1 code and it could do disassembly too, you just need to work at the C code. I made up a list of c1 instruction set to try my hand at st20emu mods, but got distracted.

YLG80

11-14-2008, 02:09 PM

Thanks JVVH,
It helps a lot, as I was still looking on how to remap the bin dump to the STx55xx like devices.
I'm also using IDA and here are the first few instructions starting from the flash base address

```
ROM:04000000 ; Segment type: Pure code
ROM:04000000 ; section ROM
ROM:04000000 ldc 40000AA0h
ROM:0400000B startp
```

ROM:0400000C ldnlp 8
ROM:0400000D eqc 9
ROM:0400000E j loc_4000011

Thanks again

Flat

11-14-2008, 04:28 PM

Hello,
The attached mak file found in an obscure location somewhere on the web could be of some help for you guys.
Regards.

YLG

I guess that obscure place is:
h**p://read.pudn.com/downloads117/sourcecode/multimedia/dvd/496566/apps/appconfig.mak__.htm

It also seems to have a dvb-ref for the 5105, but I can't figure out how to download is.
It's chinese to me :-)

YLG80

11-15-2008, 03:38 AM

Yes, you're right.

Chinese is also my problem, although, I've been successful to register.
But php is still in plain english... so I could log and grab this, which I guess will be very useful for you. (specifically the file wt_flash.c)

What do you think about that?

Unfortunately not all files can be grabbed.
Regards

---

slugworth

11-15-2008, 08:31 AM

Yes, you're right.

Chinese is also my problem, although, I've been successful to register.

If you registered you could download the entire 1.3meg rar file.
There is a link to it but they want a login name/password.

YLG80

11-15-2008, 08:40 AM

Hello,
I've registered and logged in, as my surname is written on top of the pages.
But as far as I can understand, there are VIP files and free files.
I suppose that I need to upload sources to get a DL access or to pay a fee.
That one was not Free, so I guess was VIP.
But even for the Free files, Iv'e not been able to get a full download at once. As soon as I click on the green arrow to download, it shows a page with one line only , with 3 links in Chinese.(registration, comments, upload)
Would it be in Greek it would be ok, but chinese :eyebrows:....
Anyway, that site shows lots of interesting stuff.
Rgds

YLG80

11-15-2008, 02:18 PM

OK, I've got the complete 13Mb source file.
It was a matter of uploading sources in order to receive the DL access.

---

jvvh5897

11-16-2008, 11:52 AM

It really is not that big a deal to go from c2 inst set to c1--watch out for the jab command, it is much less intuitive than the return and gcall method. I found in my early disassembly of c2 code that the emulator st20emu stepping through code gave me a much better feel for how the code worked than just disassembly ever did.

But, if your interest is DCU3 and jtagging--you really need to look at the burner that was compiled for the xv3300 box and the jctap software that runs it on a device similar to the standard buffered jtag cable. It is posted in the advanced section here.

Flat

11-17-2008, 03:12 AM

OK, I've got the complete 13Mb source file.
It was a matter of uploading sources in order to receive the DL access.

Great! Would you mind putting it up on ,say, hostclip.com. I would love to take a look at it.

Regards,
flat

YLG80

11-17-2008, 03:48 AM

Hello,
I'll PM the link.
Regards
YLG

Flat

11-17-2008, 04:09 AM

Thanks to YLG80, here it is:

h**p://www.hostclip.com/dl/a9d525a33ff36a9cd0b3ae299a8e737a

YLG80

11-17-2008, 04:17 AM

Here is the 2Mb flash dump (S29AL016D70 from Spansion = AM29LV160D)
I've compared it to another STx5105 flash file (Golvision magic Surprise) also attached
I've tried, as advised by JVVH, to use the st20 emu to trace the boot, but after a few instructions, st20emu does not recognize the next instructions.
At it least it shows the first boot steps.

YLG80

11-17-2008, 04:34 AM

@JVVH

I've setup the burner (cc a previous post in this thread)
The readme is somewhat difficult to understand ...

I've added the CPU signature in the .cpu file and started the server in a dos instance.
It shows that the CPU has been recognized and the server has started (two instances !)

But I guess I have to create a specific cfg file for my board based on an example prior to run the described command. Is that a modified MB400 file ?
But where do I have to put that file ?

Thanks for advice.
YLG

---

slugworth

11-17-2008, 04:11 PM

You will have to come up with your own version of the loader.bin also.
The one included in the burner.rar looks like a modified loader from a skyview receiver.They use a 4meg flash and compressed firmware.

thebroken

11-17-2008, 05:01 PM

Sluggie, if it boots, you can load whatever bootloader you like. Sorry for throwing this together in a hurry... sheesh, so much for helping.

---

slugworth

11-17-2008, 05:24 PM

He is trying it on a 2meg receiver that may or may not use compressed firmware.
It he manages to get it to boot it's good to know what the receiver is going to expect for any serial loading/recovery.

MSSFWP

11-18-2008, 09:38 AM

Sluggie, if it boots, you can load whatever bootloader you like.

This is a very important point here brokie, this thread was started as a Jtag project not a bootloader modification thread. Which really are two completely different things.

Brokie your fine, please continue to contribute.

As far as the flash memory size questions, or whether it uses compressed/uncompressed firmware, that's really irrellevant also in the scheme of things as long as you know where it's mapped (since the DCU looks at the raw flash information and not it's actual configuration). You also need to know whether or not it's a bottom or top boot flash. I see some of the posts reference top boot flash chips, but in this case since were really referencing the maggie/skyview for the most part it's a bottom boot (4MB) flash device for those who haven't already looked at the SST 39VF3201 datasheet provided.

---

thebroken

11-18-2008, 10:21 AM

He is trying it on a 2meg receiver that may or may not use compressed firmware.
It he manages to get it to boot it's good to know what the receiver is going to expect for any serial loading/recovery.

to communicate with the uctap, you must do

make TARGET=mytarget run
where mytarget is a jei_soc connection type in the target.cfg file.

---

jvvh5897

11-18-2008, 11:54 AM

And as for jtagging rather than using uctap--I always figured that once you get uctap working, you could sniff the comms to the box to figure out the method of building a jtagging program. Now sniffing parallel port running at the speed that a PC lets that port run can be a pain in the butt, but once a method is found doing it in stand alone program should be possible.

---

slugworth

11-18-2008, 12:57 PM

Pins 1 and 2 on the processor are dcu_trig_in and dcu_trig_out,which go to the jtag connector pin 5 and pin 3,which is interesting.
If you are looking for a schematic with the st5100, the lg_lcd_tv_42lf66-ze pdf has a good schematic.
You can get it off the net.

YLG80

11-18-2008, 01:51 PM

Thank all for the useful information.
I'm indeed trying to JTAG that receiver with the STi5105 and the above described tools.
Config is :
STi51505, 2Mb S29AL016D (in bottom mode) flash likely @ 0x40000000, 16Mb RAM unkown address page.
The dump looks OK as compared to another STi5105 system.

As said Slugworth, since the first JTAG flash dump (via Jkeys Save Mem), the receiver can no longer boot. It appears completely dead. Strange for a read only process, but could be due to protection. This is the reason why I'm tracing the bootloader.
I'm going to check tomorrow with a scope if it's not a RAM problem, as JTag Tester has not been able to find any RAM in that system, although it has 16 Mb. But perhaps the JTAG needs also the DCU3 trap routine to test the RAM.
I'll keep you updated.
Regards

---

slugworth

11-20-2008, 08:01 AM

Does it matter what version of st20 toolset you use?

I found uctap bundled with st20r1.8.
How about the operating system? would win98 be better?

---

jvvh5897

11-20-2008, 01:00 PM

Back when uctap came out, one of the writers said that any toolset should be good (but I'm not that sure of the processor support in the older ones--I seem to recall that the 5105 required writing info into the toolset to tell it how to deal with that processor).

As long as you have the right port control dll--which I think came with the program--operating system should not matter.

---

slugworth

11-20-2008, 02:13 PM

The uctap readme says to use pin17 on the jtag connector instead of 19.
If I do that,uctap doesn't detect the processor.
I also get a uctap error "the service cannot be started,either because it is disabled or because it has no enabled devices associated with it"
Uctap detects the processor then shows running 2x.

---

slugworth

11-20-2008, 07:33 PM

to communicate with the uctap, you must do

make TARGET=mytarget run
where mytarget is a jei_soc connection type in the target.cfg file.

I "set target=mybox",there is no target.cfg so I made one but st20run still can't find target.Is uctap supposed to use the config file or st20run?

---

jvvh5897

11-21-2008, 01:01 PM

both cfg and strun are used. STrun is really the program that lets you communicate to the box and it uses cfg file to tell it parts of how it needs to do it.

You use STRun to run the burner code. The burner code has the c1 code needed to communicate over the jtag port in the way needed to do the burner tasks.

---

slugworth

11-22-2008, 07:14 PM

Something else to play with.
includes the v.02 uctap
For the sti5514 processor,but you don't need the full install of stm toolset and it shows a working example of an st20run/xxx.lku situation.

---

slugworth

11-23-2008, 09:13 PM

I got past the target unknown error,now I get load error-operation not available.Variable CPUType[0] not defined,use chip command.Uctap detects the cpu but I am guessing at the other addresses in the .cpu file.Wtmecburner.lku gives the same error.

---

slugworth

11-24-2008, 02:37 PM

I hooked up a sti5518 based receiver I know jtag works with to make sure everything was copasetic,I couldn't get the old blacklist flashprg.lku to work either,same error.The burners won't talk to uctap.I tried jei_soc 127.0.0.1 or jei 127.0.0.1 no go.

---

jvvh5897

11-24-2008, 03:44 PM

Are you using regular jtag circuit or the slightly modified one needed for the uctap program that has the reset line to pin 17 of the jtag connector rather than the pin 19?

---

slugworth

11-24-2008, 04:59 PM

Are you using regular jtag circuit or the slightly modified one needed for the uctap program that has the reset line to pin 17 of the jtag connector rather than the pin 19?

won't work with the reset line to pin 17,doesn't detect the processor using that pin.Then I get target not found,check jtag connections.
For the sti5518 I used st20run -b -i mb5518um.cfg -t uctap flashprg.lku
same error all the time.

MSSFWP

11-25-2008, 05:05 AM

Well if your trying to use the .02ucTap program sw you posted Slugworth, it's configured for a 2MB flash with C2 boot configuration.
Also targets.conf needs to be updated, and the JEI socket needs to be properly configured for something other than localhost IP. ;)

Oh btw: I think pin 17 is nTRST, and won't work w/o specific jtag diagnostic equipment.

MSSFWP

11-25-2008, 06:10 AM

The full pinout on the MB400 JTAG is:

All Even (2-20) = Ground

ODD:
1) N/C
3) Error_out << 'pulled up with 10K
5) CPU_Analyse >> 'pulled up with 10K
7) N/C
9) TMS >> 'pulled up with 10K
11) TCK >> 'pulled up with 10K
13) TDI << 'pulled up with 10K
15) TDO >> 'pulled up with 10K
17) STCON_Reset# >> 'pulled up with 10K
19) TRST# >> 'pulled up with 10K

---

slugworth

11-25-2008, 07:49 AM

Well if your trying to use the .02ucTap program sw you posted Slugworth, it's configured for a 2MB flash with C2 boot configuration.
Also targets.conf needs to be updated, and the JEI socket needs to be properly configured for something other than localhost IP. ;)

Oh btw: I think pin 17 is nTRST, and won't work w/o specific jtag diagnostic equipment.

I just find it irritating that I can't get it to work on a sti5518 based receiver using different pc's and different os'es.

On a receiver that can be jtagged with other programs yet
I get the cputype[0] error,but if I then power off the receiver I get target not found;which means the burner/uctap is almost talking to the receiver.

MSSFWP

11-25-2008, 11:57 AM

I just find it irritating that I can't get it to work on a sti5518 based receiver using different pc's and different os'es.
On a receiver that can be jtagged with other programs yet.
I get the cputype[0] error,but if I then power off the receiver I get target not found;which means the burner/uctap is almost talking to the receiver.I'm not sure what version of test kit your trying this in, Someone did make a mention of the fact that your config files have to have the 5105 chip info in them.

---

thebroken

11-25-2008, 12:20 PM

I just find it irritating that I can't get it to work on a sti5518 based receiver using different pc's and different os'es.

On a receiver that can be jtagged with other programs yet.
I get the cputype[0] error,but if I then power off the receiver I get target not found;which means the burner/uctap is almost talking to the receiver.

post a copy of your cfg files, CPUType[0] is a cfg file error...

---

slugworth

11-25-2008, 03:25 PM

I'm not sure what version of test kit your trying this in, Someone did make a mention of the fact that your config files have to have the 5105 chip info in them.

V2.12 which has no jpi_ppi dll anymore.
Any change to the tap config involving the jpi and the program burps looking for a dll that doesn't exist.So I am stuck with jei_soc 127.0.0.0.1 and uctap that I don't know the addresses for.
uctapsrv.cpu STi5105 ,0x2d427041 ,0x30003004,0x80000000 ?
Uctap must show "the operation completed" 3x for any hope of success.
Most of the time I get.
C:\pvr2flash>uctapsrv.exe -p 0x378
ucTAP v 0.2 beta May 31 2006 (c) 2006 TAPDancers
ucTAP is a program to emulate ST20 Micro Connect functions over a basic JTAG in terface.
The specified service has been marked for deletion.

The specified service has been marked for deletion.

The service cannot be started, either because it is disabled or because it has n o enabled devices associated with it.

OR

The specified service does not exist as an installed service.

The specified service does not exist as an installed service.

The specified service does not exist as an installed service.

st20run -i mb400.cfg -t tap burner.lku
which is the only line that almost does something.
I think I need 5105ckg.cfg?

---

slugworth

11-25-2008, 07:07 PM

target tap tap "jei_soc 127.0.0.1" board_runtime_init
but stops at an error in the mb400.cfg file.

---

jvvh5897

11-26-2008, 11:11 AM

I would interpret that last error message as showing you haven't run a .bat file to set environment settings. Before I can compile something like STAPIDEMO I have to run a bat file that does a bunch of DOS "set" commands--think you need something similar.

I looked at the copy of burner that I dn'ed here and I see referances to .cfg files that I can find in the ST Toolset 2.1.2 but not in earlier toolsets. But for the program to see those files your env must be set correctly too.

---

slugworth

11-26-2008, 01:20 PM

All the examples of st20run xxx.lku on the net show involvement of a motherboard.cfg file which will have as a part "##includes include targets.cfg
So like st20run -i mbxxx.cfg -t tap xxx.lku
A legit install of st20r2.xx will set the environment,I double check that in windows system advanced settings- environment.The drag and drop installs of the toolset I set the environment manually myself.The pace3100.rar was a readers digest form of the toolset so that took manual environment setting.
I am concerned about the uctap errors on loading but the display shows that it is getting the idcode from the processor,so it was talking a little.
The tap jei_soc"127.0.0.1" (tgtproc) is still hazy,I don't know what command st20run expects.I assume it is a reference to a process in the mb400.cfg file.
That is the only way so far to get rid of the cputype[0] error.

MSSFWP

11-27-2008, 12:20 AM

All the examples of st20run xxx.lku on the net show involvement of a motherboard.cfg file which will have as a part "##includes include targets.cfg
So like st20run -i mbxxx.cfg -t tap xxx.lku
A legit install of st20r2.xx will set the environment,I double check that in windows system advanced settings- environment.The drag and drop installs of the toolset I set the environment manually myself.The pace3100.rar was a readers digest form of the toolset so that took manual environment setting.
I am concerned about the uctap errors on loading but the display shows that it is getting the idcode from the processor,so it was talking a little.
The tap jei_soc"127.0.0.1" (tgtproc) is still hazy,I don't know what command st20run expects.I assume it is a reference to a process in the mb400.cfg file.
That is the only way so far to get rid of the cputype[0] error.I found these on the net in some obscure place with a little searching. :)

MSSFWP

11-27-2008, 01:09 AM

V2.12 which has no jpi_ppi dll anymore.
Any change to the tap config involving the jpi and the program burps looking for a dll that doesn't exist. So I am stuck with jei_soc 127.0.0.0.1 and uctap that I don't know the addresses for.I posted the board config, as well as the standard config files, along with the updated dlls for the bin folder. I would say just extract them out to the folders they belong in, and overwrite the ones you have now, or add them in where they should already be. They weren't even that hard to find really. Just a bit of file searching involved.

btw: jpi_ppi.dll has been renamed to tm_jpi_ppi.dll in later editions. ;)

---

slugworth

11-27-2008, 05:58 PM

Since the pace3100 files work for the pace with sti5514 and same interface,I will have to try to convert all those .cfg files to work with the sti5105.
That is the only way I can avoid the cputype[0] error.
It is the only full example of st20run+xxx.lku approach I have.
All references to c2 have to be c1 and all the memory addresses have to be edited.
I may get it done before the receiver is obsolete.

---

slugworth

11-28-2008, 12:37 PM

uctap screenshot
At least it's trying to do something now.
I think it's making it to the burner.lku stage at least.

---

slugworth

11-28-2008, 06:46 PM

I believe giveio should show under device manager hidden non pnp drivers,but it doesn't.It should also show as an installed service under windows services and it doesn't.Which leads me back to post #45 there should be no error messages when loading uctap.I think I am going to have to wait until I can play with my win98 pc again. Skymax,which uses userport.sys or giveio for xp use, doesn't detect the processsor which leads me to believe uctap is starting as a server but can't talk to the lpt1 port correctly.

---

slugworth

11-28-2008, 10:19 PM

Different pc but still xp sp2.
I think it's the (tgtproc) fouling me up

uctapsrv.exe -p 0x378 -v 2
st20run -b -i mb400.cfg -t mag burner.lku
target mag tap "jei_soc 127.0.0.1" board_init
If I use board_init I get a burst of activity watching the uctap screen.
But then the process hangs at memset trap 0
Other choices for (tgtproc) yield different results,usually erroring out sooner.
I tried using a good receiver also, in case the .cfg files were designed for a good receivers response.
With no (tgtproc) I get the cputype[0] error and it says
to use the chip command,which is in the mb400.cfg anyway.

YLG80

12-03-2008, 05:11 AM

Hello,
@slugworth I've pm'd you a file that seems to work.
Here is what the burner window shows when ucTap is launched.
Unefortunately I cannot get any further as my box has been returned for repair.
I'll test it as soon as possible, but perhaps you can do some trials on your own.
This is a compilation (4Mb flash) from the burner files found in the DTV 5105 file (see link in this thread)

Screenshot of the DOS window when ucTap launched
C:\5105_flash_burner\ucTAP>tapls

C:\5105_flash_burner\ucTAP>uctapsrv -p 0x0378 -t -v 3
ucTAP v 0.1 beta Apr 12 2006 (c) 2006 TAPDancers
ucTAP is a program to emulate ST20 Micro Connect functions over a basic JTAG interface.
Operation successfull.

Operation successfull.

Operation successfull.

WARNING!! Your CPU is not known DCU_CTRL 00003004, DCU_DEVICE_ACCESS_MASK 00080000
[Of course because there is no CPU connected, but when it was connected it was showing the right ID !]

Starting a server on 9737
Starting a server on 9735
[process_request_9735][9735] data = 128
05 00 00 00 00 00 00 00
packet 05
[process_request_9735][9735] data = 128
01 00 00 00 00 00 00 00
[process_request_9735][9735] data = 128

00 00 00 00 75 63 72 2e
[process_request_9735][9735] data = 16384
71 21 58 72 e0 71 22 50
[process_request_9735][9735] data = 16384
94 74 22 3a 21 aa 74 22
[process_request_9735][9735] data = 16384

73 77 f9 a3 80 62 0c 42
[process_request_9735][9735] data = 16384
2b 55 72 e1 71 28 24 2c
[process_request_9735][9735] data = 16384
a4 12 d0 42 12 18 6c 2d
[process_request_9735][9735] data = 1064
60 ce c0 21 ac 7b 60 cd
[process_request_9735][9735] data = 16384
30 00 00 00 00 00 16 00
[process_request_9735][9735] data = 16384
00 08 00 a5 00 00 18 01
[process_request_9735][9735] data = 16384
00 00 80 0a 50 01 2a 80
[process_request_9735][9735] data = 16384
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 16384
6c ff 00 00 00 00 00 00
[process_request_9735][9735] data = 16384
00 00 00 01 22 00 04 40
[process_request_9735][9735] data = 16384
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 7952
00 00 00 00 00 00 00 00
my_write() failed; 532 -1 err=10054closing 9735
YLG80@YLG-pc connected on Wed Dec 03 12:50:18 2008

[get_hosted_poke] val = 000000ff
P[write_ack]02 00 00 00
02 00 00 00 81 01
IDCODE read -1
[write_ack]06 00 00 00
06 00 00 00 a1 01 ff ff
closing 9737
[process_request_9735][9735] data = 128
05 00 00 00 00 00 00 00
packet 05
[process_request_9735][9735] data = 128
01 00 00 00 00 00 00 00
[process_request_9735][9735] data = 128
00 00 00 00 75 63 72 2e
[process_request_9735][9735] data = 16384
71 21 58 72 e0 71 22 50
[process_request_9735][9735] data = 1136
94 74 22 3a 21 aa 74 22
[process_request_9735][9735] data = 16384
25 fa d0 23 a0 21 23 48
[process_request_9735][9735] data = 1136
71 26 9a 41 22 f0 22 40
[process_request_9735][9735] data = 16384
85 f1 48 24 f1 7d 51 f1
[process_request_9735][9735] data = 16384
62 2e 95 75 41 24 f6 a6
[process_request_9735][9735] data = 16384
22 77 40 f9 2e a3 70 26
[process_request_9735][9735] data = 16384
37 20 00 00 00 00 00 00
[process_request_9735][9735] data = 16384
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 16384
40 00 0f 00 00 00 00 04
[process_request_9735][9735] data = 16384
00 00 00 00 00 00 00 03
[process_request_9735][9735] data = 16384
a0 40 00 00 00 20 00 00

```
[process_request_9735][9735] data = 16384
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 13984
44 20 00 00 00 40 80 86
my_write() failed; 532 -1 err=10054closing 9735
YLG80@YLG-pc connected on Wed Dec 03 12:50:46 2008

[write_ack]02 00 00 00
02 00 00 00 81 01
IDCODE read -1
[write_ack]06 00 00 00
06 00 00 00 a1 01 ff ff

closing 9737
[process_request_9735][9735] data = 128
05 00 00 00 00 00 00 00
packet 05
[process_request_9735][9735] data = 128
01 00 00 00 00 00 00 00
[process_request_9735][9735] data = 128
00 00 00 00 75 63 72 2e
[process_request_9735][9735] data = 16384
71 21 58 72 e0 71 22 50
[process_request_9735][9735] data = 1136
94 74 22 3a 21 aa 74 22
[process_request_9735][9735] data = 16384
25 fa d0 23 a0 21 23 48
[process_request_9735][9735] data = 1136
71 26 9a 41 22 f0 22 40
[process_request_9735][9735] data = 16384
85 f1 48 24 f1 7d 51 f1
[process_request_9735][9735] data = 1136
62 2e 95 75 41 24 f6 a6
[process_request_9735][9735] data = 16384
81 d3 21 44 73 f9 a2 60
[process_request_9735][9735] data = 1136
81 d4 f1 72 f1 f4 c0 aa
[process_request_9735][9735] data = 12904
2f 4f 24 f6 22 f0 73 d1
[process_request_9735][9735] data = 16384
30 00 00 00 00 00 16 00
[process_request_9735][9735] data = 1136
00 08 00 a5 00 00 18 01
[process_request_9735][9735] data = 16384
44 21 99 91 00 02 00 00
[process_request_9735][9735] data = 1136
00 00 00 00 00 00 30 00
[process_request_9735][9735] data = 16384
00 00 00 00 f0 fe 08 00
[process_request_9735][9735] data = 1136
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 16384
00 00 20 00 00 00 00 00
[process_request_9735][9735] data = 1136
00 8a 50 04 38 11 45 e8
[process_request_9735][9735] data = 16384
00 a0 09 06 80 16 d0 04
[process_request_9735][9735] data = 1136
6c fd 20 00 00 00 00 00

[process_request_9735][9735] data = 16384
00 00 00 00 00 00 04 00
[process_request_9735][9735] data = 1136
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 16384
00 00 00 00 00 00 00 00
```

[process_request_9735][9735] data = 1136
00 00 00 00 00 00 0c fd
[process_request_9735][9735] data = 16384
40 00 00 00 00 00 00 00
[process_request_9735][9735] data = 1136
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 268
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 3400
00 00 00 00 08 00 00 00
[process_request_9735][9735] data = 20
07 1a 01 00 98 36 01 00
[process_request_9735][9735] data = 16
73 1c 01 00 6b 1a 01 00
[process_request_9735][9735] data = 10
74 6d 6a 65 69 2e 64 62
closing 9735
YLG80@YLG-pc connected on Wed Dec 03 12:51:34 2008

[write_ack]02 00 00 00
02 00 00 00 81 01
IDCODE read -1
[write_ack]06 00 00 00
06 00 00 00 a1 01 ff ff
closing 9737
[process_request_9735][9735] data = 128
05 00 00 00 00 00 00 00
packet 05
[process_request_9735][9735] data = 128
01 00 00 00 00 00 00 00
[process_request_9735][9735] data = 128
00 00 00 00 75 63 72 2e
[process_request_9735][9735] data = 16384
71 21 58 72 e0 71 22 50
[process_request_9735][9735] data = 1136
94 74 22 3a 21 aa 74 22
[process_request_9735][9735] data = 16384
25 fa d0 23 a0 21 23 48
[process_request_9735][9735] data = 1136
71 26 9a 41 22 f0 22 40
[process_request_9735][9735] data = 16384
85 f1 48 24 f1 7d 51 f1
[process_request_9735][9735] data = 1136
62 2e 95 75 41 24 f6 a6
[process_request_9735][9735] data = 16384
81 d3 21 44 73 f9 a2 60
[process_request_9735][9735] data = 1136
81 d4 f1 72 f1 f4 c0 aa
[process_request_9735][9735] data = 12904
2f 4f 24 f6 22 f0 73 d1
[process_request_9735][9735] data = 16384
30 00 00 00 00 00 16 00
[process_request_9735][9735] data = 1136
00 08 00 a5 00 00 18 01
[process_request_9735][9735] data = 16384
44 21 99 91 00 02 00 00
[process_request_9735][9735] data = 1136
00 00 00 00 00 00 30 00
[process_request_9735][9735] data = 16384
00 00 00 00 f0 fe 08 00
[process_request_9735][9735] data = 1136
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 16384
00 00 20 00 00 00 00 00

[process_request_9735][9735] data = 1136
00 8a 50 04 38 11 45 e8
[process_request_9735][9735] data = 16384
00 a0 09 06 80 16 d0 04
[process_request_9735][9735] data = 1136
6c fd 20 00 00 00 00 00
[process_request_9735][9735] data = 16384
00 00 00 00 00 00 04 00
[process_request_9735][9735] data = 1136
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 16384
00 00 00 00 00 00 00 00

[process_request_9735][9735] data = 1136
00 00 00 00 00 00 0c fd
[process_request_9735][9735] data = 16384
40 00 00 00 00 00 00 00
[process_request_9735][9735] data = 1136
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 268
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 3400
00 00 00 00 08 00 00 00
[process_request_9735][9735] data = 20
07 1a 01 00 98 36 01 00
[process_request_9735][9735] data = 16
73 1c 01 00 6b 1a 01 00
[process_request_9735][9735] data = 10
74 6d 6a 65 69 2e 64 62
closing 9735
YLG80@YLG-pc connected on Wed Dec 03 12:52:13 2008

[write_ack]02 00 00 00
02 00 00 00 81 01
IDCODE read -1
[write_ack]06 00 00 00
06 00 00 00 a1 01 ff ff
closing 9737

---

slugworth

12-03-2008, 09:31 AM

uctap still gives errors starting up,and the sti5105 wasn't in the uctapsrv.cpu file, I had to add it.
after setenv.bat (st20r2.12 was already loaded and env correct for that) I ran nogui.bat and get
st20run -L . -i mb400.cfg -t wtmecburner.lku
connect error-connect error unknown target wtmecburner.lku
Shouldn't the target be sti5105box like in the targets.cfg?

YLG80

12-03-2008, 12:07 PM

This is strange, because the STi5105 is included in the ucTAPsrv.cpu file included in the zip file.

I first launch ucTAP with : tapls.
Then I launch burner with : nogui sti5105box
It just give the same results, ending with an error :no target device detected.
But this is because there is no STB connected to the LPT1.
The long data burst in the ucTap window occurs immediately when launching the burner.

One remark :
Since my previous message, I 've rebooted my PC and there was a problem running gmake.
There is likely something wrong with my sentv.bat, because I had also to launch the toolkit in order to be able to run gmake.

This is what I get when I launch the burner

C:\5105_flash_burner>gmake run TARGET=sti5105box
st20run -l . -i mb400.cfg -t sti5105box wtmecburner.lku
error-kernel error : no target device detected - check target and JTAG connection

And each time the long data burst occurs in the ucTap window, which according to me means that the microconnect emulation is correctly working.

---

slugworth

12-03-2008, 01:41 PM

It doesn't like the board_runtime_init_noreset,since that string isn't in the mb400.cfg file.
I was using 30000000 as the dcu address and 00080000
for the mask.

YLG80

12-03-2008, 01:53 PM

Not sure if it can explain that error, but I've saved the stdcfg and board files uploaded by MSSWFP in the R2.1.1 related directories.
Hoewer I've not used his dll files as there was an error during the compilation.
(Could not find dll entry point)

I've used board_noreset and board_runtime_init_noreset with the same result.
I've seen that you are using the ucTap version 0.2b.
I've tried with that version with the same result.

---

slugworth

12-03-2008, 02:17 PM

I believe the mb400.cfg was geared towards the development board and has to be edited to match your actual receiver.Some receivers have 32meg of ram,euro ones have 64 or 128meg or 256m.
The only (tgtproc) that almost do something for me are the board_init or board_runtime_init.Both of those strings are in the mb400.cfg.I have different versions of that same file,I assumed they were from st20r2.2.1 but then I saw dcu_mb400.cfg which is even newer.
When it almost does something the peeks/pokes are wrong.The peeks always come back as all FF's.
The pace3100 approach for the sti5514 actually involved 2 lku files,one for booting then the other for the actual flashing.If I could dissect those maybe I could see how they approached the problem

---

slugworth

12-03-2008, 02:22 PM

From the mb400.cfg,I don't think it should be ## out.

##YK_SONG
##org
32MB RAM
memory EXTERNAL 0xC0000000 (8*_MB_ - 65*_KB_) RAM
memory mytracebuffer 0xC07EFC00 (64*_KB_) RAM
memory TRAPHANDLER 0xC07FFC00 (1*_KB_) DEBUG
memory AVMEM 0xC0800000 (16*_MB_) RAM
memory NONCACHED 0xC1800000 (8*_MB_) RAM

---

slugworth

12-03-2008, 06:36 PM

Using pin17 instead of pin19 is a must.

I tried using pin17 at the beginning of the series but I was leaving pin19 floating.
Then I tried using pin17 and 19 connected together.I just connected pin 17 instead of pin 19 and connected 19 thru a 100ohm resistor to pin12 on lpt1.At least I am getting to the wtmecburner menu now.
Definitely not an "as is" project, every .cfg file had to be edited.
Now I am seeing if it is really erasing the flash.

slugworth

12-03-2008, 10:03 PM

The program thinks the receiver has an 8meg flash and tries to erase 135 blocks but the flash never gets erased.

YLG80

12-04-2008, 08:49 AM

Thanks for that information. This is good news, as the burner seems to run correctly.
Do you know why the toolset must be running while using the burner ?
On the 8Mb flash, you're right.
In the appmakfile.mak for the DTV this is the description related to the flash:

```
## Flash part number: M28W640FS(DTT5107
8MB)/M29W800DB(MALY3_1MB)/M29W160DB(MALY3_2MB)/M29W320DB(MALY3_4MB)/
M58LW064D(MBXXX)
ifeq "$(DVD_HARDWARE)" "DTT5107"
FLASH_TYPE := M28W640FS
else
FLASH_TYPE := M29W320DB
endif
```

It uses the M28W640FS.

So I guess they have configured the wts_main.c in accordance with that flash size.
Or may be could be changed in the mb400?

---

slugworth

12-04-2008, 09:13 AM

Trying to get to the source of the problem.
You can run wtmecburner from the command line and you get the same results.So gmake isn't really needed an actually doesn't make sense since the lku is already made and you are just using the run portion of the make file.
st20run -i mb400.cfg -t sti5105box wtmecburner.lku
The settings in the uctapsrv.cpu file make no difference either,so apparently we are using uctap just for the lpt1 portion.
Wt doesn't get the flash man id or device id correct; 0x6019 and 0x6c2d and still thinks the flash is 8meg no matter what the mb400.cfg says.I manually selected the 2meg flash,same problem.I think wtmecburner has to be remade.
Burner.lku doesn't work at all,it gets stuck at "memset trap 0" and times out.

YLG80

12-04-2008, 10:26 AM

Yes I believe that we'll have to recompile the lku for other flash/ram settings

On the uctap, I agree that it's only used to emulate the ST Microconnect via the LPT. The cpu settings are used only to check if the connection is OK.
From the readme :
If ucTAP does not correctly identify the CPU, the variables DCU_CTRL_ADR and DCU_CTRL_MASK should be set via the command line options

So we don't have to worry about the DCU_CTRL_ADDRESS and MASK providing that we supply the correct CPU ID.

---

slugworth

12-04-2008, 10:54 AM

I believe one of the mb400.cfg was set for 16meg ram.
If you use that setting you get stuart_init<>=stuart_error_pio error calling pio command <0x00160005> and the program gets stuck trying to erase.
You have to un ## the 32meg setting in the mb400.cfg and ## the 16meg.

The LMI_COC_UPPER and LMI_COC_LOWER address are wrong they should be 0xC0000000+ and they are showing as 0x000C8750 and 0x00002000
I will have to look at that.

---

slugworth

12-04-2008, 12:04 PM

If you run the command line
st20run -i mb400.cfg -t sti5105box -log fooey.txt -v
The LMI settings are wrong before you even get to the wtmecburner.
The upper/lower are actually pokes that shouldn't even be on the screen.
And the LMI_STR is wrong.

YLG80

12-05-2008, 02:49 AM

Don't we have to include the appconfig.mak that I've added in the make directory. At the very beginning of the appconfig.mak it says that we should include it in the generic.mak :

The proper way to use this is to include this file
in the generic.mak file which is found in every
STAPI-tree under the <root>\make directory.
Example entry in the generic.mak:
SINGAPPL, include app configuration
include $(DVD_TARGET_PATH)/appconfig.mak
#
Determine the build environment
include $(DVD_MAKE)/environment.mak
#
Include default DVD platform configuration
include $(DVD_MAKE)/sysconf.mak

I'm working actually on the 5518 burner and have been able to rebuild a flashprg.lku.
Many environment vars have to be included in order to make it compilable.
But that version needs an older toolset version to run. I'll try with the R205 version

---

slugworth

12-05-2008, 08:38 AM

I tried tweaking the mb400_mem.cfg but wtmecburner still thinks the flash is 8meg and won't erase.
Uctap shows my_write() failed; 6 -1 err=10053 recv error: err=10053 at the end.

---

jvvh5897

12-06-2008, 02:26 PM

I know that there is a flash protect pin that might block erase/writes to the boot block, could that be a problem?--extremeview-3300 box. Can't recall if the resistor on the line enables writes or stops them--been a while since I played with that box.

---

slugworth

12-06-2008, 02:46 PM

Resistor on protects the boot block,resistor is off so you can program the bootsector.(and screw up the bootsector)boo hoo.
The uctapsrv.cpu file; if you leave the dcu_control_mask and control_device_access addresses blank,uctap comes up with it's own values which were totally different from the ones we were using.
wtmecburner still comes up with the wrong flash manu id and device code and thinks the flash is 8meg.Burner.lku doesn't work at all,gets hung up on mem set 0

---

slugworth

12-06-2008, 02:52 PM

The .map files from burner.lku and wtmecburner.lku show the flash at address 40080000,which to me doesn't seem right.
It should be 40000000
Time for a new .lku file I think.

YLG80

12-07-2008, 01:20 AM

On the 5518 version I had to erase the lku file prior to rebuild it.
Otherwhise it remains unchanged after a recompilation.

MSSFWP

12-07-2008, 06:56 AM

I know that there is a flash protect pin that might block erase/writes to the boot block, could that be a problem?--extremeview-3300 box. Can't recall if the resistor on the line enables writes or stops them--been a while since I played with that box.If i recall, it's left floating for which this part of the 3201's datasheet makes reference to.

The SST39VF1602/3202/6402 support top hardware block protection, which protects the top 32 KWord block of the device. The SST39VF1601/3201/6401 support bottom hardware block protection, which protects the bottom 32 KWord block of the device. The Boot Block address ranges are described in Table 2. Program and Erase operations are prevented on the 32 KWord when WP# is low. If WP# is left floating, it is internally held high via a pull-up resistor, and the Boot Block is unprotected, enabling Program and Erase operations on that block.

Since the maggie has it floating, it is NOT protected, at least not in "HW mode". So the next thing to look at would be the user security ID that can be programmed in. This is referenced again in the datasheet as this.

The SST39VF160x/320x/640x devices offer a 256-bit Security ID space. The Secure ID space is divided into two 128-bit segments - one factory programmed segment and one user programmed segment. The first segment is programmed and locked at SST with a random 128-bit number. The user segment is left un-programmed for the customer to program as desired.

It should be noted that this area is OTP, as refered to later in the datasheet.
Once i looked at the C files for the burner for which i admit i'm weak in understanding. I see a reference to a program sequence in the write routine with an 8 byte leader (i actually noticed this in the write routine source of several units using the 3201). "12345678" that wouldn't be the user programmable ID for the flash write routine would it ?

Nope just found it, 12345678 is an LMI base address poke. wtf: the addresses are all over the place with this thing.

---

slugworth

12-07-2008, 10:12 AM

the mb400.cfg has an if then statement that has a == I never saw before,mistake?

if (_AMERICAN_MODEL_ == "1"){NCACHE_SIZE = (1*_MB_)AVMEM_SIZE = (11*_MB_)}
the _AMERICAN_MODEL_="1" is defined at the beginning of the .cfg

---

YLG80

12-07-2008, 02:18 PM

According to me the statement is correct. == is a logical operator.

---

slugworth

12-13-2008, 12:02 PM

Wt doesn't get the flash man id or device id correct; 0x6019 and 0x6c2d and still thinks the flash is 8meg no matter what the mb400.cfg says.
The 6019 and 6c2d are actually from addresses 40000001 to 40000004 reversed, so wt isn't doing the detect id's ritual correctly.

---

alarm

02-07-2009, 11:11 AM

STiControl beta release
This is a beta test release. There are still some things I don't like and others that do not really work the way they should. That will be fixed sometime later. There is no warranty of any kind. Use it at your own risk!

The tool expects to find a STi5519 or STi5508 based Vestel dvd player environment (mainboard labeld "12MB04" or "12MB03").

Starting with version 0.4 the Cyberhome DVD 505 environment will also be recognized: STi5508 with the 2MB flash memory chip M29W160DT from STMicro and the mainboard marked with: "19SA01, Rev: C01"

If the trap handler cannot be installed properly, try to connect immediately after a reset. (nRST-button should be there on your JTAG connector). Or, press connect right after you turned on the unit (though, it's tricky to hit the right moment). The nRST line is also supported when connected to parallel port pin D4.

http://img6.imageshack.us/img6/4361/87598844hq1.png
http://img6.imageshack.us/img6/8096/27957636jy8.jpg
JTAG to parallel interface circuit diagram

If you find this too complicated, you may do it the simple but more risky way: build an interface that consists of just a cable with the parallel port plug on the one side's end and the JTAG plug on the other. Inbetween you have to put those serial inline resistors as shown in the circuit diagram.
If you do it this way, you have to pay a lot more attention: only use short cables and no extension cords. All in all, the cable from your computer's parallel plug to the dvd player's JTAG connector should not exceed 1.5 meters in length. Take care not to cause electrical discharges to this cable or to any other parts of the dvd player but the metall case!

Now connect everything, switch on the computer but leave the dvd player off until you've started STiControl. Power-on the dvd player and try to establish a logical connection by pressing the connect button. You may have to repeat this some times until you get a proper connection. This is due to the limited information I have about the decoder chip (in fact, there's almost nothing but useless propaganda publicly available from the manufacturer).

---

slugworth

03-06-2009, 09:10 PM

I can't get it to work with the command line they show,and it still has to be modded for the 4meg flash the maggie has.From st20 toolset v2.3.1

---

slugworth

03-07-2009, 09:28 AM

The readme was wrong,you need quotation marks after the .lku on the command line or it burps and stops.

st20run -b -i mb400.cfg -t mag flashprg_c1c.lku "-args -b MB400 -e"
You have to use the -i mb400.cfg or st doesn't find the target.
The .c file(s) still have to be modded for the maggie flash instead of the default 8meg flash the mb400 uses.Then the .lku remade.

YLG80

03-09-2009, 07:14 AM

Regarding post #45. (wtmecburner)
You were right, that file is missing as well as the 5105regs.cfg.:cry:
The author has decided to customize and split the toolset dcu_mb400_conf.cfg in 3 files and use them locally in the project.(one for the clock, one for the registers, one for the memory)
The existing mb400_mem.cfg is a part of the generic TS mb400 config file.

The author has also truncated the variables names found in the toolset cfg file but we

could likely rebuild a usable file.:typing:

The includes in the project mb400.cfg file (starting from line 31) can be changed so that only one include to a complete file would be enough to replace the 3 originals.

YLG80

03-09-2009, 03:13 PM

In an attempt to rebuild the flash prog tool for the 5105 I get an error just before the final lku build (linking). (see below)
Any idea on how to debug that error ?

```
--------------------------------------------------
MB400/STi5105 BOARD CONFIGURATION

SDRAM DEVICE TYPE = DDR SDRAM
SDRAM DEVICE FREQUENCY = 166 MHz
SDRAM DEVICE SIZE = 128 MBits (16 MBytes)
SDRAM DEVICE REFRESH PERIOD = 15.625 microseconds (4K/64ms)
SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.5

==========================

LMI_STR = 0x3d6d4b45
set memory size
LMI_COC_UPPER= 0x00000000
LMI_COC_LOWER= 0x00000000
LMI Config Done

Configuring registers for FMI ...
Bank[0-3] not configured fully!!
the ident from EPLD is 0x00000000
reseting flash
FMI configure done

Setup Complete
--------------------------------------------------

Serious-st20link-wts_getertxt.tco- mode: 'T450 X' linker mode: 'ST20C1 X'
st20cc: Error: in attempting to run st20link
gmake: *** [wtmecburner.lku] Error 1
```

---

jvvh5897

03-09-2009, 03:34 PM

My guess would be that you need to delete the .tco file that gives you the error and rebuild it correctly--odds are that you did a build with an old st20c1 setting

---

YLG80

03-10-2009, 02:27 AM

Thanks for your help.
I've run the gmake clean several times. It erases all tco files.
I'm trying also to fix certain problems (warnings) with variables.

Could you elaborate more on the "odd old st20C1".
Your are correct as the sources are 4 or 5 years old.
Thanks

---

slugworth

03-10-2009, 07:08 AM

The mb400.cfg should be tweaked to show 32meg ram instead of 16meg.

```
SDRAM_DEVICE_TYPE = "DDR_RAM"
## SDRAM_DEVICE_TYPE = "SDR_RAM"
##
##
SDRAM_SIZE_CONFIG = "BUILT_FOR_256MBIT"
SDRAM_FREQ_CONFIG = "166"
```

YLG80

03-10-2009, 09:09 AM

Yes sure.
But I'd like first to see if I can recompile that utility.
It is embedded in a full system with on-system debug capability, which makes it very difficult to recreate.

BTW, I've recreated the 5105ckg.cfg which was necessary to build the project.
If I find it too time consuming I will revert to the other generic flashprg prog.

---

slugworth

03-11-2009, 07:55 AM

The mb385.c file that is part of toolset v2.3.1 would be a better choice to try to modify to get it to work with the 39vf3201.
It is close to the correct ritual for reading the chip man and id.

```
/* ============================FlashROM cui_commands
defines========================== */

#define READ_MEM 0xF0 /* read flash_rom_array command (written anywhere in
mem)*/
#define COMMAND_INIT1 0xAA /* part 1 of command initialisation */
#define COMMAND_INIT2 0x55 /* part 2 of command initialisation */
#define READ_IDENT 0x90 /* read flash_rom ident command */
#define SETUP_ERASE 0x80 /* setup to do an erase command */
#define CHIP_ERASE_CONFIRM 0x10 /* confirm chip erase command */
#define PROGRAM 0xA0 /* program command */

/* ============================FlashROM Offset
defines================================ */
#define COMMAND_OFFSET1 ((0x0555)<<1) /*Offset to send COMMAND_INIT1's to */
#define COMMAND_OFFSET2 ((0x02AA)<<1) /*Offset to send COMMAND_INIT2's to */
#define MFGID_OFFSET 0x00000000 /* manufacturers id offset */
#define DEVID_OFFSET 0x00000002 /* device id offset */
#define BLOCK_PROT_OFFSET 0x00000004 /* block prot read offset */
```

---

slugworth

03-11-2009, 08:52 PM

I am at the maggie flash 100% erased stage using the edited mb385.c.

```
/* ============================FlashROM cui_commands
defines========================== */

#define READ_MEM 0xF0 /* read flash_rom_array command (written anywhere in
mem)*/
#define COMMAND_INIT1 0xAA /* part 1 of command initialisation */
#define COMMAND_INIT2 0x55 /* part 2 of command initialisation */
#define READ_IDENT 0x90 /* read flash_rom ident command */
#define SETUP_ERASE 0x80 /* setup to do an erase command */
#define CHIP_ERASE_CONFIRM 0x10 /* confirm chip erase command */
#define PROGRAM 0xA0 /* program command */

/* ============================FlashROM Offset
defines================================ */
#define COMMAND_OFFSET1 ((0x5555)<<1) /*Offset to send COMMAND_INIT1's to */
#define COMMAND_OFFSET2 ((0x2AAA)<<1) /*Offset to send COMMAND_INIT2's to */
#define MFGID_OFFSET 0x00000000 /* manufacturers id offset */
#define DEVID_OFFSET 0x00000002 /* device id offset */
#define BLOCK_PROT_OFFSET 0x00000004 /* block prot read offset
```

A full dump with jkeys shows the flash at all FF's

---

slugworth

03-12-2009, 10:52 AM

You can actually leave the 2 address blank in the .cpu file for uctap and it still works fine.Uctap will automatically come up with it's own addresses.
You can ignore any error messages in uctap,as long as it starts 2 servers.

```
STi5514, 0x1d40a041, 0x30003004, 0x00000200

STi5516, 0xXd41d041, 0x30003004, 0x00000200
STi5518, 0x1d405041, 0x3004, 0x00080000
STi5519, 0x1d415041, 0x3004, 0x00080000
STi5528, 0x1d41a041, 0x3004, 0x00080000
STi5580, 0x1d40b041, 0x3004, 0x00080000
STi5588, 0x1d41c041, 0x3004, 0x00080000
```

STi5589, 0x15400041, 0x3004, 0x00080000
STi5598, 0x15401041, 0x3004, 0x00080000
STm5700, 0x157xx041, 0x3004, 0x00080000
ST20TP3, 0x05194041, 0x3004, 0x00080000
ST20TP4, 0x25198041, 0x3004, 0x00080000
ST20GP6, 0x15196041, 0x3004, 0x00080000
ST20MC2, 0x05195041, 0x3004, 0x00080000
STi5105, 0x2d427041,
STi5100, 0x3d423041, 0x30003004, 0x00000200
STi5119, 0x3d427041, 0x30003004, 0x00000200
#
ucTAPsrv End

#

YLG80

03-12-2009, 11:16 AM

Very good. A giant step ! Have you used the new flashprg w/o mod ?
On my side :
I'm testing my burner source mods on the 5518 box.
I've been able to modify the board.cfg for the 2Mb flash with correct ID for
the Hynix 29LV160TT. The targets.cfg looks good also.

Now Burner runs up to the programming menu with no error.
I've tried a flash programming ... but it has bricked the box.
So there are still things to be fixed with the flash chip description. Likely the flash banks access.
I've debricked the box with Jkeys.

I use a PCBWIRE Multi Jtag interface and there is problem that I've not yet fixed.
That interface can be externally powered or box powered. It has 3 connectors for different stb JTAG connectors.
The one I'm using is the 20 pin connector.
It has no connection for external power. The other 2 connectors have that connection to allow the interface to be box powered.

To be able to use the burner + uCTAP, I had to connect the external power.
To be able to debrick the box, I had to remove it :stupid:
When I use that interface with external power and Jkeys, I get a startup error: cannot Identify the IRD despite of the fact that the correct IRD is displayed by Jkeys! :stupid:

If I modifiy the wiring as advised in the ucTap (pin 17 and 19), nothing works.
If I use the original wiring, uCTAP is able to reste the box and maintain the reset.

YLG80

03-12-2009, 11:20 AM

The mb385 looks good to me as it was the one being used in a previous flashprg version for the 5105.
So it's now just a matter of adjusting the flash type to the one in your stb.

This is strange (taken from you parameters copy:
#define DEVID_OFFSET 0x00000002 /* device id offset

That datasheet says that the offset is 0x00000001.

(NB Have found that it could be due to A0 system address shift to A1) see next post.)

YLG80

03-12-2009, 12:11 PM

Attached is a cpp file with SST39VF3201/SST39VF3202 parms description.

---

slugworth

03-12-2009, 01:06 PM

The mb385 looks good to me as it was the one being used in a previous flashprg version for the 5105.

So it's now just a matter of adjusting the flash type to the one in your stb.
This is strange (taken from you parameters copy:
#define DEVID_OFFSET 0x00000002 /* device id offset
That datasheet says that the offset is 0x00000001.
(NB Have found that it could be due to A0 system address shift to A1) see next post.)

vpp is not set for programming in mb385,it is set in mb382.
I am now re-writing mb382 to see if I can write with it.

YLG80

03-12-2009, 01:50 PM

I understand now why it has bricked my box.

That program is supposed to find the binary file loading adress prior to program or verify (perhaps in a command option).
If it does not find it, the program (my flashprg.c version) assumes (for a 5518) a 0x7ff50000 default burning start adresss (1Mb flash).
When I correct that error it runs ok and does no longer brick the stb.
In the flashprg 1.3 version that you are using there is no default option if it cannot find the bin loading address.
We'll have to sligthtly modify the source in order to specify a default bin file load address.

YLG80

03-12-2009, 01:57 PM

The SST39VF3201 has no Vpp pin.
The programming voltage is internally generated and triggered by a combination/sequence of the WE# and CE# .

---

slugworth

03-12-2009, 02:46 PM

I modified mb385.c but it still won't program the flash.

YLG80

03-12-2009, 03:32 PM

The erase is performed by that sequence:
The Chip-Erase operation is initiated by executing a sixbyte
command sequence with Chip-Erase command
(10H) at address 5555H in the last byte sequence. The
Erase operation begins with the rising edge of the sixth
WE# or CE#, whichever occurs first.

For example for one sector:

```
/**********************************************************
**********************/
/* PROCEDURE: Erase_One_Sector */
/* */
/* This procedure can be used to erase a total of 2048 words. */
/* */
/* Input: */
/* Dst DESTINATION address of sector to be erased */
/* */
/* Output: */
/* return TRUE: indicates success in sector-erase */
/* return FALSE: indicates timeout in sector-erase */
/**********************************************************
**********************/
BYTE Erase_One_Sector(DWORD Dst)
{
BYTE ReturnStatus=TRUE;

*(WORD *) (BaseAddrs + 0x5555 * AddrsShift) = 0x00AA; // 1st write data 0x00AA to
device addr 5555H
*(WORD *) (BaseAddrs + 0x2AAA * AddrsShift) = 0x0055; // 2nd write data 0x0055 to
device addr 2AAAH
*(WORD *) (BaseAddrs + 0x5555 * AddrsShift) = 0x0080; // 3rd write data 0x0080 to
```

```
device addr 5555H
*(WORD *) (BaseAddrs + 0x5555 * AddrsShift) = 0x00AA; // 4th write data 0x00AA to
device addr 5555H
*(WORD *) (BaseAddrs + 0x2AAA * AddrsShift) = 0x0055; // 5th write data 0x0055 to
device addr 2AAAH
*(WORD *) (BaseAddrs + Dst * AddrsShift) = 0x0030; // 6th write data 0x0030 to
device sector addr Dst
// ReturnStatus = Check_Toggle_Ready(Dst,SectorEraseTime); // wait TOGGLE bit stops
toggling
ReturnStatus = Check_Data_Polling(Dst, 0xFFFF, SectorEraseTime); // wait until DQ7
outputs 1
return ReturnStatus;
}
```

Here is the one word programming sequence:

```
/**********************************************************
**********************/
/* PROCEDURE: Program_One_Word */
/* */
/* This procedure programs one word of data into SST39VF320X. */
/* */
/* NOTE: It is necessary to first erase the sector containing the */
/* word to be programmed. */
/* */
/* Input: */
/* SrcWord: data word to be written into SST39VF320X. */
/* Dst: DESTINATION address where to be written into. */
/* */
/* Output: */
/* return TRUE: indicates success in word-program */
/* return FALSE: indicates timeout in word-program */
/**********************************************************
**********************/
BYTE Program_One_Word (WORD SrcWord, DWORD Dst)
{
BYTE ReturnStatus=TRUE;

*(WORD *) (BaseAddrs + 0x5555 * AddrsShift) = 0x00AA; // 1st write data 0x00AA to
device addr 5555H
*(WORD *) (BaseAddrs + 0x2AAA * AddrsShift) = 0x0055; // 2nd write data 0x0055 to
device addr 2AAAH
*(WORD *) (BaseAddrs + 0x5555 * AddrsShift) = 0x00A0; // 3rd write data 0x00A0 to
device addr 5555H
*(WORD *) (BaseAddrs + Dst * AddrsShift) = SrcWord; // 4th write data word into
destination address Dst
ReturnStatus = Check_Toggle_Ready(Dst, WordProgramTime); // wait TOGGLE bit stops
toggling
// ReturnStatus = Check_Data_Polling(Dst, SrcWord, WordProgramTime); // wait until
DQ7 outputs true data
return ReturnStatus;
}
```

If you can erase the chip, you should find a similar erase routine in your program. Then you should find the write/program word routine and modify it as above. :faint2: These routines are coming from the cpp file attached in a previous post.

---

slugworth 03-12-2009, 04:07 PM

read id works

```
/*Send the command required to do a READ_IDENT*/
*command_addr1 = COMMAND_INIT1;(AA to 5555)
*command_addr2 = COMMAND_INIT2;(55 to 2AAA)
```

*command_addr1 = READ_IDENT; (90 to 5555)

chip erase works
/*Perform the chip erase command*/
*command_addr1 = COMMAND_INIT1;(AA to 5555)
*command_addr2 = COMMAND_INIT2;(55 to 2AAA)
*command_addr1 = SETUP_ERASE; (80 to 5555)
*command_addr1 = COMMAND_INIT1;(AA to 5555)
*command_addr2 = COMMAND_INIT2;(55 to 2AAA)
*command_addr1 = CHIP_ERASE_CONFIRM;(10 to 5555)
program flash no workie.

/*Set up to PROGRAM to flash*/
*command_addr1 = COMMAND_INIT1;(AA to 5555)
*command_addr2 = COMMAND_INIT2;(55 to 2AAA)
*command_addr1 = PROGRAM;(A0 to 5555)

/*Write the word to addr*/
*addr = flashdata;

/*Wait until status is stable - i.e. operation complete*/
wait_stable(
The original flash is the M29KW032E,which uses the same routines
with a different offset (555 instead of 5555 and 2AA instead of 2AAA.

YLG80

03-12-2009, 04:09 PM

In MB385, the routine starts on line 238 up to to line 289.
It uses the following sequence to program a word :
/*Set up to PROGRAM to flash*/
*command_addr1 = COMMAND_INIT1;
*command_addr2 = COMMAND_INIT2;
*command_addr1 = PROGRAM;

where :
#define COMMAND_INIT1 0xAA /* part 1 of command initialisation */
#define COMMAND_INIT2 0x55 /* part 2 of command initialisation */
and
#define PROGRAM 0xA0 /* program command */

This looks correct for your chip regarding to what is described in the DS although the command_offset1
and 2 are not correct.(see below)

But just before that instruction block the command_addr sequence is likely not correct.
It uses the following instructions:
volatile char* command_addr1 = (char*) (getBaseOfBank(bank) | COMMAND_OFFSET1);
volatile char* command_addr2 = (char*) (getBaseOfBank(bank) | COMMAND_OFFSET2);

where :
#define COMMAND_OFFSET1 ((0x0555)<<1) /*Offset to send COMMAND_INIT1's to */
#define COMMAND_OFFSET2 ((0x02AA)<<1) /*Offset to send COMMAND_INIT2's to */

The COMMAND_OFFSET1 should be 0x5555 i.o 0x0555
and COMMAND_OFFSET2 should be 0x2AAA i.o 0x02AA

(in the Flash chip used in MB385, i.e M29KW032E, the first nibble in the command_offset is "don't care" which is not the case in the SST39VF320x)

So you should try changing the values at the beginning of the file and recompile.
Howefully the timing will be correct...

The write-word sequence for your chip is shown in a table in the datasheet:

Good luck, it should work....

YLG80

03-12-2009, 04:30 PM

The 0x5555 and 0x2AAA are not the command init, but the command_offset.
The original command_init values are looking correct.
The sequence you need to setup to write a word is :
0x5555 (command_offset)- -> data 0xAA (command_init)
0x2AAA (command_offset)- -> data 0x55 (command_init)
0x5555 (command_offset)- -> data 0xA0 (command_init)
then and only then, the Word address (WA) and the data to be written.

BTW how do your create a so nice blue area with your inclusions?

---

slugworth

03-12-2009, 05:37 PM

I was able to program once with a puny 1k file of all zeroes,which leads me to believe it's a timing problem in the code.

---

slugworth

03-12-2009, 08:11 PM

You know what this means boys and girls.
Thanks to all the people that helped- across the globe.

MSSFWP

03-13-2009, 03:22 AM

The 0x5555 and 0x2AAA are not the command init, but the command_offset.
The original command_init values are looking correct.
The sequence you need to setup to write a word is :
0x5555 (command_offset)- -> data 0xAA (command_init)
0x2AAA (command_offset)- -> data 0x55 (command_init)
0x2AAA (command_offset)- -> data 0x55 (command_init)
0x5555 (command_offset)- -> data 0xA0 (command_init)
then and only then, the Word address (WA) and the data to be written.

BTW how do your create a so nice blue area with your inclusions?Your on the right track bro. &5555 is the entry vector for the built in flash chip command/programming mode interpreter on the 3201 flash chip. &2AAA is the flash chips internal I/O control register that sets the toggling status of the DQx lines. ;)

---

slugworth

03-13-2009, 07:14 AM

You don't need the full toolset install,I used the pvr2flash.rar to install the st20 core files and deleted the files not needed.The drawback is the flash program only works with slow pc's- 733mhz P3 vintage.That was with a simple jtag hookup,I'm not sure about a modified buffered one.

---

slugworth

03-13-2009, 08:59 AM

The st20 core files,instead of downloading the full toolset.
Unzip to c:\ preserving the directory structure.
Then you will just need the .lku and .bat files.

YLG80

03-13-2009, 12:28 PM

:stoned:
The standard lku would not work without having been recompiled with your working cfg files.

---

alarm

03-13-2009, 01:35 PM

.All good evening.
I from Europe also do not understand English, I write by means of the translator. I so

have understood, what there are good news and proceeding from your discussion follows, what the processor 5105 and 5119 are absolutely identical?
And still a question where it is possible to take the instruction for st20_core.zip.
Yours faithfully Alex

---

slugworth

03-13-2009, 02:05 PM

.All good evening.
I from Germany also do not understand English, I write by means of the translator. I so have understood, what there are good news and proceeding from your discussion follows, what the processor 5105 and 5119 are absolutely identical?
And still a question where it is possible to take the instruction for st20_core.zip.

Yours faithfully Alex

For jtagging,the sti5105 and sti5119 should be close enough.
I am still working on instructions,I just wanted to get the core files out quickly.
Basically you unzip to c:\ and get dosdrop.exe loaded then you can make sure uctap is working.The jtag connector is the standard 20pin except the wire that normally goes to pin19 on the receiver now goes to pin17. pin19 on the receiver goes to pin12 on the db25.When you drop to dos in the c:\maggie folder and run tap.bat, your processor should be detected.

---

alarm

03-13-2009, 02:54 PM

At me the processor 5119, and is defined as 5512.?????

Alex_Trask

03-13-2009, 05:35 PM

Good Job boys
I wanted to thank you for the work done to make possible this method.

---

slugworth

03-13-2009, 08:36 PM

What I used to fix my maggie.
The previous st20 core file with value added.

---

alarm

03-14-2009, 03:34 AM

@slugworth
Thanks still time. But unfortunately in a folder stdcfg there is no file in format CFG for the chip 5119?

YLG80

03-14-2009, 05:31 AM

The Sti5119 is a 5105 upgrade.
You should run Jkeys in order to fetch the Sti5119 CPUID and add that Id to uCTAPsrv.cpu if not present.
The Sti5119 CPU ID is 0X3D427041

The line to be added to ucTapsrv.cpu should look like that:
STi5119,[TAB]0x3D427041,[TAB],0x30003004,[TAB]0x80000000

Reverting to my post #91 and the problems with the JTAG interface (erratic)
INTERFACE
The PCBWIRE MULTI interface can work without power even when the 20 pin JTAG connector is used, because the buffer is powered throught the LPT lines pull up resistors.

Not very good !
It's not advised to use the interface like that, because the buffer could no be powered enough. On one PC the buffer 74HC244 had only 2.6V instead of 3.3V. With an old Acer laptop the buffer was correctly powered through the 2.2K pull-up resistors.

LAPTOP CPU SPEED

The erratic behaviour was due to the PC CPU speed. (already mentioned by Slugworth)

I use a Win2K 850 Mhz laptop with 100MHz bus speed laptop for jtagging.
At 850Mhz, I cannot get stable results with JKEYS and uCTAPsrv.
If I slow down the CPU speed (using SpeedStep) to 700Mhz, both programs have a stable behaviour.

At lower CPU speed with an externally powered interface, I've successfully used my own STi5105 flashprg build together with ucTAP to verify the Flash (Spansion S29AL016D70B) content of my Sti5105 box vs a local file.
Result is 100% OK.
The ucTAPsrv windows showed a successfull DCU3 setup with no error.

(No need to swap any pin on my interface as advised in the original uCTAP readme.txt.)
After that flashprg verification, the STB needs to be hard reset to restart.

---

alarm

03-14-2009, 07:29 AM

I have made so 17 пин from platinum of a receiver have submitted on 19pin a cable.
19pin from platinum of a receiver has submitted on 12 pin a cable.
The others 4 pin coincide, as well as in usual 20 pin Jtag?

RECEIVER kabel
db25 pin2 to receiver pin9 -----TMS---pin9
pin3 pin11-----TCK---pin11
pin4 pin13-----TDI---pin13
pin5 pin17-----TRST--pin19

pin12 pin19-----GND---pin12
pin13 pin15-----TDO---pin15
pin24 ground (even pins on receiver are ground

But process does not go?

---

slugworth

03-14-2009, 07:32 AM

UcTAPsrv windows showed a successfull DCU3 setup with no error.
(No need to swap any pin on my interface as advised in the original uCTAP readme.txt.)

I had to use pin17 instead of pin19 or the receiver registers don't get programmed.Uctap detects the processor correctly using pin19,but the rest of the process fails.That is with a simple jtag.

---

slugworth

03-14-2009, 07:37 AM

I have made so 17 пин from platinum of a receiver have submitted on 19pin a cable.
19pin from platinum of a receiver has submitted on 12 pin a cable.
The others 4 pin coincide, as well as in usual 20 pin Jtag?

RECEIVER kabel
db25 pin2 to receiver pin9 -----TMS
pin3 pin11-----TCK
pin4 pin13-----TDI
pin5 pin17-----TRST
pin12 pin19
pin13 pin15-----TDO
pin24 ground (even pins on receiver are ground

But process does not go?

The original readme neglected to mention you have to use 100ohm resistors from receiver to pc.

---

slugworth

03-14-2009, 07:44 AM

A regular jtag cable will detect the processor in uctap,assuming the jtag wiring is correct and your receiver has the standard 20pin connector.
Once correctly detected you can rewire for pin17 usage instead.
This also assumes you are getting no windows related errors.

---

alarm

03-14-2009, 08:40 AM

A regular jtag cable will detect the processor in uctap,assuming the jtag wiring is correct and your receiver has the standard 20pin connector.
Once correctly detected you can rewire for pin17 usage instead.
This also assumes you are getting no windows related errors.

I do so:
1) . the program dos also I press button Activate further in the appeared window I press îê.
2) .I start ïðîãàðàìì Installer and I press button Install, further I achieve an inscription: Service has successfully started. Further I close program Installer.
3) . receiver in a network and The program tap also I receive a following window:

Further process does not go, where my mistake?

---

slugworth

03-14-2009, 09:47 AM

If you look at uctapsrv.cpu with notepad,you will notice if uctap detects the processor as 0x00000000 the default is sti5512,which is what I think is happening.
It sounds like the jtag is still wired wrong.

STi5510, 0x2d4ad041, 0x3004, 0x00080000
STi5512, 0x1d401041, 0x3004, 0x00080000
STi5512A, 0x3d406041, 0x3004, 0x00080000
STi5512M, 0x3d407041, 0x3004, 0x00080000
STi5512N, 0x00000000, 0x3004, 0x00080000
STi5514, 0x1d40a041, 0x30003004, 0x00000200
STi5516, 0xXd41d041, 0x30003004, 0x00000200
STi5518, 0x1d405041, 0x3004, 0x00080000
STi5519, 0x1d415041, 0x3004, 0x00080000
STi5528, 0x1d41a041, 0x3004, 0x00080000
STi5580, 0x1d40b041, 0x3004, 0x00080000
STi5588, 0x1d41c041, 0x3004, 0x00080000
STi5589, 0x15400041, 0x3004, 0x00080000
STi5598, 0x15401041, 0x3004, 0x00080000
STm5700, 0x157xx041, 0x3004, 0x00080000
ST20TP3, 0x05194041, 0x3004, 0x00080000
ST20TP4, 0x25198041, 0x3004, 0x00080000
ST20GP6, 0x15196041, 0x3004, 0x00080000
ST20MC2, 0x05195041, 0x3004, 0x00080000
STi5105, 0x2d427041, 0x30003004, 0x00000200
STi5100, 0x3d423041, 0x30003004, 0x00000200
STi5119, 0x3d427041, 0x30003004, 0x00000200

---

slugworth

03-14-2009, 09:51 AM

mb193.c had a delay routine I added to the mb385.c and got it to work on a pc that wouldn't work before.

```
/* ============================FlashROM Offset defines============================== */
#define COMMAND_OFFSET1 ((0x5555)<<1) /*Offset to send COMMAND_INIT1's to */
#define COMMAND_OFFSET2 ((0x2AAA)<<1) /*Offset to send COMMAND_INIT2's to */

#define MFGID_OFFSET 0x00000000 /* manufacturers id offset */
#define DEVID_OFFSET 0x00000002 /* device id offset */
#define BLOCK_PROT_OFFSET 0x00000008 /* block prot read offset */


/**
```

```
* Wait a bit
*/
static void ProcWait( int timeDelay )
{
int i, j, k;
for (i=0; i<timeDelay; i++)
{
for (j=0; j<5; j++)
{
k++;
}
}
}

------------------------------------------------------------------------------
/*Set up to PROGRAM to flash*/
*command_addr1 = COMMAND_INIT1;
*command_addr2 = COMMAND_INIT2;
*command_addr1 = PROGRAM;
/*Write the word to addr*/
*addr = flashdata;
{ProcWait (1);}
/*Wait until status is stable - i.e. operation complete*/
wait_stable(
#if 0
bank
#endif

);
```

---

slugworth

03-14-2009, 10:30 AM

New .lku for magnum with delay added.
Tested on a 1.5ghz athlon.

---

alarm

03-14-2009, 11:39 AM

Today I used all varinty Jtag-kabel, 20 pin, 6pin 3.3v, 6pin ohne 3.3v.
Perhaps you need a processor at 115 foot building.
I can not understand the chip on edentichnye you get, and I do not ..!!!

---

slugworth

03-14-2009, 12:06 PM

If you use jkeys it should show the processor id.
It should match the processor id in the tapsrv.cpu file.

---

alarm

03-14-2009, 12:28 PM

When I used a 19pin TRST processor is determined
Now when I use your schema 17 pin TRST, 19PIN GND
PROZESSOR-FFFFFFF???

---

slugworth

03-14-2009, 12:34 PM

pin19 to db25 pin12, not ground

you will need jkeys to get a full flash dump from your receiver.
You can't save flash with the DCU3 jtag YOU MUST DO IT WITH JKEYS.
Do this first.If you erase the flash you are kaput.

---

slugworth

03-14-2009, 12:40 PM

I think you should mention the receiver brand/model and the flash chip type.
We love close up pictures of the mainboard in the receiver also.

YLG80

03-14-2009, 01:06 PM

@Slugworth
I suppose this is the correct wiring and type of interface you are using ?

---

slugworth

03-14-2009, 01:19 PM

With a 100ohm resistor in series like the other connections.

---

alarm

03-14-2009, 01:33 PM

I think you should mention the receiver brand/model and the flash chip type.

We love close up pictures of the mainboard in the receiver also.

FaVal mercury S100.

Flash S29AL016D70TF102
Prozessor Omega STI 5119ALC

YLG80

03-14-2009, 01:40 PM

Ok thanks.
Here is the correct interface wiring with R6 added.
I would advise to use a buffered interface rather than this one that connects 5V signals (LPT) to the 3.3V logic of the STB via the resistors.

Could be harmful for the CPU.

---

slugworth

03-14-2009, 02:06 PM

You should be able to save mem with this data added to your jkeys.def file.
Manually select faval from the model list.
IRD, 6, "FaVal", 15, 1, 1, 1, 2, 2, 0, 0, 0, 0,0x40010151, 0, "BA-", 3, 4
IRDFlash, 6, "Flash 1(S29AL016D)", 0x2249, 0x40000000, 0x200000, 2, 2, 0

Flash, 6, "S29AL016D", 0x2249, 0x200000, 1, 1, 0, 1, 35, 1
Sector, 6, 1, 0 , 0x4000
Sector, 6, 1, 0x4000 , 0x2000
Sector, 6, 1, 0x6000 , 0x2000
Sector, 6, 1, 0x8000 , 0x8000
Sector, 6, 31,0x10000, 0x10000

---

alarm

03-14-2009, 02:06 PM

All realized his mistake! Thank you very much!
To be involved 7pin:
kabel-receiver
19-19
17-TRST-17
15-TDO-15
13-TDI-13
11TCK-11
9-TMS-9
20-GND-20

---

alarm

03-14-2009, 02:21 PM

Many thanks to all the distinguished experts of this forum!
Especially a huge thank slugworth.

Respekt.

---

alarm

03-15-2009, 04:24 AM

Greetings to all!
Compound tested at 100%. What problem? Thanks.

YLG80

03-15-2009, 05:28 AM

You have likely not formatted the data correctly in the ucTAPsrv.cpu
file and / or not loaded giveio.sys

You have to respect exactly the data format as mentioned in a previous post.
Like this, respecting the columns

```
# ucTAPsrv CPU definitions file v 0.0
#
# NAME, ID, SD, DCU_CONTROL, ASK_CONTROL_DEVICE_ACCESS
STi5510, 0x2d4cd041, 0x3004, 0x00080000
STi5512S, 0x1d401041, 0x3004, 0x00080000
STi5512A, 0x3d406041, 0x3004, 0x00080000
STi5512M, 0x3d407041, 0x3004, 0x00080000
STi5512N, 0x00000000, 0x3004, 0x00080000
STi5514, 0x1d40a041, 0x30003004, 0x00000200
STi5516, 0xXd41d041, 0x30003004, 0x00000200
STi5518, 0x1d405041, 0x3004, 0x00080000
STi5519, 0x1d415041, 0x3004, 0x00080000

STi5528, 0x1d41a041, 0x3004, 0x00080000
STi5580, 0x1d40b041, 0x3004, 0x00080000
STi5588, 0x1d41c041, 0x3004, 0x00080000
STi5589, 0x15400041, 0x3004, 0x00080000
STi5598, 0x15401041, 0x3004, 0x00080000
STm5700, 0x157xx041, 0x3004, 0x00080000
ST20TP3, 0x05194041, 0x3004, 0x00080000
ST20TP4, 0x25198041, 0x3004, 0x00080000
ST20GP6, 0x15196041, 0x3004, 0x00080000
ST20MC2, 0x05195041, 0x3004, 0x00080000
STi5105, 0x2d427041, 0x30003004, 0x00000200
STi5100, 0x3d423041, 0x30003004, 0x00000200
STi5119, 0x3d427041, 0x30003004, 0x00000200
#
# ucTAPsrv End
#
```

Be sure to load the giveio.sys driver prior to run ucTAPsrv.
Otherwhise ucTAP will not have direct access to the LPT port.

---

alarm

03-15-2009, 05:52 AM

How to reformat? Why Jkey program not to see the processor as well?
When I connect:
19TRST
15TDO
13TDI
11TCK
9TMS
29GND

I get this:

---

shirkan

03-15-2009, 06:36 AM

i need JTAG scheme for Opticum 7000 CX plus /5119/. Please

---

alarm

03-15-2009, 06:45 AM

When I connect:
19TRST
15TDO
13TDI
11TCK
9TMS
29GND

---

alarm

03-15-2009, 07:24 AM

You have likely not formatted the data correctly in the ucTAPsrv.cpu file and / or not loaded giveio.sys
You have to respect exactly the data format as mentioned in a previous post.
Like this, respecting the columns
[
Be sure to load the giveio.sys driver prior to run ucTAPsrv.
Otherwhise ucTAP will not have direct access to the LPT port.

ucTAP will not have direct access to the LPT port:

YLG80

03-15-2009, 08:41 AM

You should double check if giveio.sys or other parallel port driver that give direct access to the parallel port is loaded.
With XP SP2 you can only use the parallel driver (ECP) via the API.

Go to your control panel, system, peripherals and check if your LPT port is showing ECP.
If not, you should reboot your computer and go to the BIOS settings and set the Parallel port to ECP.

Check also if you can find giveio.sys in your system :
c:\windows\system32\drivers (or sometimes in c:\windows\system32

If you are not sure, there is a driver loader in slugworth package named maggie.zip.
Run the file installer provided in that archive.It will load a driver.

Be sure also to check if your firewall does not block the communication via LPT.
Click on : Start, Control Panel, firewall and choose the exceptions tab.

Check if JKEYS is allowed to use the parallel port.
If not sure, temporarly disable your firewall and try again JKEYS.

---

slugworth

03-15-2009, 09:22 AM

He had jkeys working in 1 post.
Get jkeys working and get a full good 2meg flash dump.
Then worry about getting uctap working.
The installer and userport.sys should have been in c:\maggie

---

slugworth

03-15-2009, 09:23 AM

You only have to install the driver once.

---

slugworth

03-15-2009, 10:43 AM

another uctap connection pic.

---

alarm

03-15-2009, 06:41 PM

Nothing mean???? Scheme as you have. Two computer, the result is zero.

---

slugworth

03-15-2009, 06:52 PM

Whatever happened to c:\maggie?
All the files should have been in that directory and sub directories.
Did you unzip the file to c:\ ?
I think you should start over.
The procedure is based on all files being in the correct places.

MSSFWP

03-16-2009, 07:21 AM

Whatever happened to c:\maggie?
All the files should have been in that directory and sub directories.
Did you unzip the file to c:\ ?
I think you should start over.
The procedure is based on all files being in the correct places.
First of all...good job guys.

One thing i did notice slugworth. When i tried your setup for the first time, i had to reload the userport.sys driver even though the service was reported as running so that the paths were displayed correctly, then i had to stop and restart the service to get it to work right.

Probably one of those dumb Windblows things, but whatever the case that's what fixed it.

---

slugworth

03-16-2009, 07:40 AM

First of all...good job guys.

One thing i did notice slugworth. When i tried your setup for the first time, i had to reload the userport.sys driver even though the service was reported as running so that the paths were displayed correctly, then i had to stop and restart the service to get it to work right.

Probably one of those dumb Windblows things, but whatever the case that's what fixed it.
If you unzipped to c:\ originally the userport and installer.exe will be in the c:\maggie directory.
You must have the proper directory structure or the setenv.bat file will be all wrong.PC users that have the migs can edit the xp environment settings,but since you only have to use the program once and for novice users it was better to do it in a batch file.
Don't forget,when I wrote the readme it was on a pc that already had userport loaded from the skymax jtag days.I didn't have a virgin pc to try it on.I have no way to test it on vista either.
If you get errors in uctap,you can ignore all the errors before the micro detection stage.As long as it detects the sti5105 and starts 2 servers you are good to go.The same file should work on euro receivers with the same micro and flash chip.You would have to supply your own firmware .bin file instead of the v145.bin supplied naturally.
You can also just program in the 1st 64k of the flash,(the bootstrap section +the area where the revs are stored)I did this on the first version then loaded the rest of the flash with the serial loader.

---

slugworth

03-16-2009, 07:47 AM

Again,not really a factor since you only use the program once.Novice users should just stick with the setenv.bat usage.
If you want the driver to start whenever the computer is restarted proceed with the following steps.
1. In the Control Panel, open System and go to the hardware tab.

2. Click on the Device Manager button. This will open a new window.

3. In the Device Manager window, click on the View menu and select "Show hidden devices." This will reveal a Non-Plug and Play Drivers icon in the file tree.

4. Expand the Non-Plug and Play Drivers tree.

5. Find and right click userport and select Properties from the popup menu. This will bring up a window of the userport Properties.

6. In the Properties window, select the Driver tab.

7. Select Automatic from the dropdown box for the type.

This change will take effect after you reboot the computer.
Now the userport.sys will load whenever you run windows.

---

alarm

03-16-2009, 08:39 AM

Tell me please, what do you use the length Jtag-kabel?

---

slugworth

03-16-2009, 09:13 AM

23inch or around 58cm for simple non buffered jtag cable.

---

alarm

03-16-2009, 09:30 AM

Nothing.Resivery processor ALI, STi 5518 Okay all, and this as if someone has cursed !!!!!?????

---

slugworth

03-16-2009, 09:49 AM

In this episode the programming stopped before it was complete.
If that happens,hit ctrl+c and terminate batch job.
Then run pro.bat again.

---

spartacussat

03-16-2009, 12:26 PM

I just think alarm is trying to fool you guys or mislead this thread.
So, please Slugworth, instead of losing time and energy in explaining to "alarm" what he seems to refuse to understand, can you write a step by step tutorial on "How to JTAG an STi5105/STi5119 device" (even on the faster P4 computers and Wndows XP). I guess this would be very helpful to everybody. Thank you very much for the results

---

slugworth

03-16-2009, 12:39 PM

When doing DCU3 research you will need to know;
1. the processor type in the receiver,like sti5105,sti5517,sti5119 ect.
You need to know this for sure,if your receiver has a heatsink on the processor you may have to take it off to get the processor type.
2. the type and number of the flash chip, like 29lv160, sst39vf3201, ect.
3. get the pdf file for your flash off the net.
4. the address the receiver uses for the flash chip.
Once you have that info you can play with making you own flashburner.lku with st20 toolset v2.3.1 and the sample .c files that come with it.
I still haven't seen a full dump from a sti5119 verifying the flash type and address,or you would be jtagging a sti5119 by now.I don't have that type receiver so I have to rely on others for the details.
The hardware steps can be verified by getting a full flash dump with jkeys.

---

slugworth

03-16-2009, 01:11 PM

st20 toolset v2.3.1 so you can start playing at home.

http://rapidshare.com/files/209990006/stm-st20.231-2.3.1-MSWin32-x86.exe

---

jvvh5897

03-16-2009, 02:26 PM

I suspect you could use the ID that jkeys gives you to determine the processor without having to pull a heatsink--course that assumes that folks have done that ID to processor

cross ref for you and posted the info. And that you have good reason to think that jkeys will work at least well enough to get you the ID # and that you have some info on the jtag cable pin out required.

Anyone trying to do a jtag on a box for the first time is likely to be facing many challenges and expecting a how-to really is expecting others to do all the hard work for you.

---

alarm

03-16-2009, 03:30 PM

Program for EJTAG ALI processor is edentichny cmd.exe and I am using this program has restored more than one dozen receivers with Ali protsessorom.U you unfortunately do not have such a receiver and we can not do predprinyat.Ya think that we still add something, or a short leg on the mass of the processor, or file an additional strain on the flash .....????
Questions, questions, questions ....????

MSSFWP

03-16-2009, 04:21 PM

Program for EJTAG ALI processor is edentichny cmd.exe and I am using this program has restored more than one dozen receivers with Ali protsessorom.U you unfortunately do not have such a receiver and we can not do predprinyat.Ya think that we still add something, or a short leg on the mass of the processor, or file an additional strain on the flash .....????
Questions, questions, questions ....????Well first of all, communication through EJTAG and normal JTAG communications are two entirely different things, and should not be confused with each other. They may have the same end result, but they use different ways to communicate with the CPU/flash.

Second of all in this project, on this unit you should be using a buffered JTAG device. I'm not saying that a standard Jtag will not work, but serious write errors can occur.

It should be noted that with a buffered JTAG device the cable length should not matter.

Another note: If using a buffered JTAG device, the computer speed/OS versions don't seem to be an issue either. I've tried this on 6 different PC's running 4 different OS's at speeds varying from 700Mhz to 3.0Ghz. It worked fine in all occations. ;)

---

slugworth

03-16-2009, 05:09 PM

Program for EJTAG ALI processor is edentichny cmd.exe and I am using this program has restored more than one dozen receivers with Ali protsessorom.U you unfortunately do not have such a receiver and we can not do predprinyat.Ya think that we still add something, or a short leg on the mass of the processor, or file an additional strain on the flash .....????
Questions, questions, questions ....????

Jkeys with the original cable should get you the processor id.
If not, your receiver jtag connector may be wired different or what you think is a receiver jtag connector isn't a jtag connector.
I don't have a sti5119 receiver so you have to be the pioneer.

---

slugworth

03-16-2009, 05:16 PM

Second of all in this project, on this unit you should be using a buffered JTAG device. I'm not saying that a standard Jtag will not work, but serious write errors can occur.
In most cases you just have 1 dead receiver and you use the program once,so a simple jtag is good enough.If you are having serious problems and can't write the full flash,you can just program the 1st 64k (bootstrap section) and program the rest of the receiver fully with the serial port.
People with buffered 100% ribbon type cables would have no way to modify the cable anyway.

---

slugworth

03-16-2009, 07:00 PM

nmake program to be used with the st20 toolset.
A command line program,so copy to the toolset examples\flash folder and drop to dos in that folder.Then just type nmake and it will make the default lku files.Edit the appropriate mbxxx.c file to match your receiver if it has as a processor other than the sti5105/39vf3201.then run nmake to make .lku files other than the defaults.

---

altcode

03-16-2009, 08:56 PM

error on prgram need help

---

slugworth

03-16-2009, 09:06 PM

error on prgram need help

jtag cable not wired right or cable not connected correctly.
uctap is seeing the processor id as 0x00000000 instead of the correct id.
jkeys should show the correct id if the cable/connection is correct.
STi5512N, 0x00000000, 0x3004, 0x00080000
STi5514, 0x1d40a041, 0x30003004, 0x00000200
STi5516, 0xXd41d041, 0x30003004, 0x00000200
STi5518, 0x1d405041, 0x3004, 0x00080000
STi5519, 0x1d415041, 0x3004, 0x00080000
STi5528, 0x1d41a041, 0x3004, 0x00080000
STi5580, 0x1d40b041, 0x3004, 0x00080000
STi5588, 0x1d41c041, 0x3004, 0x00080000
STi5589, 0x15400041, 0x3004, 0x00080000

STi5598, 0x15401041, 0x3004, 0x00080000
STm3700, 0x157xx041, 0x3004, 0x00080000
ST20TP3, 0x05194041, 0x3004, 0x00080000
ST20TP4, 0x25198041, 0x3004, 0x00080000
ST20GP6, 0x15196041, 0x3004, 0x00080000
ST20MC2, 0x05195041, 0x3004, 0x00080000
STi5105, 0x2d427041, 0x30003004, 0x00000200
STi5100, 0x3d423041, 0x30003004, 0x00000200
STi5119, 0x3d427041, 0x30003004, 0x00000200

MSSFWP

03-16-2009, 09:26 PM

In most cases you just have 1 dead receiver and you use the program once,so a simple jtag is good enough.If you are having serious problems and can't write the full flash,you can just program the 1st 64k (bootstrap section) and program the rest of the receiver fully with the serial port.
People with buffered 100% ribbon type cables would have no way to modify the cable anyway.
I understand where your coming from bro, and i agree. Just for the record ATM i'm only testing the configuration you come up with in various os's and speeds to see if it will work in all systems, so far that part is right. I've only tested it on a single puked SV1000, which i deliberately sent a "dirty bin" to (with the serial cable) afterwards to make it puke again after first verifying the boot repair.

BTW:
I have a home brew buffered quad mode with 8 spring loaded "hair clip" test leads on the end of the cable. I've debricked a lot of donated electronic "trash" with that thing, i'll swear by it. But I understand that a lot of folks buy them off of ebay though. ;)

---

slugworth

03-16-2009, 10:15 PM

BTW:
I have a home brew buffered quad mode with 8 spring loaded "hair clip" test leads on the end of the cable. I've debricked a lot of donated electronic "trash" with that thing, i'll swear by it. But I understand that a lot of folks buy them off of ebay though. ;)
For my conexant jtag I definitely needed a buffered jtag, but I had no 74hc244,so I

made a buffered jtag out of transistors/resistors from rat shack.
I ripped the idea from the old zx computer app note.

---

alarm

03-17-2009, 02:18 AM

I have sleduyuschee.ya can not erase the flash????

---

alarm

03-17-2009, 02:33 AM

Dear slugworth give me all you izvesnye program
Jkey, who is working with a 5105.S whom you worked and you got to erase flesh.
program flash, I can program Pionero v 4.0 (FINAL), but I can not erase, even though I use 4 different programs.

---

alarm

03-17-2009, 03:14 AM

This is correct?

---

slugworth

03-17-2009, 06:43 AM

You can only read with jkeys,not erase/program.
You will need a .lku modified for the sti5119 and that size flash
and run the .lku under st20.
Good luck.

---

slugworth

03-17-2009, 06:45 AM

This is correct?

Region user specified 40000000
save mem

Alex_Trask

03-17-2009, 06:55 AM

For my conexant jtag I definitely needed a buffered jtag, but I had no 74hc244,so I made a buffered jtag out of transistors/resistors from rat shack.
I ripped the idea from the old zx computer app note.

If you need a buffer 74HC244 i can send it to you for free ;-)

---

slugworth

03-17-2009, 07:17 AM

Still waiting to see a 2meg flash dump from a sti5119.

---

alarm

03-17-2009, 07:56 AM

My receiver is dead and I can not get damp.U I have the original software, but there is a dump from another receiver with edentichnoy flash and a processor 5119.

---

slugworth

03-17-2009, 09:09 AM

I need a dump from the bad receiver to verify the flash address.
Even if the flash was blank you still should be able to dump.

---

alarm

03-17-2009, 09:24 AM

I need a dump from the bad receiver to verify the flash address.
Even if the flash was blank you still should be able to dump.

Took the dump with a dead receiver.

---

slugworth

03-17-2009, 09:38 AM

Did you manage to erase the flash?
The dump was all ff's which means the flash is erased or 40000000 is the wrong address for a sti5119
Don't use the default address in jkeys.

---

alarm

03-17-2009, 11:38 AM

I can not write the dump in the receiver.

---

slugworth

03-17-2009, 12:24 PM

You have to dump with jkeys at address 40000000

Then look at the dump with a hex editor.
It should start with hex 20 20 20 24
If not,try address 7FE00000

---

alarm

03-17-2009, 03:03 PM

Address 40000000 is not correct, worked with the addresses 7FE00000.
Connect the 19 pin 3.3v.Programmy Pionero and Jkey started working korekt.Protsessor determined, I can write the flash and to count, but when you open the dump is obtained from the receiver, 0.Ne one of the programs does not know the CPU Code.I can not connect the receiver with your program maggie.

thanks for their support, we will wait for good news in Europe, all silent.

---

slugworth

03-17-2009, 03:49 PM

If the flash is really at 7FE00000 like you say,this pro.bat should work.
Rename your 2meg flash file to 2m.bin and put in the maggie folder.
st20run -b -i mb400.cfg -t mag flashprg_c1c.lku "-args -b MBSDM -p 2m.bin -f"

---

alarm

03-17-2009, 05:21 PM

If the flash is really at 7FE00000 like you say,this pro.bat should work.
Rename your 2meg flash file to 2m.bin and put in the maggie folder.
st20run -b -i mb400.cfg -t mag flashprg_c1c.lku "-args -b MBSDM -p 2m.bin -f"

rename the file and that's what got:no:

---

slugworth

03-17-2009, 05:39 PM

so run the bat I posted
copy/paste the code in my previous post and save as pro.bat and then run
You still don't have the files in c:\maggie I see.
It's like pulling teeth with some people.

---

alarm

03-17-2009, 06:10 PM

I renamed and copied to your bin file in maggie does not understand what next to do ...

---

slugworth

03-17-2009, 06:49 PM

DOS uses batch files.
tap.bat starts uctap- you minimize the screen then run pro.bat in DOS.
(the edited one for sti5119)the original is for 4meg flash.

YLG80

03-18-2009, 02:19 AM

I'm still questioning myself on the ucTAP interface configuration with pin 19 of the JTAG connector being connected to pin 12 of the LPT DB25 connector.
I've made tests with that configuration and it does not work with my STB, although moving pin 19 to pin 17 on the JTAG connector looks correct.

Thanks to that mod on pin 17, the STB can be reset via the inferface w/o having to cycle power while launching ucTAP.

On pin 12 of the LPT connector (see picture attached):
Connecting pin 12 of de DB25 to pin 19 of the JTAG connector means connecting two inputs together with no signal to drive them.
nTRST is an input to the DCU and DB25 pin 12 is also an input to LPT status port register.
This is not consistent.

I would assume that pin 19 (nTRST) of the JTAG connector should be driven by another LPT output pin.

This is what is shown in the ST20 Connect interface manual.(picture attached)

YLG80

03-18-2009, 05:02 AM

I've probed the pins with a scope while launching JKEYs and ucTAP.
Pin 6 of the DB25 LPT interface is used by ucTAP and not by JKEYS.
When starting JKEYS, pin 6 goes to LOW and remains LOW even after stopping JKEY's.
ucTAP raises that pin HIGH first and then triggers that pin low (short pulse) which leads me to conclude that this pin should be used to drive pin 19 of the JTAG connector.

I've again modified my buffered JTAG in accordance with that conclusion.

Now it runs OK but with one condition :
running JKEYS once before running ucTAP. (When launching JKEYS, the reset signal is released and the STB becomes operational. When quitting JKEYS, the STB returns to

reset state. Thus this pin signal should be inverted)

ucTAp can run OK because JKEYS resets pin 6 to LOW prior to run ucTAP. ucTAP alone leaves that pin HIGH when we quit the flash program.

To fix that issue, the signal going from LPT DB25 pin 6 to pin 19 should be inverted for example by means of a transistor driver or a simple logic inverter.

The beauty of that configuration, is that ucTAP properly resets the STB.
I've scoped a full ucTAP/flashprg session (verify) and it shows that pin 6 is used to reset the STB upon starting the flashprg verification.

---

slugworth

03-18-2009, 07:26 AM

For people with working receivers,if you use the command line
st20run -b -i mb400.cfg -t mag flashprg_c1c.lku
without any arguments after the .lku the screen will just show the receiver being set up and the old help screen that was part of the original program.Then you can play with jtag connector mods and editing the various .cfg files to see how the process reacts without having to worry about wasting your receiver.

---

slugworth

03-18-2009, 07:43 AM

I'm still questioning myself on the ucTAP interface configuration with pin 19 of the JTAG connector being connected to pin 12 of the LPT DB25 connector.
Near the beginning of this thread I wasted a lot of time trying to get the setup working not using pin17 or using pins 17 and 19 tied together or leaving pin 19 floating and using pin17.The original uctap readme said to use pin17 and tie pin19 high thru a resistor.I metered pin 12 on the db25 and it seemed ideal,close by and it was at +3.3v so that's why I decided to use that pin.

If you have a working receiver/setup already,you can play with the various ideas presented here.Not something you want to do on a dead receiver.

YLG80

03-18-2009, 07:46 AM

Here is a flash dump of a 5119 STB:
http://dump.elektroda.net/main4400.html
Click on LINBOX 5818 to download the firmware in bin file (4Mb)

---

slugworth

03-18-2009, 08:39 AM

You might be able to mount the transistor on the db25 and tap the +3.3v off some of the db25 pins.

---

slugworth

03-18-2009, 08:46 AM

tckdiv= is mention in some target.cfg files.

That is another thing to play with.
##target mag tap "jei_soc 127.0.0.1 tckdiv=" board_runtime_init
target mag tap "jei_soc 127.0.0.1 tckdiv=4" board_runtime_init
Again,I would only play on known working setups.

---

slugworth

03-18-2009, 08:59 AM

Old flashprg.c and .h files circa 1998 mentioned a dump mode,but never incorporated it into the program.
It would be nice to get that feature put in.

```
char *modestr(ROMTOOL_MODE m)
{
static char *modes[] = {
"NULL_MODE",
"ERROR_MODE",

"ERASE",
"PROGRAM",
"VERIFY",
"DUMP"
};
return(modes[m]);
}
```

```
/* Type defs for filenames and parameters */
typedef enum romtool_mode {NULL_MODE, ERROR_MODE, ERASE, PROGRAM, VERIFY,
DUMP} ROMTOOL_MODE;
typedef enum romtool_report {NULL_REPORT, ERROR_REPORT, FULL, REDUCED, NONE}
ROMTOOL_REPORT;
typedef enum romtool_board {EVAL5500, STB3} ROMTOOL_BOARD;
```

---

slugworth

03-18-2009, 09:38 AM

You can make shortcuts to the .bat files on your desktop,get uctap running and to run the other
bat file double click on the shortcut.
Makes it easier when you are using a breakout box to monitor the db25 lines.
Keep your eye on the prize.

YLG80

03-18-2009, 10:16 AM

Yes, I've read all possible posts on that interface wiring with ucTAP, including yours.
I'm running tests on a Russian GS-FTA6900 and I'm lucky enough to have a full schematics set for that STB. It's a running STB.
It depends perhaps on the wiring inside the different STB.
The STRONG SRT4356 has exactly the same interface although the NJPI-RST circuit is different. Going directly, via a jumper, to the CPU.

In my STB that signal goes to an AND gate with nRST.(74HC08)
Anyway, to run ucTAp with a standard JKEYS compatible interface, it's necessary to move the connection from pin 19 to pin 17 of the interface.
Pin 19 must be set LOW before launching ucTAP.
ucTAP will then raise that pin to HIGH and pulse it LOW to start a JTAG session.
Afterwards it will remain high except if we initiate different processes within the same

flashprg session. (verify, erase, program)

I've double checked pin 12 of the DB25. When the status register is polled, there is a short pluse on that pin if there is a pullup resistor connected (2.2KOhms).
This is perhaps the reason why it works well in your configuration.
With my buffered interface it does not produce any pulse at the output of the buffer.(74HC244)

YLG80

03-18-2009, 10:27 AM

from Slugworth

tckdiv= is mention in some target.cfg files.
That is another thing to play with.
Code:

##target mag tap "jei_soc 127.0.0.1 tckdiv=" board_runtime_init
target mag tap "jei_soc 127.0.0.1 tckdiv=4" board_runtime_init

Again,I would only play on known working setups.

Yes, thanks to point this out.
Very interesting.
From the manual :

Target boards that do not work at the default TCK rate of 25 MHz should work with a tckdiv set at either two or four.
Values between 16 - 64 can work for very slow targets such as emulators.

But this can be tricky if we don't know whether the problem is related to the interface or the timing ...
As you mentioned, it's better to debug first the interface with a known good STB.:faint2:

---

slugworth

03-18-2009, 11:09 AM

As I mentioned before,a lot of the tweaks should be done on setups already working,to see how much effect it has on performance.I added in a breakout box to monitor signals,doubling the length of my simple jtag cable with no ill effects.

---

slugworth

03-19-2009, 12:46 PM

I modified mb400.cfg to weed out stuff not needed for jtagging.
It uses the original dcu_mb400_conf.cfg from toolset v2.3.1 and no separate 5105ckg.cfg.
Should make it easier for people modifying the procedure for other processors.
Should still be considered a work in progress.
the 699.bin is the skymag v700 beta I call 699

Alex_Trask

03-19-2009, 01:37 PM

Today morning ucTAP was able to start the service without errors on a Skybox STi5512SWE.
Now ucTAP is not able to start the service using the same way.

Why this happens.

---

slugworth

03-19-2009, 02:03 PM

Today morning ucTAP was able to start the service without errors on a Skybox STi5512SWE.

Now ucTAP is not able to start the service using the same way.

Why this happens.

One of the great mysteries of electronics.
As long as it detects the processor and starts 2 servers you should be ok.

Alex_Trask

03-19-2009, 04:16 PM

Why ucTAP detect my STi5119 as a STi5105?

---

slugworth

03-19-2009, 05:11 PM

ucTAPsrv.cpu file wrong?
STi5105, 0x2d427041, 0x30003004, 0x00000200
STi5100, 0x3d423041, 0x30003004, 0x00000200
STi5119, 0x3d427041, 0x30003004, 0x00000200

Alex_Trask

03-19-2009, 05:15 PM

It include these processors

Alex_Trask

03-19-2009, 05:23 PM

Sti5119ALC SST39VF3201 Full Flash

Using jkeys with normal buffered jtag interface, at start address 40000000

Don't use this file in your receivers, it is channel locked. Only a regular conax smart card can open channel list.

Password Protected File

---

slugworth

03-19-2009, 06:25 PM

It include these processors
Doesn't really matter,you don't even need the id in the .cpu file and it
should still work.It is mostly for diagnostics to make sure the tap is wired right.The same goes for the addresses after the micro id,not needed.

---

alarm

03-19-2009, 06:26 PM

@Alex_Trask
Can-can you here at the forum to put the program on which you worked?

Thank you!

---

slugworth

03-19-2009, 06:31 PM

Sti5119ALC SST39VF3201 Full Flash

Using jkeys with normal buffered jtag interface, at start address 40000000

Don't use this file in your receivers, it is channel locked. Only a regular conax smart card can open channel list.

Password Protected File
The main firmware starts at address 0x70020 in the dump.
You can verify against a normal ods2000c v123.pgm file.

---

slugworth

03-19-2009, 06:33 PM

Why ucTAP detect my STi5119 as a STi5105?

delete the st5105 from the .cpu file

Alex_Trask

03-19-2009, 07:40 PM

delete the st5105 from the .cpu file

Right, I have delete the sti5105 and all others using 0x30003004, 0x00000200 leaving only STi5119 and was detected.

You think I can erase and write flash using a .lku file for STi5119 and SST39VF3201.

If yes, the next step is: finding a full flash compatible with conax embedded stb.

---

slugworth

03-19-2009, 08:16 PM

Run ho.bat with the files from the c:\maggie setup to verify the maggie lku with your receiver.It should setup the receiver then kick into the commandline options for the burner.No erase or program with ho,so it is safe.

---

slugworth

03-19-2009, 08:22 PM

the next step is: finding a full flash compatible with conax embedded stb.
Actually,just find a bootstrap section from a clone receiver,then load the rest of the file with the clones' serial loader.
Usually the bootstrap section is the 1st 64k of the flash but may vary from receiver to receiver/processor to processor.

---

slugworth

03-19-2009, 08:32 PM

Don't be alarmed by the lack of sst39vf3201 in the ho screen.
I never got around to adding it yet.

Alex_Trask

03-20-2009, 05:49 AM

@Alex_Trask
Can-can you here at the forum to put the program on which you worked?

Thank you!

I used jkeys.exe with a normal buffered jtag interface on WinXP sp3 with 3GHz processor.

The differences are between jkeys.def
I used the latest jkeys with this jkeys.def but in your case it can't work because your flash is different.

Alex_Trask

03-20-2009, 08:14 AM

I can not make the files .lku and others for sti5119 and 39vf3201 so ho.bat doesn't work.

There is some file that I can try to use?

---

alarm

03-20-2009, 10:06 AM

What am I doing wrong???

---

alarm

03-20-2009, 10:13 AM

What am I doing wrong???

Where do I make a mistake?

Alex_Trask

03-20-2009, 11:19 AM

Where do I make a mistake?

See the attached image.
1- If processor is detected- don't touch the red areas.
2- The green areas are OK
3- Put the value 7FE00000
4- Click only on blue area "Save Mem"

Don't try flash programming or other things with jkeys because jkeys can't progam or erase flash on a STi5119 based receiver.

If this method can't save flash, the problem is hardware or software between PC and Receiver

---

alarm

03-20-2009, 11:53 AM

Thank you.
Which program should I erase and program flash?

---

alarm

03-20-2009, 12:04 PM

4- Click only on blue area "Save Mem"

Done and that's what happened.:no::no::no:

Alex_Trask

03-20-2009, 12:23 PM

Done and that's what happened.:no::no::no:

These errors are not caused by jkeys.
I think that is an PC error or something else, but not a jkeys error.

One of my friends have the same problem with the same programs, jtag interface and receiver as me, but I can read flash and he can't read flash.

I don't know where the problem is.
Try changing PC, Operating System, jtag interface and download fresh jkeys.

---

slugworth

03-20-2009, 12:32 PM

A 4meg flash would never be at address 7FE00000, more likely at address 40000000 like stated in a previous post.

---

alarm

03-20-2009, 12:52 PM

Width instead 16Bits exhibited 32Bits.V Box 2 exhibited Delta instead of 4.

---

alarm

03-20-2009, 01:11 PM

A 4meg flash would never be at address 7FE00000, more likely at address 40000000 like stated in a previous post.

slugworth
You are absolutely right! Asked 40000000 address and received IRDmem file. And then?

Alex_Trask

03-20-2009, 01:14 PM

A 4meg flash would never be at address 7FE00000, more likely at address 40000000 like stated in a previous post.

I thought that the flash is 2meg, but trying the address 400000000 dones't cost.

@alarm
Anyway I think that the problem is you, or your PC, not jkeys.

---

slugworth

03-20-2009, 01:32 PM

slugworth
You are absolutely right! Asked 40000000 address and received IRDmem file. And then?

The default length in jkeys for dumping is 80000 which is 512k size.
The length has to be goosed to 400000 to get a full 4meg dump.
Try again.

Alex_Trask

03-20-2009, 03:23 PM

The default length in jkeys for dumping is 80000 which is 512k size.
The length has to be goosed to 400000 to get a full 4meg dump.
Try again.

He want image to understand

Alex_Trask

03-20-2009, 03:42 PM

I can not make the files .lku and others for sti5119 and 39vf3201 so ho.bat doesn't work.
There is some file that I can try to use?
http://www.ftatalk.com/attachment.php?attachmentid=41121&d=1237558518

---

alarm

03-20-2009, 05:30 PM

The default length in jkeys for dumping is 80000 which is 512k size.
The length has to be goosed to 400000 to get a full 4meg dump.
Try again.

I dropped a full dump: 40000000 address, length 400000
This dump can not use my receiver is dead.

slugworth Alex_Trask
Please help me run the program I was not maggie prisleduyu commercial purposes. It is very difficult to communicate to english.U I have all the necessary files, but I do not understand how they need to invest in the maggie???? You are professionals and I am an amateur.

---

slugworth

03-20-2009, 06:54 PM

The .lku would have to be made for that flash instead of the sst39VF3201.

---

slugworth

03-20-2009, 11:42 PM

Experimental Sti5119 2meg lku.
Unzip to c:\maggie and run ho then go.

You will have to edit the pro.bat and add in the name of your 2meg.bin

---

slugworth

03-21-2009, 08:38 AM

http://www.ftatalk.com/attachment.php?attachmentid=41121&d=1237558518

The mb400.cfg is set for a receiver with 32meg flash.If you have more or less
the mb400.cfg will have to be edited to match.

---

alarm

03-21-2009, 01:46 PM

Experimental Sti5119 2meg lku.
Unzip to c:\maggie and run ho then go.
You will have to edit the pro.bat and add in the name of your 2meg.bin

@slugworth
Using your file today 5119meg_lku.
In the main folder replacing files maggie go, ho, pro, to edentichnye files from a folder 5119meg_lku. In the main folder for my maggie dump receiver renaming it, as you told me in 2m.bin.folder called maggie2. Use processor ALI is konekt, run maggie2 nothing.
Spread its folder maggie2, please, what I am doing wrong???
Thank you!

---

altcode

03-21-2009, 06:28 PM

No luck on Maggie
With jeys it dect it as Stx5155ABL Id 0x2D427041
Maggie is STx5105 using jvvh def file
When runing tap .bat St 5512N ??
Jtag cable pin 17 19 comb uctap-jtag post 144 no go
tryed 833 computer same thing
once when ran tap,bat it it was ST 5105 but no go
Then tryed 500 compter with the normal jtag cable and after

trying over and over it will show the 5105 over the 5512N
when it was 5105 then changed to the uctap cable and ran
the files had errors but it did bring the Maggie back from the dead
Thanks slugworth

---

alarm

03-22-2009, 07:57 AM

You can actually leave the 2 address blank in the .cpu file for uctap and it still works fine.Uctap will automatically come up with it's own addresses.
You can ignore any error messages in uctap,as long as it starts 2 servers.
STi5514, 0x1d40a041, 0x30003004, 0x00000200
STi5516, 0xXd41d041, 0x30003004, 0x00000200
STi5518, 0x1d405041, 0x3004, 0x00080000
STi5519, 0x1d415041, 0x3004, 0x00080000
STi5528, 0x1d41a041, 0x3004, 0x00080000
STi5580, 0x1d40b041, 0x3004, 0x00080000
STi5588, 0x1d41c041, 0x3004, 0x00080000
STi5589, 0x15400041, 0x3004, 0x00080000
STi5598, 0x15401041, 0x3004, 0x00080000
STm5700, 0x157xx041, 0x3004, 0x00080000
ST20TP3, 0x05194041, 0x3004, 0x00080000
ST20TP4, 0x25198041, 0x3004, 0x00080000
ST20GP6, 0x15196041, 0x3004, 0x00080000
ST20MC2, 0x05195041, 0x3004, 0x00080000
STi5105, 0x2d427041,
STi5100, 0x3d423041, 0x30003004, 0x00000200
STi5119, 0x3d427041, 0x30003004, 0x00000200
#
ucTAPsrv End
#

Delete all addresses, except for 5119. I can not????
#
ucTAPsrv CPU definitions file v 0.0
#
format is NAME, ID, SD, DCU_CONTROL, MASK_CONTROL_DEVICE_ACCESS

#
#

STi5119, 0x3d427041, 0x30003004, 0x00000200
#
ucTAPsrv End
#

YLG80

03-22-2009, 12:06 PM

Here are the ID codes for the Sti7100xxx (where xxx are the cut versions) and STi7109xxx.

Sti7100cut3x, 0x2D424041, 0x30003004, 0x00000200
Sti7100cut20, 0x1D424041, 0x30003004, 0x00000200
Sti7100cut1x, 0x0D424041, 0x30003004, 0x00000200
Sti7109cut1x, 0x0D42c041, 0x30003004, 0x00000200
Sti7109cut2x, 0x1D42c041, 0x30003004, 0x00000200
Sti7109cut30, 0x2D42c041, 0x30003004, 0x00000200

With these codes, ucTAP can connect normally to the stb.
However, i don't know yet the correct codes for SD and DCU_CONTROL.:biggrin1:

YLG80

03-22-2009, 02:35 PM

No luck on Maggie
With jeys it dect it as Stx5155ABL Id 0x2D427041
Maggie is STx5105 using jvvh def file

This is because there is a mistyped character (5155 i.o 5105) in your jkeys.def file.
The CPU ID was correct although the translation in human readable format was not.:laugh:

---

slugworth

03-22-2009, 03:11 PM

Here are the ID codes for the Sti7100xxx (where xxx are the cut versions) and STi7109xxx.

Sti7100cut3x, 0x2D424041, 0x30003004, 0x00000200
Sti7100cut20, 0x1D424041, 0x30003004, 0x00000200
Sti7100cut1x, 0x0D424041, 0x30003004, 0x00000200
Sti7109cut1x, 0x0D42c041, 0x30003004, 0x00000200
Sti7109cut2x, 0x1D42c041, 0x30003004, 0x00000200
Sti7109cut30, 0x2D42c041, 0x30003004, 0x00000200

With these codes, ucTAP can connect normally to the stb.
However, i don't know yet the correct codes for SD and DCU_CONTROL.:biggrin1:
I think uctap ignores the 0x0,0x1,0x2,0x3 in the micro id's
so the first 3 will be detected as the 1st one in the list.
The 0x30003004 and 0x00000200 aren't needed in the .cpu.

---

effesse

03-22-2009, 03:36 PM

Dear sirs, is it possible to use uCtap with Stx 5118 ALB (0x2D427041 in jKeys)? My receiver use a M29W160ET (bot boot) and a 10 pin jtag header similato to Nec/Altera standard (same pinout used by Altera ByteBlaster II/MV). I'm still not able to read the full 2MB of the firmware; any procedures I done with other softwares I always got files full of 'FFFFFFFF' or '00000000'!?! What do I need/download to use a working uCtap environment on my PC?

Many thanks in andvance, kind regards, Stefano.

Alex_Trask

03-22-2009, 04:25 PM

Dear sirs, is it possible to use uCtap with Stx 5118 ALB (0x2D427041 in jKeys)? My receiver use a M29W160ET (bot boot) and a 10 pin jtag header similato to Nec/Altera standard (same pinout used by Altera ByteBlaster II/MV). I'm still not able to read the full 2MB of the firmware; any procedures I done with other softwares I always got files full of 'FFFFFFFF' or '00000000'!?! What do I need/download to use a working uCtap environment on my PC?

Many thanks in andvance, kind regards, Stefano.

I don't know how uctap can save the firmware but you can try jkeys using these addresses.
1- Don't touch red areas.
2- Green areas must be same as image.
3- Put addresses and click only "SaveMem".

Regards
*credo che funzionera' :shakehands:

---

effesse

03-22-2009, 04:45 PM

:beer: Yup!!! Ha funzionato con la più semplice delle Jtag e senza Wall! Spiegami perché 40000000 per favore? A presto, Stefano. :beer:

Alex_Trask

03-22-2009, 05:02 PM

:beer: Yup!!! Ha funzionato con la più semplice delle Jtag e senza Wall! Spiegami perché 40000000 per favore? A presto, Stefano. :beer:
Translated:
It worked with the simplest of Jtag, without Wall.exe! Explain why 40000000 please? See you, Stefano

I think here is allowed only english.

I think Sti5118ALB uses DCU3.
DCU3 uses these addresses to map the flash.
Don't try jkeys to erase or write the flash because jkeys doesn't make DCU3
For more read entire topic.

Regards

---

effesse

03-22-2009, 05:15 PM

I think here is allowed only english.

I supposed you were Italian, tnx a lot for all and good night, I'm quite sleeping :-S !

---

alarm

03-22-2009, 06:23 PM

:cry:No one can help me? Today was a day of the program maggie, no konekt ,,,,,,,, :no:
The scheme of the receiver to the processor 5119ALC.

---

effesse

03-23-2009, 08:01 AM

The mb400.cfg is set for a receiver with 32meg flash.If you have more or less
the mb400.cfg will have to be edited to match.

Dear sir, I'm just now able to read my SSt M29W160ET (2MB top boot flash memory) of

my 0x2D427041 (STx5118 ALB Cpu) via a 10 pin header on my satellite receiver modifing the .cpu file and using the the 5119_2m_c1c.lku file.

May you help me erasing and rewriting my 2MB flash memory with uCtap and St20Run softwares, please? I need help modifing the mb400.cfg file and I don't know where to learn about St20Run commands.

I think you will be able to flash my M29W160ET via my simple Jtag cable (I have a buffered one too). I made by myself an Altera Byte Blaster MV/II Jtag interface too because the 10 pin header on the receiver complains with the 'Nec / Altera Byte Blaster MV/II' pinouts (I use the multimeter to follow the wiring pads between the 10 pin header and the STx 5118 ALB Cpu) but I really don't know if that Altera Jtag is strictly usefull to my job. :-(

Kind regards, Stefano.

---

slugworth

03-23-2009, 08:56 AM

Jtag would have to be modified for uctap/10pin usage.
Convert the 10pin jtag pinouts on paper and rewire the jtag for uctap.

Alex_Trask

03-23-2009, 10:21 AM

After "Setup Complete", ho.bat shows "load error-kernel error : target is not responding"

I think it is useless for me to try to modify these files :(:(:(

YLG80

03-23-2009, 02:06 PM

Dear sirs, is it possible to use uCtap with Stx 5118 ALB (0x2D427041 in jKeys)? My receiver use a M29W160ET

As Alex said, you should be able to dump the memory content with JKEYS. If you get only ff's, that likely means that you've not specified the correct flash start address i.e 0X40000000
JKEYS was originally made for the Sti551x that uses another memory mapping in the 7Fxxxxxx region.

Spiegami perché 40000000 per favore?

0X40000000 because the STI51xx have usually flash roms mapped at that address. This makes the difference with the other CPU's like STi5518. (not to confuse with Sti5118)

YLG80

03-23-2009, 02:14 PM

@Alarm,
It's sems that you have stability problems with your interface and ucTPAP

Try to slow down your PC. (if it's a laptop, try to setup the Energy Managment to Battery preservation option. It slows down the CPU))

Try also to reset your LPT port prior to launch ucTAP.
I must do that on my side.
Just launch JKEYS "once for nothing" before and quit the program prior to start ucTAP.
Jkeys resets certain LPT lines that are not reset by ucTAP.

Since I'm doing that, I've no longer any problem to get connected to the CPU with ucTAP.

Are you sure that your connection between the CPU JTAG and the JTAG interface is not too long?
What type of interface are you using? Direct connection via 100 Ohms resitsors, like slugworth ?

Keep trying ! You should succeed.:laugh:

YLG80

03-23-2009, 02:17 PM

Hello Alex,
Have you setup the environment before running ho.bat.
I guess that Slugworth has included a setenv.bat or similar file.

A - Launch ucTAPsrv.exe (one dos window)
B - Open a second DOS window and setup the environment (sentenv.bat or similar)
C - Launch ho.bat

YLG80

03-23-2009, 02:32 PM

I believe that your Altera cannot be used as is.
However if your rewire the pins so that the correct signals from the LPT port are connected to the JTAG pin's, you should be able to use it.
I attach what I believe is your Altera interface, and a drawing of my interface with the 10 pin's connector surrounded in red.
You just need to connect the correct JATG signals to the appropriate DB25 pin.
Altera (left) PCBWire (right)

Alex_Trask

03-23-2009, 04:25 PM

Hello Alex,
Have you setup the environment before running ho.bat.

I guess that Slugworth has included a setenv.bat or similar file.

A - Launch ucTAPsrv.exe (one dos window)
B - Open a second DOS window and setup the environment (sentenv.bat or similar)
C - Launch ho.bat

Thank you YLG
I think I have no right file.cfg, file_mem.cfg and file.lku for a 4meg flash as is SST39VF3201.

I am not good to edit these files :(

---

alarm

03-23-2009, 05:35 PM

Today, even tried on 2 computers, the result is 0!
I think the next problem, my receiver is not available on other models debug.V sure the jumper to debug. In all of my failures???

YLG80

03-24-2009, 02:04 AM

@alarm
Could you remind me what type of JTAG interface are you using?

YLG80

03-24-2009, 02:14 AM

@Alex

I think I have no right file.cfg, file_mem.cfg and file.lku for a 4meg flash as is SST39VF3201.

Are you sure of your flash chip model ?
(from the datasheet)

Organized as 1M x16: SST39VF1601/1602
2M x16: SST39VF3201/3202
4M x16: SST39VF6401/6402

If your chip is a SST39VF3201, then Slugworth last build is OK for you.
That chip is a 2Mb and not a 4Mb

YLG80

03-24-2009, 03:39 AM

I've decompiled ucTAPsrv and it uses a tcp protocol to simulate the ST20 micro connect. For those using a computer with a Firewall, I would advise to setup an exception (for ex. XP Firewall, exception tab) for port 0X378 if they are using that LPT port and an exception for the ucTAPsrv.exe application.
Or simply disable the firewall when using ucTAPsrv in order to allow it to pass through the door:date:.

I guess that people using other Antivirus with firewall, like Fsecure, should also allow communication to pass through the parallel port.
Fsecure can be temporarely disabled.
This could explain your communication problems with the STB.

Alex_Trask

03-24-2009, 04:41 AM

@Alex

Are you sure of your flash chip model ?
(from the datasheet)

Organized as 1M x16: SST39VF1601/1602
2M x16: SST39VF3201/3202
4M x16: SST39VF6401/6402

If your chip is a SST39VF3201, then Slugworth last build is OK for you.
That chip is a 2Mb and not a 4Mb

You are not right YLG!

SST39VF3201 is a 4MByte flash for sure.

Thanks anyway

---

alarm

03-24-2009, 04:41 AM

@alarm
Could you remind me what type of JTAG interface are you using?

I use 6 Rin Jtag scheme slugworth. Cable 15cm.

---

effesse

03-24-2009, 04:46 AM

@Alex

Are you sure of your flash chip model ?
(from the datasheet)

Organized as 1M x16: SST39VF1601/1602
2M x16: SST39VF3201/3202
4M x16: SST39VF6401/6402

If your chip is a SST39VF3201, then Slugworth last build is OK for you.

That chip is a 2Mb and not a 4Mb

@ylg80

I think flash with 160 is 16Mbit so 2MByte,
flash with 320 is 32Mbit so 4MByte,
flash with 640 is 64Mbit then 8MByte,
correct me if I'm wrong, please.

I have a M29W160ET flash and it's just 2MByte.

Best regards, Stefano.

---

effesse

03-24-2009, 04:58 AM

I believe that your Altera cannot be used as is.
However if your rewire the pins so that the correct signals from the LPT port are connected to the JTAG pin's, you should be able to use it.
I attach what I believe is your Altera interface, and a drawing of my interface with the 10 pin's connector surrounded in red.
You just need to connect the correct JATG signals to the appropriate DB25 pin.
Altera (left) PCBWire (right)

This is my Jtag 10 pin header pinout, checked with multimeter, PcbWire is intended for other kind of CPUs but not Stx 5118 ALB. However tnx for the tips.

Best regards, Stefano.

YLG80

03-24-2009, 05:05 AM

:stoned:
I'm confused ?????
Are you thinking to 4M x 8 bit bytes.
This a 16 bits byte flash chip.

---

effesse

03-24-2009, 05:24 AM

:stoned:
I'm confused ?????
Are you thinking to 4M x 8 bit bytes.
This a 16 bits byte flash chip.

So I'm right:

1M x 16 = 16Mbit = 2MByte (39vf160)
2M x 16 = 32Mbit = 4MByte (39vf320)
4M x 16 = 64Mbit = 8Mbyte (39vf640)

Any one can correct me, please ?

Kind regards, Stefano.

Alex_Trask

03-24-2009, 05:28 AM

:stoned:
I'm confused ?????
Are you thinking to 4M x 8 bit bytes.
This a 16 bits byte flash chip.

I don't know what you want to say but you must read the datasheet "TABLE 10".

Regards :)

I quote Stefano: you are right!

---

effesse

03-24-2009, 06:03 AM

I don't know what you want to say but you must read the datasheet "TABLE 10".

Regards ;)

I quote Stefano: you are right!

.1. I know primarily that 1 byte is always equal to 8 bits.

.2. I know the 32 in the device name means 32M(bits) then I divide 32M into 8 bits (how I said at point .1) so (32 / 8) = 4M(Bytes)

.3. I'm pretty sure of this for about last 10 years of eprom personal interesting.

Sorry for my bad English but I'm Italian, ciao.

YLG80

03-24-2009, 07:33 AM

:laugh:You are right.
With the set top box I'm always thinking to 16 bits words when I'm playing with flash chips !
If you look on the Net, you will see that there are several discussions like that ie on Wikipedia.
So I'm Ok with 4 MB with a byte of 8 bits.

---

slugworth

03-24-2009, 07:44 AM

If you get toolset v2.3.1 the examples folder has examples for various processors and flash devices.It's a matter of tweaking the correct mbxxx.c file to match the flash in your receiver then make a new .lku file.
The maggie burner was based on the edited mb385.c file.

---

slugworth

03-24-2009, 07:50 AM

I forgot to mention you also have to edit the flashprg.c file before you make the new .lku,to reflect the correct starting address.

Alex_Trask

03-24-2009, 08:10 AM

If you get toolset v2.3.1 the examples folder has examples for various processors and flash devices.It's a matter of tweaking the correct mbxxx.c file to match the flash in your receiver then make a new .lku file.
The maggie burner was based on the edited mb385.c file.

To make new .lku file in examples folder of st20 toolset, nmake.exe need all mbxxx.c evalxxx.c etc... Doesn't work with only one file.c
How can I create or edit mbxxx.cfg and mbxxx_mem.cfg from mbxxx.c?

There are too many definitions that I don't understand!

---

slugworth

03-24-2009, 09:29 AM

To make new .lku file in examples folder of st20 toolset, nmake.exe need all mbxxx.c evalxxx.c etc... Doesn't work with only one file.c
How can I create or edit mbxxx.cfg and mbxxx_mem.cfg from mbxxx.c?

There are too many definitions that I don't understand!
The .mkf file,flashprg.c flashprg.h mb385.c all have to be edited.
Nobody said it was easy.

MB400.cfg was the last model in the toolset,so sti51xx are all variations of that config file.

Alex_Trask

03-24-2009, 11:59 AM

The .mkf file,flashprg.c flashprg.h mb385.c all have to be edited.
Nobody said it was easy.
MB400.cfg was the last model in the toolset,so sti51xx are all variations of that config file.

I have made a lot of .lku files, but same results: go error-address space not stopped
Maybe the .lku is right!

So I tried 100 or more times to edit .cfg files with bad results.

*All times same error as in images

---

slugworth

03-24-2009, 12:27 PM

ram is still showing as 32 meg

---

slugworth

03-24-2009, 12:31 PM

the mb400.cfg has to be edited to match the ram in your receiver.
You take away the ## in front of the line you want to use.

```
SDRAM_DEVICE_TYPE = "DDR_RAM"
## SDRAM_DEVICE_TYPE = "SDR_RAM"

##
## SDRAM_SIZE_CONFIG = "BUILT_FOR_512MBIT"
SDRAM_SIZE_CONFIG = "BUILT_FOR_256MBIT"
SDRAM_FREQ_CONFIG = "166"
## SDRAM_SIZE_CONFIG = "BUILT_FOR_128MBIT"
## SDRAM_FREQ_CONFIG = "133"
if (SDRAM_DEVICE_TYPE == "SDR_RAM") {
SDRAM_CAS_CONFIG = "SDRAM_CAS_2_0"
} else {
if (SDRAM_FREQ_CONFIG == "133") {
SDRAM_CAS_CONFIG = "SDRAM_CAS_2_0"
} else {
SDRAM_CAS_CONFIG = "SDRAM_CAS_2_5"
}
}
FLASH_CONFIG = 4
USE_TRACE = 0
##
##
## includes
## include "5105regs.cfg" ## register offset definitions
include "5105ckg.cfg" ## Clock configuration
## include "mb400_mem.cfg" ## LMI and EMI configuration
include "targets.cfg"

## used mb300.h
_KB_ = 1024
_MB_ = (1024 * 1024)
LMI_BASE = 0xC0000000
if (SDRAM_SIZE_CONFIG=="BUILT_FOR_256MBIT") {
LMI_SIZE = (32*_MB_)
NCACHE_SIZE = ( 4*_MB_)
AVMEM_SIZE = (16*_MB_)
}
if (SDRAM_SIZE_CONFIG=="BUILT_FOR_128MBIT") {
LMI_SIZE = (16*_MB_)
NCACHE_SIZE = ( 2*_MB_)
```

NCACHE_SIZE = (2*_MB_)

AVMEM_SIZE = (8*_MB_)
}
if (SDRAM_SIZE_CONFIG=="BUILT_FOR_512MBIT") {
LMI_SIZE = (64*_MB_)
NCACHE_SIZE = (8*_MB_)
AVMEM_SIZE = (32*_MB_)
}
TH_SIZE = (1*_KB_)
if (USE_TRACE == 1) {
TRACE_SIZE = (64*_KB_)
} else {
TRACE_SIZE = (0)
}
512mbit is a receiver with 64meg ram and so on and so forth.

Alex_Trask

03-24-2009, 12:34 PM

My ram is 128Mb K4S281632I-UC60 but on mb400.cfg there are not options to set a 128Mb ram.

I am wrong?

---

slugworth

03-24-2009, 12:46 PM

No option,you will have to add it and follow the pattern for the rest of the settings; 32 is double the 16, 64 is double the 32 etc.
so it would be "built_for_1024mbit"

Alex_Trask

03-24-2009, 12:52 PM

I have just made the mb400.cfg but mb400_mem.cfg it's more complicate.

Now I'm trying.

---

v2neo

03-24-2009, 02:04 PM

Great job on maggy guys. Iv got an st5107 arion box with m28w640fs, from what i understand the core of the cpu is similar to 5105, i dont have experiance with either of these. However using jkeys with simple jtag using ping 9,11,13,15,19 i can get cpu 0x0D432041 and skymax reports Init JTAG
IDCODE : 0D432041
Fail DCU, Plesae Restart or Reset CPU Box

was wondering if i switch pin 19 to 17 if i can just read memeory using jkeys? does someone have a def for this?

thanx

Alex_Trask

03-24-2009, 02:32 PM

Great job on maggy guys. Iv got an st5107 arion box with m28w640fs, from what i understand the core of the cpu is similar to 5105, i dont have experiance with either of these. However using jkeys with simple jtag using ping 9,11,13,15,19 i can get cpu 0x0D432041 and skymax reports Init JTAG
IDCODE : 0D432041
Fail DCU, Plesae Restart or Reset CPU Box

was wondering if i switch pin 19 to 17 if i can just read memeory using jkeys? does someone have a def for this?

thanx

@v2neo

Read http://www.ftatalk.com/showpost.php?p=1671490&postcount=239

Bytes 200000 for 2Mb flash
Bytes 400000 for 4Mb flash etc

Use normal jtag interface or buffered one!

@slugworth

Thank you, I fixed sdram but the final result is the same!
Please tell me what can I change now.

Regards

---

slugworth

03-24-2009, 04:13 PM

Great job on maggy guys. Iv got an st5107 arion box with m28w640fs, from what i understand the core of the cpu is similar to 5105, i dont have experiance with either of these. However using jkeys with simple jtag using ping 9,11,13,15,19 i can get cpu 0x0D432041 and skymax reports Init JTAG
IDCODE : 0D432041
Fail DCU, Plesae Restart or Reset CPU Box

was wondering if i switch pin 19 to 17 if i can just read memeory using jkeys? does someone have a def for this?

thanx

it's a variation of the 5100, c2 core but needs DCU3 jtag.
look at the mb390.cfg that is part of toolset v2.3.1

---

amrinder71

03-24-2009, 04:21 PM

slugsworth can u help me here:http://www.ftatalk.com/showthread.php?t=271070 i need your pro coding help

---

Alex_Trask

03-24-2009, 05:50 PM

I'm posting the files I use to jtag STi5119ALC - SST39VF3201

If you have a moment, please look at these files.

If you see anything wrong tell me.

Thanks, Sorry for my bad english.

---

v2neo

03-24-2009, 06:00 PM

it's a variation of the 5100, c2 core but needs DCU3 jtag.
look at the mb390.cfg that is part of toolset v2.3.1

to clarify what i was asking. Do u jtag pin 17 or pin 19 using jkeys to read mem? using pin19 i can get cpu but to read mem i have to use pin 19 or 17?

reason i ask is i attempted to read with pin19 but i get error reading ird (dcu peek) msg

im just tryign to read the mem now not trying to erase/rewrite yet

thanx

Alex_Trask

Alex_Trask

03-24-2009, 06:15 PM

Yes, you must use pin19.
If you get dcu peek error, you can try this:
Open wall.exe (search for it) and leave it open, then open jKeys to Save Mem

I don't know if I'm right but trying doesn't cost!
Regards

---

v2neo

03-24-2009, 06:21 PM

Yes, you must use pin19.
If you get dcu peek error, you can try this:
Open wall.exe (search for it) and leave it open, then open jKeys to Save Mem

I don't know if I'm right but trying doesn't cost!
Regards

tryed doing as u sugested with wall first runnig than jkeys (i still get the same error) but tryed to read anyways form 40000000 and (200000 and 400000) both failed to read/save

thanx

Alex_Trask

03-24-2009, 06:33 PM

tryed doing as u sugested with wall first runnig than jkeys (i still get the same error) but tryed to read anyways form 40000000 and (200000 and 400000) both failed to read/save

thanx

I think that there are communication problems or stability between pc and stb.
I don't know what, but something is wrong. The problem I think is not jkeys.

Try different OS, different PC, jtag etc if you can

*Your flash is 8Mb right? if yes you must use Byte 800000 for full flash dump

---

slugworth

03-24-2009, 06:36 PM

I'm posting the files I use to jtag STi5119ALC - SST39VF3201

If you have a moment, please look at these files.

If you see anything wrong tell me.

Thanks, Sorry for my bad english.

The inits to detect the flash are wrong,but that would show up later as a problem.
Add flashprg.map to the build.mkf file to output a map file for troubleshooting.

```
#
# Makefile
#
# Copyright (C) STMicroelectronics Ltd. 2000-2004
#
# Makefile for the application
#

CCUSERFLAGS = -fdisable-device -g -O0

EXPORTS = flashprg_c1c.lku flashprg_c1c.dbg flashprg.map

OBJSC1 = flashprg.tc1.mb400.tc1
```

OBJSC1 = flashprg.tc1 mb400.tc1

OBJSC2 = flashprg.tc2 mb400.tc2

CLEAN = $(EXPORTS) $(OBJSC1) $(OBJSC2)

build: $(EXPORTS)

flashprg_c1c.lku flashprg_c1c.dbg: $(OBJSC1)
$(CC) -p STi5105MB400 $(LDFLAGS) -o $o $@ $(OBJSC1) -M flashprg.map

---

slugworth

03-24-2009, 06:39 PM

The delay is missing from the mb400.c file,so it probably wouldn't work on a fast pc.

---

v2neo

03-24-2009, 06:44 PM

I think that there are communication problems or stability between pc and stb.
I don't know what, but something is wrong. The problem I think is not jkeys.

Try different OS, different PC, jtag etc if you can

*Your flash is 8Mb right? if yes you must use Byte 800000 for full flash dump

ill try on different pc. But just so u know same pc with same jtag was used many times on 301-10 and 301-13 and others..... so dont think its pc or jtag itself

thanx

Alex_Trask

03-24-2009, 06:56 PM

ill try on different pc. But just so u know same pc with same jtag was used many times on 301-10 and 301-13 and others..... so dont think its pc or jtag itself

thanx

@v2neo
Same identical problem has one of my friends, he never solved

@slugworth
I'm using WinXP Pro SP2 on Intel celeron @797MHz 248MB RAM to do jtag.

Can I slowdown it more?

---

slugworth

03-24-2009, 07:05 PM

mb400.c modded,works with magnum receiver.
Use MB400 in the commandline instead of the old MB385.
There are still errors in the map file.
As an example the flash shows as 8meg size.

Alex_Trask

03-24-2009, 07:45 PM

mb400.c modded,works with magnum receiver.
Use MB400 in the commandline instead of the old MB385.

The mb400.c that I modded works with magnum receiver?

There are still errors in the map file.
As an example the flash shows as 8meg size.

Should change addresses?

---

slugworth

slugworth

03-24-2009, 07:55 PM

The mb400.c that I modded works with magnum receiver?

No,I had to mod your mod.
Nice trimming job tho.
If you you compare you will see what I did.

---

slugworth

03-24-2009, 07:58 PM

Should change addresses?
Somewhere the flash size is defined as 8meg,that has to be found and changed.

---

slugworth

03-24-2009, 08:02 PM

All this assumes the sti5119 locations are the same as the sti5105; if they are different you will be dead in the water.

Alex_Trask

03-24-2009, 08:18 PM

I hope not, I spent a month with this receiver.:typing::ranger::faint:

Don't say that.:tape2:

---

slugworth

03-24-2009, 08:47 PM

I hope not, I spent a month with this receiver.:typing::ranger::faint:

Don't say that.:tape2:

How many years did I spend?

---

v2neo

03-24-2009, 08:52 PM

sorry to butt in again. But i tryed 2 other pc's with same result.... could there be anything else??? rmember im just trying to READ using jkeys thats all

thanx

---

slugworth

03-24-2009, 09:09 PM

sorry to butt in again. But i tryed 2 other pc's with same result.... could there be anything else??? rmember im just trying to READ using jkeys thats all

thanx

Most jkeys errors are due to a bad ground or jtag wires too long.
If the address is correct you should dump with no error.

---

slugworth

03-24-2009, 09:17 PM

dcu_mb400.cfg has to be edited to get the map flash to 4meg.
memory EXTERNAL (_ST_global_EXTERNAL_BASE) (_ST_global_EXTERNAL_SIZE) RAM

_ST_global_FLASH_BASE = 0x40000000
_ST_global_FLASH_SIZE = (8*M)

memory FLASH (_ST_global_FLASH_BASE) (_ST_global_FLASH_SIZE) ROM
memory STEM0 0x41000000 (32*M) DEVICE
memory STEM1 0x43000000 (32*M) DEVICE
bootiptr (_ST_global_FLASH_BASE)

heap EXTERNAL
stack EXTERNAL (24*K)

PlaceDebugTrapHandler EXTERNAL

---

v2neo

03-24-2009, 09:25 PM

Most jkeys errors are due to a bad ground or jtag wires too long.
If the address is correct you should dump with no error.

so even if i can read using same jtag on 301-13 it could be wires too long?? i will try buffered and shorter cable.

also the error i get "error reading from ird (dcu peek0) is due to jtag cable u think?

thanx

Alex_Trask

03-25-2009, 08:09 AM

I'm becoming crazy!!!!.
I get always the same error!

---

slugworth

03-25-2009, 08:56 AM

Without the actual receiver to play with,not much I can do.

---

alarm

03-25-2009, 09:19 AM

Without the actual receiver to play with,not much I can do.

I can send you a receiver processor 5119.YA need your home address where you live.

There is no possibility to make a mini-program for the processor 5119.s which could only be erased, flash?

Alex_Trask

03-25-2009, 05:40 PM

I'm always trying, but I'm not a good coder!:smash:

Here there are a lot of files that must be modified but I do not know which files

exactly!!!!:frusty:

---

dadolino

03-26-2009, 02:19 AM

Hi guys,

i have also the problem to erase and to flash the S29AL016D via St5119 cpu. The cpu is detect correct with ST20 also with jkeys. With St20 an go.bat it starts but it dont erase the flash. With pro.bat the same. Have you any idea what is wrong? The jtag interface seems to be right. I have made some pictures so maybe it helps.:twitcy:

Alex_Trask

03-26-2009, 06:42 AM

Up to now nobody has jtagged a STi5119 with good results.

I'm making 100+ tests per day but nothing good.
I am not sure, but it seems that st20 toolset 2.3.1 does not support STi5119.

---

slugworth

03-26-2009, 07:33 AM

try the .lku from the original toolset v2.3.1 with different -b options on the commandline

to see how the receiver reacts.Like instead of -b MB385 try the others on the no list.

---

slugworth

03-26-2009, 07:55 AM

sorry to butt in again. But i tryed 2 other pc's with same result.... could there be anything else??? rmember im just trying to READ using jkeys thats all

thanx

Add an extra ground wire from receiver to pc.
I had the same problem last night on a previously working setup.
I added a jumper wire with clips from receiver chassis to pc chassis and it dumped fine after that.

Alex_Trask

03-26-2009, 08:07 AM

Excuse me, I did not understand!

Are you saying to change this? -b MB385 to -b MB282, or MB400 or any MBxxx?

Thanks

---

slugworth

03-26-2009, 08:50 AM

Excuse me, I did not understand!

Are you saying to change this? -b MB385 to -b MB282, or MB400 or any MBxxx?

Thanks

doesn't hurt to try.

Alex_Trask

03-26-2009, 09:44 AM

Unfortunately does not work, exactly same error.

Maybe I must search something about st20 toolset and STi5119.

Alex_Trask

03-26-2009, 08:02 PM

Look at this flashprg.map slugworth.

I'm going to test the lku file and in a few minuts i will edit this topic to add the result.
Nothing, I get the same error

YLG80

03-29-2009, 02:49 AM

@dadolino
Are you using Slugworth setup to try to erase your flash?
If yes, you would have to revert to the original MB385 flash erase and write definitions.
Slugworth had to change them for the maggie flash chip.
Basically the maggie has a flash command offset with 4 digits like 0x5555 or 0xAAAA although yours has a command offset on 3 digits like 0x555.
So, with the original MB385 definitions, it should work for you.
Of course you would need to recompile the flashprg.lku.
You should use the modified slugworth version as he added some extra timing delay in order to accomodate with fast pc's.

---

slugworth

03-29-2009, 08:40 AM

For some 2meg flashes the sdm.c may be a better one to mod.
Don't forget to add the delay and edit the flash address/size in the .c files you mod.(don't

forget the flashprg.c addresses)

---

slugworth

03-31-2009, 09:40 AM

The original flash programmer v1.0 from toolset v1.6.2 had a crude menu system,since at that time there were only 2 development boards to play with.Once they started adding more boards,they discarded the menu and went to a commandline only format.
My goal is to get a menu system that will cover different size flash without having to edit the commandline.The 2meg option is just a tease at this point and the lku is still buggy.I can erase but not program.The menu should also have an exit selection in case you want to bail out gracefully.Adding a dump function is another goal,instead of relying on jkeys all the time.

YLG80

04-06-2009, 05:29 AM

I'm really getting mad with this trap handler ...

When I look into the various STi5105 or Sti5119 firmwares, I always find the following routine in the bootloader, romload section :

```
OM:04000B54 sub_4000B54: ; CODE XREF: sub_4000B54+41p
ROM:04000B54 stl 1
ROM:04000B55 ldl 1
ROM:04000B56 ldnl 0BFC8h
ROM:04000B5A eqc 2
ROM:04000B5B cj loc_4000B6D
ROM:04000B5D ldc 1Fh
ROM:04000B5F bitmask
ROM:04000B61 not
ROM:04000B62 ldc 3004h
ROM:04000B66 ldnl 0
ROM:04000B67 and
ROM:04000B68 ldc 3004h
ROM:04000B6C stnl 0
```

((0x3004 is the DCU Control Register ... in a ST20 C2 CPU....)

I've found that routine in the Magic Surprise 5105 and in my own STB with Sti5105. This can be found around address 40000Bxx in the romload section.

Same in the 5119 firmware recently posted on ftatalk.

This would mean that the DCU Control register in a Sti51xx is also located in 0x3004

????
I'm wondering if it has something to do with little or big-endian notation.

Why are they showing 0x30000000 as a DCUBAse in the TS chip.cfg file?

I've also disassembled and fully commented the c1trap.bin available in the ST20 TS. Surprising ! Have a look on the beginning of that trap routine:

```
c1trap.bin Format : Binary file
Base Address: 0000h Range: 80000140h - 800002F7h Loaded length: 01B7h
Commented by YLG80
Processor : st20
================================================================
============================

Segment type: Regular
section RAM
------------------------------------------------------------------------
ldc 3004h ; DC_CONTROL_REG
; has to be replaced by 0x30000004 with Sti5105 ?????
```

```
ldlp 0 ; start of DCULD
```

```
/* there is a memory arbiter/DCU2 bug whereby an access to memory can linger while
the access completes and a DCU read during this period results in the wrong value
being returned. Workround is to get the CPU-memory link quiet before accessing DCU.
C1 needs nops after the access, particularly for target pokes. (DCUST/DCULD)
*/
ldnl 0
adc 0
rot
ldnl 0
nop
nop
nop
nop
nop ; end of DCUST/DCULD
ldc 60000h ;(DC_MASK_CONTROL_INHIBIT_TRAP |
DC_MASK_CONTROL_INHIBIT_UNTIL_NEXT_FLOW)
or
ldc 3004h ; DC_CONTROL_REG
; has to be replaced by 0x30000004 with Sti5105
ldlp 0 ; again DCUST
ldnl 0
adc 0
rot
stnl 0
nop
nop
nop
nop
nop ; end of DCUST/DCULD
; /* Increment re-entry count */
```

Here below the beginning of the c2trap.bin routine :

```
RAM:80000188 ; ---------------------------------------------------------------------------
RAM:80000188 stl 0Ch
RAM:80000189 stl 0Dh
RAM:8000018A stl 0Eh
RAM:8000018B ldc 3004h
RAM:8000018F ldlp 0
RAM:80000190 ldnl 0
RAM:80000191 adc 0
RAM:80000192 pop

RAM:80000194 ldnl 0
RAM:80000195 ldc 60000h
RAM:8000019A or
RAM:8000019C ldc 3004h
RAM:800001A0 ldlp 0
RAM:800001A1 ldnl 0
RAM:800001A2 adc 0
RAM:800001A3 pop
RAM:800001A5 stnl 0
RAM:800001A6 ldl 1Eh
RAM:800001A8 adc 1
RAM:800001A9 stl 1Eh
RAM:800001AB ldpri
```

That would mean that the only differences between the C2 and C1 trap routines are to be found in a few CPU instructions that are different.

Any thoughts on that mystery?:drama:

YLG80

04-06-2009, 11:43 AM

Whatever is to be found in the C1trap:deal:, the DCU_CONTROL_REGISTER is actually in 0X30000004. :biggrin1:

When I read 0x3004 in an STi5105 it always returns 0.

When I read 0x30000004 it returns 0x00000016 (STB in idle state)

When I write 0x20000000 into 0x30000004, it returns 0X20000010 (DCU_HOSTED_BUSY bit is ON).

When I write 0x60000 into 0x30000004, the CPU is no longer responding. Cannot restart it via JTAG

Normally, if the traphandler routine was correctly loaded and executed, I should read 60000. I'm going to modify the trap routine so that it writes 0x60000 in 0x30000004 instead of 0x3004.
I still have to figure out how to assemble code for a C1 core ...:der:
To be continued ...:hello:

PS I use a jtag routine based on the nice T. Vlad program.

---

jvvh5897

04-07-2009, 02:12 PM

To disassemble C1 code, you can use IDA Pro's ST20C1 setting, I think you can figure out what base address to use. On those small trap handlers, you could get away with just about any addr you want, but on bigger files you have better results with correct base addr.

You can assemble code from C with the ST Toolsets. If you want to write in assembly language, you can do that under C if you want to (you can see examples of that in the files you are using for the trap handlers--some of it is written in asm under C).

---

rmssf

04-07-2009, 03:26 PM

Hi, very interesting thread here.

There is one thing though that confuses me, if the 5119 DCU memory space goes from 0x30000000 to 0x30000FFF, how come in the 'ucTAPsrv.cpu' file you use 0x30003004 for 'DCU_CONTROL'? And what is 'MASK_CONTROL_DEVICE_ACCESS' used for?

Thanks and good work.

---

slugworth

04-07-2009, 04:25 PM

Hi, very interesting thread here.

There is one thing though that confuses me, if the 5119 DCU memory space goes from 0x30000000 to 0x30000FFF, how come in the 'ucTAPsrv.cpu' file you use 0x30003004 for 'DCU_CONTROL'? And what is 'MASK_CONTROL_DEVICE_ACCESS' used for?

Thanks and good work.
At the time the .cpu file settings were unknown.
The 0x30003004 was just a guess,based on the sti5514 DCU3 jtag settings which were the only example of a working DCU3 jtag.As it turns out the settings in the .cpu file are basically for troubleshooting to show you the uctap is detecting the cpu correctly,the dcu_control and mask_control settings in that file can be left blank.The ucTAPsrv will come up with its own settings,which are transparent to the jtagging process.
If you'll notice one of the settings for the sti5514 is an id of 0x00000000,which I relabel "ain't right" instead of sti5514 for my .cpu file to show the uctap isn't detecting the cpu properly.A micro would never have an id of all zeroes.

---

star14

04-08-2009, 12:04 PM

Important type

mb400.cfg ->

```
## SDRAM_DEVICE_TYPE = "DDR_RAM"
SDRAM_DEVICE_TYPE = "SDR_RAM"
##
##
## SDRAM_SIZE_CONFIG = "BUILT_FOR_256MBIT"
## SDRAM_FREQ_CONFIG = "166"
SDRAM_SIZE_CONFIG = "BUILT_FOR_128MBIT"
SDRAM_FREQ_CONFIG = "133"
```

to me a positive result

YLG80

04-11-2009, 03:13 AM

@jvvh thanks for your post.
I've managed to write a new DCU3trap handler using the TS.
Rather easy, although I had to convert the C2 code to C1.
My new trap handler which includes all flash prog routines fits into the maximum allowed space of 1Kb.

I've deeply analyzed and commented the ucTap capture (from Slugworth) of his successfull maggie session.
Although it clears up the SRAM from 0x8000001c up to 0x80000038 (INTERNAL SRAM), the traphandler is finally loaded into EXTERNAL SDRAM.(0xC07FFC00)

This makes the traphandler setup process much more complex as we have to program the
SDRAM registers according to the SDRAM type present in the STB.
(CAS 2.0 or 2.5, bank select, memory access speed, refresh cycle and so on.)

In the capture, this is all made with several pokes to setup the LMI registers.
The LMI setup is completed just before the memset/memset trap instruction found in the capture.
After that the DUC3_Controller flags are changed from 0x3C11 to 0c3C50 in order to activate the trap
and allow the host application to run (i.e flashprg).

I believe it would be much more easy to install the trap in INTERNAL RAM (for example starting from 0x80000140)

despite of the fact that the Sti5105 has only 2 Kb of INTERNAL SRAM.
In stddefs.cfg, there is a directive to setup the traphandler in internal.

---

slugworth

04-11-2009, 07:49 AM

Map from the flashprg v1.0 dc1/mb292a.cfg based modified for sti5105.
The prog detects and erases the flash,but burps when it tries to program.

```
MAP OUTPUT FILE
---------------

LKU file : flashprg.lku

Segment Map
-----------

Name Address Size Type
-------------- ---------- -------- --------
SYSTEM 0x80000000 128 RESERVED
INTERNAL 0x80000080 896 RAM
TRAPHANDLER 0x80000400 1024 DEBUG
```

EXTERNAL 0xc0000000 33553408 RAM

LMI 0xe0000000 16777216 DEVICE
EMI 0x20200000 4096 DEVICE
INTLEVELCTRL 0x20800000 4096 DEVICE
LPC 0x20808000 4096 PERIPHERAL
IRB 0x20818000 4096 PERIPHERAL
PIO0 0x20820000 4096 DEVICE
PIO1 0x20821000 4096 DEVICE
PIO2 0x20822000 4096 DEVICE
PIO3 0x20823000 4096 DEVICE
ASC0 0x20830000 4096 DEVICE
ASC1 0x20831000 4096 DEVICE
SSC0 0x20840000 4096 DEVICE
SSC1 0x20841000 4096 DEVICE
DCU 0x30000000 4096 DEVICE
DCACHE 0x30001000 4096 DEVICE
ICACHE 0x30002000 4096 DEVICE
PWM 0x30003000 4096 DEVICE
INT_CONTROLLER 0x30004000 4096 DEVICE
FLASH 0x40000000 4194304 ROM
SIMFLASH 0x41000000 4194304 DEVICE

Section/Region Map

Name Address Size Type Segment SB
---------- ---------- -------- -------- ------------- --
def_code 0xc0000000 18496 CODE EXTERNAL
def_bss 0xc0004840 5324 BSS EXTERNAL **
def_data 0xc0005d0c 332 INIT EXTERNAL
<shutdown_section> 0xc0005e58 12 CONST EXTERNAL
<startup_section> 0xc0005e64 12 CONST EXTERNAL
bootdata 0xc0005e70 40 BOOTDATA EXTERNAL
stack 0xc0005e98 16764596 STACK EXTERNAL
heap 0xc1002d4c 16764596 HEAP EXTERNAL

YLG80

04-12-2009, 02:42 AM

Thanks for that very interesting post.
It looks like if the trap is installed in SRAM in 0x80000400.
Perhaps it cannot program the chip because of the flash chip config, i.e commands, block/word programming and so on.

Does somebody knows how and where to define the Workspace when starting a debug session with a trapHandler?
Do I have to define the Wptr, Iptr @ the beginning of the SRAM, 0x80000000?

---

slugworth

04-12-2009, 06:55 AM

As far as I got with a menu type flashprg before I got distracted by other fun projects.
Based on the original flashprogrammer v1.0 from back when there was only 2 development boards to choose from.
I find that makefile format easier to work with.
The program will erase the flash on a maggie,but when it comes time to program it only does 1.9k and dies.

---

slugworth

04-12-2009, 09:13 AM

Perhaps it cannot program the chip because of the flash chip config, i.e commands, block/word programming and so on.

I think it's more of a speed/buffer size problem.

MSSFWP

04-12-2009, 05:30 PM

Map from the flashprg v1.0 dc1/mb292a.cfg based modified for sti5105.
The prog detects and erases the flash,but burps when it tries to program.

I bet it does....have you compared it with the actual 5105 burner MAP file? There are a few noticable differences.

Segment Map

Name Address Size Type
--- --- --- ---
SYSTEM 0x80000000 128 RESERVED
INTERNAL 0x80000080 1920 RAM
EXTERNAL 0xc0000000 6289408 RAM
TRAPHANDLER 0xc05ff800 1024 DEBUG
TRAPHANDLER 0xc05ffc00 1024 DEBUG
NONCACHED 0xc0600000 2097152 RAM
AVMEM 0xc0800000 8388608 RAM
LMI 0xe0000000 16777216 DEVICE
EMI 0x20200000 4096 DEVICE
INTLEVELCTRL 0x20800000 4096 DEVICE
LPC 0x20808000 4096 PERIPHERAL
IRB 0x20818000 4096 PERIPHERAL
PIO0 0x20820000 4096 DEVICE
PIO1 0x20821000 4096 DEVICE
PIO2 0x20822000 4096 DEVICE
PIO3 0x20823000 4096 DEVICE
PIO4 0x20824000 4096 DEVICE
PIO5 0x20825000 4096 DEVICE
ASC0 0x20830000 4096 DEVICE
ASC1 0x20831000 4096 DEVICE
ASC2 0x20832000 4096 DEVICE
ASC3 0x20833000 4096 DEVICE
SSC0 0x20840000 4096 DEVICE
SSC1 0x20841000 4096 DEVICE
SSC2 0x20842000 4096 DEVICE
DCU 0x30000000 4096 DEVICE
DCACHE 0x30001000 4096 DEVICE
ICACHE 0x30002000 4096 DEVICE
PWM 0x30003000 4096 DEVICE
INT_CONTROLLER 0x30004000 4096 DEVICE
FLASH 0x40010000 2621440 ROM
ST_FMI_CI 0x7f400000 4194304 DEVICE
ST_FMI_C 0x7f800000 2097152 DEVICE
ST_FMI_D 0x7fc00000 2097152 DEVICE

Section/Region Map

Name Address Size Type Segment SB
--- --- --- --- --- --
os20_th_data 0x80000080 272 INIT INTERNAL
os20_root_tdesc 0x80000190 36 BSS INTERNAL
os20_int_util_text 0x800001b4 84 CODE INTERNAL
os20_int_simple_text 0x80000208 56 CODE INTERNAL
os20_int_moderate_text 0x80000240 112 CODE INTERNAL
def_code 0xc0000000 23324 CODE EXTERNAL
def_data 0xc0005b1c 1216 INIT EXTERNAL **
system_section 0xc0005fdc 5767168 BSS EXTERNAL
def_bss 0xc0585fdc 30716 BSS EXTERNAL
p_code_section 0xc058d7d8 9120 CODE EXTERNAL

def_const 0xc058fb78 1004 CONST EXTERNAL

os20_th_code 0xc058ff64 608 CODE EXTERNAL
os20_task_queue 0xc05901c4 704 BSS EXTERNAL
os20_int_complex_text 0xc0590484 200 CODE EXTERNAL
<shutdown_section> 0xc059054c 12 CONST EXTERNAL
<startup_section> 0xc0590558 12 CONST EXTERNAL
bootdata 0xc0590564 40 BOOTDATA EXTERNAL
stack 0xc059058c 227644 STACK EXTERNAL
heap 0xc05c7ec8 227640 HEAP EXTERNAL

Eboda

04-14-2009, 03:48 AM

Is there any clue about check sum on 5105 firmware's?

Ther is some portion that are editable and some not.

I want to change background picture, but this is under check sum calculation.

Thank you!

YLG80

04-14-2009, 06:33 AM

In my STB with Sti5105, the firmware is organized as follows:
Starting from address 0 up to 0x2C890 is the bootloader that can be edited or disassembled.
Then starting from 0x30000 is the firmware that will be loaded with the bootloader.
A part of that firmware is compressed using the Adler ZLIBC.
So you cannot edit that part directly, without decompressing it.

Usually these are OS20 compressed files.
In my firmware there are approximately 35 compressed files . I've decompressed these files one by one with ZLIBC, dos version.
These files are screen layout, languages, graphics, channel list etc ...

There is also a checksum in the header file.
It's going to be difficult to change the content of a compressed file, recompress it, integrate it into the firmware and change the checksum.

Eboda

04-14-2009, 06:56 AM

Can you take a look at my firmware, besides yours?

I think, mine has no such many part compresed. The background mpg is clear.

---

slugworth

04-14-2009, 12:59 PM

I bet it does....have you compared it with the actual 5105 burner MAP file? There are a few noticable differences.

The goal was to get the settings as close to the old dcu settings as possible.

YLG80

04-14-2009, 01:21 PM

That map comparison is very interesting. The first one shows a trap handler installed in INTERNAL ram, which could be much more simple to install.
The second one is strange as it shows two traphandlers.
Upon retruning home, I'm going to try Slugworh configuration with the traphandler declared in INTERNAL static RAM. This could reduce drastically the number of pokes to setup the traphandler and also to standardize the trap installation whatever is the SDRAM memory amount. This would simplify my tentative to program the traphandler setup in Delphi.

YLG80

04-14-2009, 01:27 PM

@EBODA

I've quickly checked your firmware.
The bootloader is not included in the file.
I've disassembled it with IDA with no problem. A very large portion has been automatically disassembled unlike with my own firmware.
At first instance it looks like there is no compressed files.
I cannot find the same headers TZIP/ZIPR/ZIPD as in my firmware.

MSSFWP

04-14-2009, 07:59 PM

The goal was to get the settings as close to the old dcu settings as possible.I'm not sure what you mean bro.

That's the map file from the 5105 burner kit made for the maggie specifically.
I didn't make it, but you know who did, and you know already that it does work.

In fact i'm playing with it on another unit (not a maggie or SV) that also uses the 5105 with a 39LF/VF3201 flash and it booted it up just fine even though it used a different bin/coding. When i get a chance, i want to put a header in the Satopia with the 5119 and the 2MB chip it has to see if the boot coding is the same. Want to bet they are. ;)

MSSFWP

04-14-2009, 08:28 PM

That map comparison is very interesting. The first one shows a trap handler installed in INTERNAL ram, which could be much more simple to install.
The second one is strange as it shows two traphandlers.

Upon retruning home, I'm going to try Slugworh configuration with the traphandler declared in INTERNAL static RAM. This could reduce drastically the number of pokes to setup the traphandler and also to standardize the trap installation whatever is the SDRAM memory amount. This would simplify my tentative to program the traphandler setup in Delphi.I noticed the dual listing for traphandler too, and winched when i saw it. We haven't had too much discussion on the matter, but it has been pretty evident that the old style DCU control (starting at 80000000) is at least partially enabled at the same time as the new style DCU3 control. They are NOT the same though and in this case it isn't mapped there anyway. Could it be possible that the traphandler for both DCU and DCU3 are loaded at the same time?

The only person that could answer that question correctly is probably gonna lay low so i guess we'll find out their thinking sooner or later though testing. ;)

---

slugworth

04-14-2009, 09:13 PM

uctapsrv -p 0x0378 -v 3 >DCU3.txt
tap.bat for redirecting the output to a file instead of the screen.
Peeks and pokes monitor from the 181.zip lku

---

slugworth

04-15-2009, 06:38 AM

That's the map file from the 5105 burner kit made for the maggie specifically.
I didn't make it, but you know who did, and you know already that it does work.
I never found the elusive "kit" and the 3 examples of burner.lku I have never worked.
Probably why this is a 34 page thread and not a 2 page thread.

YLG80

04-15-2009, 01:26 PM

@Slugworth
You 181menu.zip files contains the mbMB400.cfg with the explanation on the 2 trap instances reported in the .map file.
You have first that directive :

```
if (_ST_no_traphandler_placement != 1) {
PlaceDebugTrapHandler INTERNAL
```

}

and second this one :

PlaceDebugTrapHandler EXTERNAL

Due to limitations in the amount of INTERNAL memory on the STi5105, we
cannot use the default OS20 placement of code/data.
OS20_config.place_scheduler_code=""
}

This can explain the two trap instances.
This is confirmed in your tap capture that shows that the External memory is setup as in your first and second capture.
But there is only one "memset trap 0" in your last capture, after the FMI and LMI setup, so I guess the trap was only installed in SDRAM.
Perhaps the second PlaceDebugTrapHandler command has overwritten the first one.

I attach the commented Slugworth very first tap capture up to memset trap and trap activation.
This is very similar to the last capture except for the SDRAM addresses after the memset trap statement. (I have no info on that statement :s:)

YLG80

04-15-2009, 01:32 PM

@MSSFWP
I'm not sure what you mean bro.

Sulgworth is trying to simplify the DCU3 traphandler setup as much as he can.
I was wondering if he could not setup the trap in the INTERNAL SRAM instead of in SDRAM in order to reduce the number of pokes and to immune the program from the SDRAM type that will likely be very different in different STB's.
This would make the setup very similar to the one used in DCU2 devices.

---

slugworth

04-15-2009, 04:37 PM

The makefile only included dc1.cfg and mb292a.cfg,so the trap only shows once in my map file.The mb400.cfg that you run the lku with is different and I didn't think to edit that also.

---

slugworth

04-15-2009, 05:55 PM

I changed the traphandler to internal in the mb400.cfg also,and took a dump of an erase episode.

MSSFWP

04-16-2009, 03:24 AM

@MSSFWP

Sulgworth is trying to simplify the DCU3 traphandler setup as much as he can.
I was wondering if he could not setup the trap in the INTERNAL SRAM instead of in SDRAM in order to reduce the number of pokes and to immune the program from the SDRAM type that will likely be very different in different STB's.
This would make the setup very similar to the one used in DCU2 devices.yeah i knew what slugworth was referencing to, but that's not really what i meant.

Anyway, to answer your question as best as possible, look at the reference comments in the various board config files to sum it up for you.## Performance may be improved by placing all of this next group in
INTERNAL, but that would exceed the space left by many test harnesses
after their internal_partition allocation So apparently it IS possible to place the code internally. But some things are probably going to be misplaced or over-wrote hence the need to place them externally in SDRAM.

YLG80

04-17-2009, 04:51 AM

@MSSFWP
Thanks for that information on INTERNAL setup.
I've also seen something saying that a trap in INTERNAL could generate problems with the cache associated.
But not sure if it would be a problem for programming flash chips.

On the various Slugworth trials (thanks to him for that hard work)

As I mentioned before, I cannot find any trace of a DCU3c1trap.bin or c1trap.bin or whatever trap included in the lib directory in the ucTap captures.

This is a key information to know if we want to progress further in the trap setup.

So I've recompiled his last 181menu after having first removed almost everything from the TS directory (except the bin files of course).
Crazy task.
This was done in order to see which files were claimed at build and link time. I've added the claimed files one by one during the compilation until the program was fully compiled w/o any error.

THERE IS NO xxxxTRAP.BIN NEEDED AT COMPILED TIME, so it does not use any .bin available in the lib directory.

[added later : the trap handler bin file is loaded at runtime by the chip command, invoking the sti5105chip.conf
file that is located in the stdfcg]

As far as I understand, the flashprg uses exactly the same routine used by the standard STB bootlader when it needs to copy the ROM into RAM.
(see the cstartup in the src directory for the details)
It uses the settrap.c, btrap.c included in the cstart and cstart2 libraries.
This is mentioned in the btrap.c :

Install a default breakpoint trap handler

Note: I've found the cstart and cstart2, together with romload and romload2 routines in the bootloader of my stb.

---

slugworth

04-17-2009, 06:58 AM

The config file or modified config file always contains the statement "chip STi5105" which leads to other cfg files.
So even though the files aren't in your build directory,if they are part of the st20 toolset/path they will be included.
The uctap dumps may not be detailed enough to show the trap being loaded.

YLG80

04-17-2009, 08:01 AM

All files were removed from the TS structure and not in the PATH prior to start the new build.
Only claimed files were reinstalled during the build up. The trapxxxx.bin were never claimed by the compiler.
I'm now verifying that this file is not required at runtime but I have problems to redefine the target for the 181menu.

---

effesse

04-18-2009, 10:14 AM

Hi all, I would like to rewrite the Bootloader (1st 64kb of flash) on my IRD

CPU: STx 5118 ALB - 216 pin
DEVICE ID: 2D427041

FLASH CHIP: M29W160ET (2MB)

SDRAM: A2V28S40TCP (16MB)
TUNER: USA-0288-6K-A-HF (chip ST STx0288 + ST86000)

I'm able to read address from 0x40000000 lenght 0x10000 correctly, via jKeys too (thanks to Alex) and via this command too:

st20run -i product.cfg -t htiramdebug (from Pvr2Flash in this thread)

I would like to Erase the whole flash and rewrite atleast the 1st 64Kb of it. When I use the files in the maggie.zip archive I reach this:

C:\maggie>st20run -b -i mb400.cfg -t mag 5119_2m_c1c.lku

```
----------------------------------------------------
MB400/STi5105 BOARD CONFIGURATION

setup Clock Generator for ST_5105
CKG_CLOCK_SELECT_CFG = 68
--> Mfactor = 5 Nfactor = 37 Pfactor = 0
PLL Request --> PLL Frequency = 400 . 0 MHz
PLL_A Initialised --> PLL Frequency = 399 . 600 MHz
--> Mfactor = 7 Nfactor = 86 Pfactor = 0
PLL Request --> PLL Frequency = 664 . 0 MHz
PLL_B Initialised --> PLL Frequency = 663 . 428 MHz
Setup PLL dividers for ST_5105 chip
CPU = 200 MHz Phase = 0
LMI = 166 MHz Phase = 0
BLITTER = 100 MHz Phase = 0
SYSTEM = 100 MHz Phase = 0
FDMA = 200 MHz Phase = 0
VIDEO = 50 MHz Phase = 0
FLASH = 100 MHz Phase = 0
Setup PLL Frequency Synthesizers for ST_5105 chip
CKG_PIX_CLK_SETUP0 = 0xffffffff
CKG_PLLA_CONFIG0 = 0xffffffff
CKG_PLLA_CONFIG1 = 0xffffffff
CKG_PLLB_CONFIG0 = 0xffffffff
CKG_PLLB_CONFIG1 = 0xffffffff
ST_5105 Clock Generator Successfully setup

==Informations of SDRAM==
SDRAM DEVICE TYPE = DDR SDRAM

SDRAM DEVICE FREQUENCY = 166 MHz
SDRAM DEVICE SIZE = 256 MBits (32 MBytes)
SDRAM DEVICE REFRESH PERIOD = 7.8 microseconds (8K/64ms
SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.5
============================

LMI_STR = 0x3d6d4b45
set memory size
LMI_COC_UPPER= 0xffffffff
LMI_COC_LOWER= 0xffffffff
LMI Config Done

Configuring registers for FMI ...
Bank[0-3] not configured fully!!
the ident from EPLD is 0x000000ff
reseting flash
FMI configure done


Setup Complete
----------------------------------------------------
```

... and the run halts.

What I'm wrong? Is any data correct?

How I have to operate to Erase the whole flash and rewriting atleast the 1st 64Kb of it?

Does it exists actually some .lku files with a menu from which can I choose the size and type of Flash to Erase and rewrite atleast 64kb of data?

Many thanks in advance to all for replies, kind regards, Stefano.

---

slugworth

04-18-2009, 11:50 AM

The amount of ram in the mb400.cfg has to match the receiver.

---

effesse

04-18-2009, 12:40 PM

The amount of ram in the mb400.cfg has to match the receiver.

Tnx Slugworth, I start modifing mb400.cfg but I don't know if other strings are correct, help me please.

Best regards, Stefano.

---

slugworth

04-18-2009, 01:08 PM

I have mine set for ddr ram.
Yours isn't setting the clock freqs for some reason.

MSSFWP

04-18-2009, 01:31 PM

5105/5119 similar chips using C1 code.

The 5118 is comparable to the 5100 i believe which uses C2 code.

---

slugworth

04-18-2009, 02:17 PM

I believe the lku would complain if you were trying to run it on a C2 core.
I know C2 lku complain when they detect the 5105,which is C1.

---

effesse

04-18-2009, 04:20 PM

When I try to use flashprg_c1c.lku this is the output screen:

```
C:\maggie>ho

C:\maggie>rem st20run -b -i mb400.cfg -t mag 5119_2m_c1c.lku

C:\maggie>st20run -b -i mb400.cfg -t mag flashprg_c1c.lku

----------------------------------------------------
MB400/STi5105 BOARD CONFIGURATION

setup Clock Generator for ST_5105
CKG_CLOCK_SELECT_CFG = 68
--> Mfactor = 5 Nfactor = 37 Pfactor = 0
PLL Request --> PLL Frequency = 400 . 0 MHz
PLL_A Initialised --> PLL Frequency = 399 . 600 MHz
--> Mfactor = 7 Nfactor = 86 Pfactor = 0
PLL Request --> PLL Frequency = 664 . 0 MHz
PLL_B Initialised --> PLL Frequency = 663 . 428 MHz
Setup PLL dividers for ST_5105 chip
CPU = 200 MHz Phase = 0
```

LMI = 166 MHz Phase = 0

BLITTER = 100 MHz Phase = 0
SYSTEM = 100 MHz Phase = 0
FDMA = 200 MHz Phase = 0
VIDEO = 50 MHz Phase = 0
FLASH = 100 MHz Phase = 0
Setup PLL Frequency Synthesizers for ST_5105 chip
CKG_PIX_CLK_SETUP0 = 0xffffffff
CKG_PLLA_CONFIG0 = 0xffffffff
CKG_PLLA_CONFIG1 = 0xffffffff
CKG_PLLB_CONFIG0 = 0xffffffff
CKG_PLLB_CONFIG1 = 0xffffffff
ST_5105 Clock Generator Successfully setup

==Informations of SDRAM==
SDRAM DEVICE TYPE = DDR SDRAM
SDRAM DEVICE FREQUENCY = 166 MHz
SDRAM DEVICE SIZE = 128 MBits (16 MBytes)
SDRAM DEVICE REFRESH PERIOD = 15.625 microseconds (4K/64ms)
SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.5
==========

LMI_STR = 0x3d6d4b45
set memory size
LMI_COC_UPPER= 0xffffffff
LMI_COC_LOWER= 0xffffffff
LMI Config Done

Configuring registers for FMI ...
Bank[0-3] not configured fully!!
the ident from EPLD is 0x000000ff
reseting flash
FMI configure done

Setup Complete

But when I use flashprg.lku from the 5518burner the screen is:

C:\maggie>ho

C:\maggie>rem st20run -b -i mb400.cfg -t mag 5119_2m_c1c.lku

C:\maggie>st20run -b -i mb400.cfg -t mag flashprg.lku

MB400/STi5105 BOARD CONFIGURATION

setup Clock Generator for ST_5105
CKG_CLOCK_SELECT_CFG = 68
--> Mfactor = 5 Nfactor = 37 Pfactor = 0
PLL Request --> PLL Frequency = 400 . 0 MHz
PLL_A Initialised --> PLL Frequency = 399 . 600 MHz
--> Mfactor = 7 Nfactor = 86 Pfactor = 0
PLL Request --> PLL Frequency = 664 . 0 MHz
PLL_B Initialised --> PLL Frequency = 663 . 428 MHz
Setup PLL dividers for ST_5105 chip
CPU = 200 MHz Phase = 0
LMI = 166 MHz Phase = 0
BLITTER = 100 MHz Phase = 0
SYSTEM = 100 MHz Phase = 0
FDMA = 200 MHz Phase = 0
VIDEO = 50 MHz Phase = 0

FLASH = 100 MHz Phase = 0

Setup PLL Frequency Synthesizers for ST_5105 chip
CKG_PIX_CLK_SETUP0 = 0xffffffff
CKG_PLLA_CONFIG0 = 0xffffffff
CKG_PLLA_CONFIG1 = 0xffffffff
CKG_PLLB_CONFIG0 = 0xffffffff
CKG_PLLB_CONFIG1 = 0xffffffff
ST_5105 Clock Generator Successfully setup

==Informations of SDRAM==
SDRAM DEVICE TYPE = DDR SDRAM
SDRAM DEVICE FREQUENCY = 166 MHz
SDRAM DEVICE SIZE = 128 MBits (16 MBytes)
SDRAM DEVICE REFRESH PERIOD = 15.625 microseconds (4K/64ms)
SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.5
==========

LMI_STR = 0x3d6d4b45
set memory size
LMI_COC_UPPER= 0xffffffff
LMI_COC_LOWER= 0xffffffff
LMI Config Done

Configuring registers for FMI ...
Bank[0-3] not configured fully!!
the ident from EPLD is 0x000000ff
reseting flash
FMI configure done

Setup Complete
---

load error-invalid lku file : Attempt to load flashprg.lku (built for a C2 core)
onto a target with a C1 core

so, my Stx 5118 Alb CPU is C1 or C2 core?

Tnx in advance for replies, best regards, Stefano.

YLG80

04-19-2009, 01:17 AM

I confirm what Slugworth wrote on the lku complaining if the stb CPU core is not the one expected by the lku.
I've done the test yesterday and the lku setup did stop on a core mismatch error.

YLG80

04-19-2009, 01:40 AM

I believe that the build you are using simply does not recognize the 5119 CPU.
I would suggest to recompile the lku after having added the following to the chip.cfg (in stdcfg directory):

_ST_AddDevice "STi5118" 0xd405 0x0000 "st20c1" "CacheControlC100" "C1Interrupts" "DCU3" (0x30000000) "c1th" "DCU3c1trap.bin" "STi5105chip.cfg" "STi5105chip" 2 3 "cache_map_c1_c100%os"
_ST_AddDevice "STi5119" 0xd415 0x0000 "st20c1" "CacheControlC100" "C1Interrupts" "DCU3" (0x30000000) "c1th" "DCU3c1trap.bin" "STi5105chip.cfg" "STi5105chip" 2 3 "cache_map_c1_c100%os"

(Add these lines by the end of the chip.cfg file)

Just to be clean, add also these lines at the beginning of the chip.cfg in the Addhelp chip

section

STi5118,\n\
STi5119,\n\

This is necessary for the compiler "chip" command and this is likely the reason why your memory is not correctly setup.
That declaration would somewhat fool the compiler and make it think that an Sti5119 is like a Sti5105 which is correct.
(In the 5118 DS they say that this CPU is s/w and binary compatible with the 5105)
Could you post the result if you try this.

---

effesse

04-19-2009, 02:24 PM

I believe that the build you are using simply does not recognize the 5119 CPU.
I would suggest to recompile the lku after having added the following to the chip.cfg (in stdcfg directory):

_ST_AddDevice "STi5118" 0xd405 0x0000 "st20c1" "CacheControlC100" "C1Interrupts" "DCU3" (0x30000000) "c1th" "DCU3c1trap.bin" "STi5105chip.cfg" "STi5105chip" 2 3 "cache_map_c1_c100%os"
_ST_AddDevice "STi5119" 0xd415 0x0000 "st20c1" "CacheControlC100" "C1Interrupts" "DCU3" (0x30000000) "c1th" "DCU3c1trap.bin" "STi5105chip.cfg" "STi5105chip" 2 3 "cache_map_c1_c100%os"

(Add these lines by the end of the chip.cfg file)

Just to be clean, add also these lines at the beginning of the chip.cfg in the Addhelp chip section

STi5118,\n\
STi5119,\n\

This is necessary for the compiler "chip" command and this is likely the reason why your memory is not correctly setup.
That declaration would somewhat fool the compiler and make it think that an Sti5119 is like a Sti5105 which is correct.
(In the 5118 DS they say that this CPU is s/w and binary compatible with the 5105)
Could you post the result if you try this.

Thank you YLG80. I will try to modify the chip.cfg file soon. But I don't have any idea about how to compile a .lku file! I really don't know anything about commands to compile .lku files. I downloaded and installed ST20R2.3.1 , OSPlusR3.0.3 and STMCR1.3.1 from ST ftp site, but I'm actually not able to use them. Can you drive me to compile the flashprg_c1c.lku for my necessity?

Many thanks in advance for all and for any further tips, kind regards, Stefano.

---

slugworth

04-19-2009, 02:41 PM

By default toolset 2.3.1 will make 5 lku files.

The makefile included with the example compiles several versions of the lku (linked unit). These are:

flashprg_c2b0.lku - For c2 cores using bank 0 of RAM
flashprg_c2b1.lku - For c2 cores using bank 1 of RAM
flashprg_5100.lku - For the STi5100 on the MB390 board
flashprg_dc1.lku - For c1 cores, specifically the ST20DC1
flashprg_c1c.lku - For c1 cores with RAM at 0xc0000000 such as the STm5700,STi7710 and STi5105

and STi5105
to compile you may simply run your make utility from the flash directory.

the c1c was tweaked for the magnum flash,I would try the default one or the dc1 to see how it reacts.I wouldn't worry about flash at this stage.

YLG80

04-20-2009, 12:13 AM

My mistake. You don't really need to recompile the lku.
The chip command is included in the mb400.cfg in the maggie package.
That file is executed by the st20run command, so any mod will be taken into account.
You can add the lines in that file starting from line 131 and change the "chip STi5105" to "chip Sti5119".
Then you will be sure that the C1 core will be selected for your CPU together with the right trap handler.

---

jvvh5897

04-20-2009, 01:18 PM

I usually specify the chip type c1 or c2 in the command line call to compile and link--think I saw that in the batch to generate the traps in the toolset too.

YLG80

04-20-2009, 02:42 PM

Yes you're right it can, but it was not specified in the maggie package batch files.
The chip is also used to assign a correct traphandler for the core.
If it does not find the CPU ID it could assign a DCU2 handler.

Note the assignment to TrapHandler[spaceid] is AFTER the chip command. This is because the chip command will set the value of TrapHandler[spaceid] to the default traphandler for that chip.
Alternatively, a new chip definition could be added to chip.cfg.

chip command. If no chip command is executed then by default the toolset builds for and expects to find a DCU2 diagnostic controller.
./..
Users must use the chip command before linking and running code on devices with DCU3 diagnostic controllers. This ensures that the correct DCU type and DCU3 base address are used.

---

effesse

04-21-2009, 12:19 PM

My mistake. You don't really need to recompile the lku.
The chip command is included in the mb400.cfg in the maggie package.
That file is executed by the st20run command, so any mod will be taken into account.
You can add the lines in that file starting from line 131 and change the "chip STi5105" to "chip Sti5119".
Then you will be sure that the C1 core will be selected for your CPU together with the right trap handler.

I modded the chip.cfg and run again ho.bat that's output:

```
C:\maggie>ho

C:\maggie>st20run -b -i mb400.cfg -t mag flashprg_c1c.lku

-----------------------------------------------------
MB400/STi5105 BOARD CONFIGURATION

setup Clock Generator for ST_5105
CKG_CLOCK_SELECT_CFG = 68
--> Mfactor = 5 Nfactor = 37 Pfactor = 0
PLL Request --> PLL Frequency = 400.0 MHz
```

PLL Request --> PLL Frequency = 400 . 0 MHz
PLL_A Initialised --> PLL Frequency = 399 . 600 MHz

--> Mfactor = 7 Nfactor = 86 Pfactor = 0
PLL Request --> PLL Frequency = 664 . 0 MHz
PLL_B Initialised --> PLL Frequency = 663 . 428 MHz
Setup PLL dividers for ST_5105 chip
CPU = 200 MHz Phase = 0
LMI = 166 MHz Phase = 0
BLITTER = 100 MHz Phase = 0
SYSTEM = 100 MHz Phase = 0
FDMA = 200 MHz Phase = 0
VIDEO = 50 MHz Phase = 0
FLASH = 100 MHz Phase = 0
Setup PLL Frequency Synthesizers for ST_5105 chip
CKG_PIX_CLK_SETUP0 = 0xffffffff
CKG_PLLA_CONFIG0 = 0xffffffff
CKG_PLLA_CONFIG1 = 0xffffffff
CKG_PLLB_CONFIG0 = 0xffffffff
CKG_PLLB_CONFIG1 = 0xffffffff
ST_5105 Clock Generator Successfully setup

==Informations of SDRAM==
SDRAM DEVICE TYPE = DDR SDRAM
SDRAM DEVICE FREQUENCY = 166 MHz
SDRAM DEVICE SIZE = 128 MBits (16 MBytes)
SDRAM DEVICE REFRESH PERIOD = 15.625 microseconds (4K/64ms)
SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.5
==========

LMI_STR = 0x3d6d4b45
set memory size
LMI_COC_UPPER= 0xffffffff
LMI_COC_LOWER= 0xffffffff
LMI Config Done

Configuring registers for FMI ...
Bank[0-3] not configured fully!!
the ident from EPLD is 0x000000ff
reseting flash
FMI configure done

Setup Complete

I noticed the lines with all Zero values and with '0xffffffff' values and the 'Bank[0-3] not configured fully!!' line.
Further the Ident from EPLD i suppose not have to be '0x000000ff' for my SSt M29W160ET Flash chip. However in jKeys too I always got 'manuf/device=FFFFFFFF/FFFFFFFF'. I suppose something must have gone twisted here and in jKeys too!

What I have to do for someone who wants to help me? What kind of file I have to post for making to help me better?

Thank you very much again, kind regards, Stefano.

---

slugworth

04-21-2009, 12:50 PM

I would check the mb400_mem.cfg file to make sure that is edited also.

YLG80

04-22-2009, 05:19 AM

@efesse : your flash is not properly configured !

banks [0-3] not fully configured.

My STB DTS1601 with Sti5105 has a 2Mb flash.
(the flash chip is an S29AL016D = AM29LV160)
Here is the config to modify in the MB400_mem.cfg in order to get to "all banks [0-3] fully configured" status
This is to be modified in the InitFMI procedure
The code below has been modified in the original Toolset dcu_mb400_conf.cfg file but can be easily adapetd for the maggie mb400_mem.cfg. (simply remove the _ST_STB5105_ in front of the variables
with a text editor)

```
##-------------------------------------------------------------------------
## Procedure Name : ST_5105_InitFMI --> Need to be updated _ UPDATED FOR DTS1601
## Purpose : Initialise FMI for ST_5105 chip on MB400 board
##-------------------------------------------------------------------------
proc ST_5105_InitFMI {

## Ensure all FMI control registers are unlocked
## at reset the state of these regs is 'undefined'

poke -d (_ST_STB5105_FMI_LOCK) 0x00000000
poke -d (_ST_STB5105_FMI_STATUSLOCK) 0x00000000

## Number of FMI Banks : Enable all banks
poke -d (_ST_STB5105_BANKS_ENABLED) 0x00000004

## NOTE: bits [0,7] define bottom address bits [22,29] of bank
## Bank 0 - 16MBytes Atapi Configured as 16-bit peripheral
## bits [22:29] = 00000011
poke -d (_ST_STB5105_BANK_0_TOP_ADDRESS) 0x00000003 ##0x00000003 ## 0x40000000 - 0x40FFFFFF

## Bank 1 - 32MBytes Stem0/DVBCI/EPLD Configured as 16-bit peripheral
## bits [22:29] = 00001011
poke -d (_ST_STB5105_BANK_1_TOP_ADDRESS) 0x0000000B ##0x0000000B ## 0x41000000 - 0x42FFFFFF

## Bank 2 - 32MBytes Stem1 Configured as 16-bit peripheral
## bits [22:29] = 00010011
poke -d (_ST_STB5105_BANK_2_TOP_ADDRESS) 0x00000013 ## 0x43000000 - 0x44FFFFFF

## Bank 3 - 8MBytes S29AL016D or S29LV160 Flash
## Note only the top 8Mbytes is populated from 0x7F800000
## bits [22:29] = 11111111
poke -d (_ST_STB5105_BANK_3_TOP_ADDRESS) 0x00000017 ## 0x45000000 - 0x7FFFFFFF

##-------------------------------------------------------------------------
## Program bank functions
##-------------------------------------------------------------------------

##-------------------------------------------------------------------------
## Bank 0 - MB390 bank A = 2MBytes Flash (S29AL016D in DTS1601)
##-------------------------------------------------------------------------
poke -d (_ST_STB5105_FMICONFIGDATA0_BANK0) 0x041016d1 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA1_BANK0) 0x9d200000 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA2_BANK0) 0x9d220000 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA3_BANK0) 0x00000000 ## like in DTS1601

##-------------------------------------------------------------------------
```

```
## Bank 1 - MB390 bank B = 32MBytes STEM0/DVBCI
##-------------------------------------------------------------------------
```

```
## Bank 1: Configured for 4Mbx16
poke -d (_ST_STB5105_FMICONFIGDATA0_BANK1) 0x00101699 ## like in DTS1601
## BE not active during rd - 16bit
poke -d (_ST_STB5105_FMICONFIGDATA1_BANK1) 0xBE426200 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA2_BANK1) 0xBE426200 ## like in DTS1601
##poke -d (_ST_STB5105_FMICONFIGDATA1_BANK2) 0x9d200000
##poke -d (_ST_STB5105_FMICONFIGDATA2_BANK2) 0x9d220000
poke -d (_ST_STB5105_FMICONFIGDATA3_BANK1) 0x00000000 ## like in DTS1601

##-------------------------------------------------------------------------
## Bank 2 - MB390 bank C = 32MBytes STEM1
##-------------------------------------------------------------------------

## STANDARD STEM CFG
poke -d (_ST_STB5105_FMICONFIGDATA0_BANK2) 0x001016d1 ## like in DTS1601
## BE not active during rd - 16bit
poke -d (_ST_STB5105_FMICONFIGDATA1_BANK2) 0x9d200000 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA2_BANK2) 0x9d220000 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA3_BANK2) 0x00000000 ## like in DTS1601

## STANDARD STEM CFG
## poke -d (STI5528_EMI_BANK2_DATA0) 0x001016f9 ##This line for 8bit bus
## poke -d (STI5528_EMI_BANK3_DATA0) 0x001016d1 ## 16bit
## poke -d (STI5528_EMI_BANK2_DATA1) 0x9d200000
## poke -d (STI5528_EMI_BANK2_DATA2) 0x9d220000
## poke -d (STI5528_EMI_BANK2_DATA3) 0x00000000

##-------------------------------------------------------------------------
## Bank 3 - MB390 bank D = EPLD
##-------------------------------------------------------------------------
##poke -d (_ST_STB5105_FMICONFIGDATA0_BANK3) 0x001016f9 ## 8bit
##poke -d (_ST_STB5105_FMICONFIGDATA1_BANK3) 0x9d200000
##poke -d (_ST_STB5105_FMICONFIGDATA2_BANK3) 0x9d220000
##poke -d (_ST_STB5105_FMICONFIGDATA3_BANK3) 0x00000000
poke -d (_ST_STB5105_FMICONFIGDATA0_BANK3) 0x00108791 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA1_BANK3) 0x44618F00 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA2_BANK3) 0x44618F00 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA3_BANK3) 0x0000000A ## like in DTS1601


## ------- Program Other FMI Registers -------- ##
## Synchronous Flash runs @ 1/3 bus clk

poke -d (_ST_STB5105_FMI_GENCFG) 0xC0000000 ## like in DTS1601
poke -d (_ST_STB5105_FMI_FLASHCLKSEL) 0x00000002 ## like in DTS1601
poke -d (_ST_STB5105_FMI_FLASHCLKSEL) 0x00000002 ## Flash clock 1/3 STbus CLK
poke -d (_ST_STB5105_FMI_CLKENABLE) 0x00000001 ## Enable Flash

poke -d (_ST_MB400_FLASH_EPLD_CONTROL) 0x03030303 ## Flash reset & VPP high
## read the status cfg
x = (peek (_ST_STB5105_FMI_STATUSCFG) -d -q) & 0x1F
if(x != 0x0F){
write "Bank[0-3] not configured fully!!" -x (x)
} else {
write All Banks[0-3] have been configured
}
```

However I still cannot get access to the flashprg menu with my stb.
I've also to adapt the SDRAM config (16Mb instead of 32Mb)
By the end of the setup it tries to get access to memory beyond the upper limit. But the related pokes are no longer in the mb400_conf.cfg.
Still searching where is the error because the SDRAM looks correctly configured.
I'm peeking the correct values out of the registers in my running stb and I adapt those

I'm peeking the correct values out of the registers in my running stb and I adapt these

values in the config files.
Very hard work .... but in progress.

YLG80

04-22-2009, 06:55 AM

I have modified my post #341 regarding the traphandler bin file :
"The trap handler bin file is loaded at runtime (so not included in any lku) by the chip command, invoking the sti5105chip.conf file that is located in the stdfcg directory"

I believe that this is THE file to analyze for setting up a trap with another program.

It's in that setup portion that I still have an error when I try to use flashprg.
Despite of the fact that the SDRAM is correctly setup in mb400_mem.cfg, the chip command continues to try to use memory above the 16Mb limit.
This is perhaps due to the fact that the MB400 board has 32Mb RAM.

---

kicsigee

04-22-2009, 09:44 AM

Hi all!

i have a Faval s100. when i upload a wrong software, then the receiver's displayed 88:88. now the receiver not works. i tried many programs to fix my mistake but i can't. i used the maggie software the copmuter-receiver contact is ok, so the jtag cable is fine, but i don't have enought information to use the maggie software. Please help! my mother is really want to watch tv, but because of me, she can't :)

---

kicsigee

04-22-2009, 09:46 AM

http://kepfeltoltes.hu/090415/nemtom1_www.kepfeltoltes.hu_.jpg
http://kepfeltoltes.hu/090415/nemtom_www.kepfeltoltes.hu_.jpg

---

kicsigee

04-22-2009, 10:36 AM

http://kepfeltoltes.hu/090415/nemtom1_www.kepfeltoltes.hu_.jpg
http://kepfeltoltes.hu/090415/nemtom_www.kepfeltoltes.hu_.jpg

i tried the jtag 5119 maggie by slugworth

YLG80

04-22-2009, 11:17 AM

Are you sure that you cannot simply upload the firmware via the RS232, if any RS232 connector available. Perhaps the bootloader is still OK.

---

kicsigee

04-22-2009, 11:40 AM

Are you sure that you cannot simply upload the firmware via the RS232, if any RS232 connector available. Perhaps the bootloader is still OK.

i don't know. the software update is not working because when i turn on the receiver,and still push the standby button, i don't see the "LOAD" word on the display. my informations is not 100%sure. do you think it is possible to change the firmware?

---

slugworth

04-22-2009, 12:28 PM

I would jkeys jtag dump the entire flash, to see what the damage is.
You can use the same jtag cable as what you use with uctap.
Make sure you dump@address 40000000 for the entire flash size.

---

kicsigee

04-23-2009, 12:13 AM

dear slugworth!

The package, you made (jtag 5119 maggie) is worked for you? my faval s100 is

compatible whit this program? because the " go.bat" erase is stopped many times. i linked a picture 4 comments upper. please help! i don't know what is the problem!

---

slugworth

04-23-2009, 06:49 AM

I don't have a 5119, so that was just a guess at making a 5119 file.

and don't call me dear.

---

alarm

04-23-2009, 10:13 AM

I suggested slugworth send a 5119 receiver for the experiments, but never received a response to its proposal.

---

slugworth

04-23-2009, 10:25 AM

I suggested slugworth send a 5119 receiver for the experiments, but never received a response to its proposal.

I like to stay anonymous and I think I am going to play with other projects instead.
I originally set out to fix my receiver,now it's time to move on to other things.

---

effesse

04-24-2009, 03:52 AM

@efesse : your flash is not properly configured !

My STB DTS1601 with Sti5105 has a 2Mb flash.
(the flash chip is an S29AL016D = AM29LV160)
Here is the config to modify in the MB400_mem.cfg in order to get to "all banks [0-3] fully configured" status
This is to be modified in the InitFMI procedure
The code below has been modified in the original Toolset dcu_mb400_conf.cfg file but can be easily adapetd for the maggie mb400_mem.cfg. (simply remove the _ST_STB5105_ in front of the variables
with a text editor)

```
##-------------------------------------------------------------------------
## Procedure Name : ST_5105_InitFMI --> Need to be updated _ UPDATED FOR DTS1601
## Purpose : Initialise FMI for ST_5105 chip on MB400 board

##-------------------------------------------------------------------------
proc ST_5105_InitFMI {

## Ensure all FMI control registers are unlocked
## at reset the state of these regs is 'undefined'

poke -d (_ST_STB5105_FMI_LOCK) 0x00000000
poke -d (_ST_STB5105_FMI_STATUSLOCK) 0x00000000

## Number of FMI Banks : Enable all banks
poke -d (_ST_STB5105_BANKS_ENABLED) 0x00000004

## NOTE: bits [0,7] define bottom address bits [22,29] of bank
## Bank 0 - 16MBytes Atapi Configured as 16-bit peripheral
## bits [22:29] = 00000011
poke -d (_ST_STB5105_BANK_0_TOP_ADDRESS) 0x00000003 ##0x00000003 ## 0x40000000 - 0x40FFFFFF

## Bank 1 - 32MBytes Stem0/DVBCI/EPLD Configured as 16-bit peripheral
## bits [22:29] = 00001011
```

poke -d (_ST_STB5105_BANK_1_TOP_ADDRESS) 0x0000000B ##0x0000000B ##
0x41000000 - 0x42FFFFFF

Bank 2 - 32MBytes Stem1 Configured as 16-bit peripheral
bits [22:29] = 00010011
poke -d (_ST_STB5105_BANK_2_TOP_ADDRESS) 0x00000013 ## 0x43000000 -
0x44FFFFFF

Bank 3 - 8MBytes S29AL016D or S29LV160 Flash
Note only the top 8Mbytes is populated from 0x7F800000
bits [22:29] = 11111111
poke -d (_ST_STB5105_BANK_3_TOP_ADDRESS) 0x00000017 ## 0x45000000 -
0x7FFFFFFF

##---
Program bank functions
##---

##---
Bank 0 - MB390 bank A = 2MBytes Flash (S29AL016D in DTS1601)
##---
poke -d (_ST_STB5105_FMICONFIGDATA0_BANK0) 0x041016d1 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA1_BANK0) 0x9d200000 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA2_BANK0) 0x9d220000 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA3_BANK0) 0x00000000 ## like in DTS1601

##---
Bank 1 - MB390 bank B = 32MBytes STEM0/DVBCI
##---
Bank 1: Configured for 4Mbx16
poke -d (_ST_STB5105_FMICONFIGDATA0_BANK1) 0x00101699 ## like in DTS1601
BE not active during rd - 16bit
poke -d (_ST_STB5105_FMICONFIGDATA1_BANK1) 0xBE426200 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA2_BANK1) 0xBE426200 ## like in DTS1601
##poke -d (_ST_STB5105_FMICONFIGDATA1_BANK2) 0x9d200000
##poke -d (_ST_STB5105_FMICONFIGDATA2_BANK2) 0x9d220000
poke -d (_ST_STB5105_FMICONFIGDATA3_BANK1) 0x00000000 ## like in DTS1601

##---
Bank 2 - MB390 bank C = 32MBytes STEM1
##---

STANDARD STEM CFG
poke -d (_ST_STB5105_FMICONFIGDATA0_BANK2) 0x001016d1 ## like in DTS1601

BE not active during rd - 16bit
poke -d (_ST_STB5105_FMICONFIGDATA1_BANK2) 0x9d200000 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA2_BANK2) 0x9d220000 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA3_BANK2) 0x00000000 ## like in DTS1601

STANDARD STEM CFG
poke -d (STI5528_EMI_BANK2_DATA0) 0x001016f9 ##This line for 8bit bus
poke -d (STI5528_EMI_BANK3_DATA0) 0x001016d1 ## 16bit
poke -d (STI5528_EMI_BANK2_DATA1) 0x9d200000
poke -d (STI5528_EMI_BANK2_DATA2) 0x9d220000
poke -d (STI5528_EMI_BANK2_DATA3) 0x00000000

##---
Bank 3 - MB390 bank D = EPLD
##---
##poke -d (_ST_STB5105_FMICONFIGDATA0_BANK3) 0x001016f9 ## 8bit
##poke -d (_ST_STB5105_FMICONFIGDATA1_BANK3) 0x9d200000
##poke -d (_ST_STB5105_FMICONFIGDATA2_BANK3) 0x9d220000
##poke -d (_ST_STB5105_FMICONFIGDATA3_BANK3) 0x00000000
poke -d (_ST_STB5105_FMICONFIGDATA0_BANK3) 0x00108791 ## like in DTS1601

```
poke -d (_ST_STB5105_FMICONFIGDATA1_BANK3) 0x44618F00 ## like in DTS1601
poke -d (_ST_STB5105_FMICONFIGDATA2_BANK3) 0x44618F00 ## like in DTS1601
```

```
poke -d (_ST_STB5105_FMICONFIGDATA3_BANK3) 0x0000000A ## like in DTS1601


## ------- Program Other FMI Registers -------- ##
## Synchronous Flash runs @ 1/3 bus clk
poke -d (_ST_STB5105_FMI_GENCFG) 0xC0000000 ## like in DTS1601
poke -d (_ST_STB5105_FMI_FLASHCLKSEL) 0x00000002 ## like in DTS1601
poke -d (_ST_STB5105_FMI_FLASHCLKSEL) 0x00000002 ## Flash clock 1/3 STbus CLK
poke -d (_ST_STB5105_FMI_CLKENABLE) 0x00000001 ## Enable Flash

poke -d (_ST_MB400_FLASH_EPLD_CONTROL) 0x03030303 ## Flash reset & VPP high

## read the status cfg
x = (peek (_ST_STB5105_FMI_STATUSCFG) -d -q) & 0x1F
if(x != 0x0F){
write "Bank[0-3] not configured fully!!" -x (x)
} else {
write All Banks[0-3] have been configured
}
```

However I still cannot get access to the flashprg menu with my stb.
I've also to adapt the SDRAM config (16Mb instead of 32Mb)
By the end of the setup it tries to get access to memory beyond the upper limit. But the related pokes are no longer in the mb400_conf.cfg.
Still searching where is the error because the SDRAM looks correctly configured.
I'm peeking the correct values out of the registers in my running stb and I adapt these values in the config files.
Very hard work .... but in progress.

This is what appened when I did your modifications:

```
--------------------------------------------------
MB400/STi5105 BOARD CONFIGURATION

setup Clock Generator for ST_5105
CKG_CLOCK_SELECT_CFG = 68
--> Mfactor = 5 Nfactor = 37 Pfactor = 0
PLL Request --> PLL Frequency = 400 . 0 MHz
PLL_A Initialised --> PLL Frequency = 399 . 600 MHz
--> Mfactor = 7 Nfactor = 86 Pfactor = 0
PLL Request --> PLL Frequency = 664 . 0 MHz
PLL_B Initialised --> PLL Frequency = 663 . 428 MHz
Setup PLL dividers for ST_5105 chip
CPU = 200 MHz Phase = 0
LMI = 166 MHz Phase = 0
BLITTER = 100 MHz Phase = 0
SYSTEM = 100 MHz Phase = 0
FDMA = 200 MHz Phase = 0
VIDEO = 50 MHz Phase = 0
FLASH = 100 MHz Phase = 0
Setup PLL Frequency Synthesizers for ST_5105 chip
CKG_PIX_CLK_SETUP0 = 0xffffffff
CKG_PLLA_CONFIG0 = 0xffffffff
CKG_PLLA_CONFIG1 = 0xffffffff
CKG_PLLB_CONFIG0 = 0xffffffff
CKG_PLLB_CONFIG1 = 0xffffffff
ST_5105 Clock Generator Successfully setup

==Informations of SDRAM==
SDRAM DEVICE TYPE = DDR SDRAM
SDRAM DEVICE FREQUENCY = 166 MHz
```

SDRAM DEVICE SIZE = 128 MBits (16 MBytes)
SDRAM DEVICE REFRESH PERIOD = 15.625 microseconds (4K/64ms)

SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.5
==========

LMI_STR = 0x3d6d4b45
set memory size
LMI_COC_UPPER= 0xffffffff
LMI_COC_LOWER= 0xffffffff
LMI Config Done
Bank[0-3] not configured fully!! 0x0000001f
the ident from EPLD is 0x000000ff
reseting flash

FMI configure done

Setup Complete

I hope you and Slugworth can do something good with this procedure to help many users with DCU3 and 2MB Flash broken receivers!

Thanks a lot again, best regards, Stefano.

YLG80

04-24-2009, 12:20 PM

I'm at the same point.

Do you get an error by the end of the ucTAP session (st20run application error) or just nothing?
I get an error-address.
I've exactly the same config(16Mb and 2Mb flash) except Sti5105 i.o your Sti5118 CPU.

BTW :
your Flash is still not fully configured, but I believe it's not the problem for now. First trying to find why the RAM is incorrectly setup or used

Bank[0-3] not configured fully!! 0x0000001f

should become

Bank[0-3] fully configured!! 0x0000000f

your LMI is also not correctly configured (LMI_COC_UPPER and LOWER)
Still trying to fix the issue that occurs when the trap is supposed to be placed in memory.

---

slugworth

04-24-2009, 12:44 PM

You have cfg files to deal with at make lku stage,then again when you try to run the lku. I would try one of the generic lku files from the 2.3.1 toolset and see how that reacts with tweaks you make to mb400.cfg.Don't worry about flash size at this stage.

YLG80

04-24-2009, 02:05 PM

I'm currently working on the original flashprg from the TS with the original cfg files.
A few minutes ago I've tried with the original maggie package and have exactly the same error (see picture) although everything looks fine (EPLD, LMI_COC are OK ...)
2Mb or 4Mb flash in the cfg does not make the difference (for effesse)
I've sent my build to effesse to try.

YLG80

04-24-2009, 02:13 PM

A note : as the STB I'm working on (DTS1601) is (still :laugh:) fully functional,

I've peeked all FMI values and I've replaced them (when necessary) in the MB400_mem.cfg

I've perhaps also to check the values in the 5105ckg.cfg but I've doubts on what it has to do with the SDRAM memory.
Efesse and myself are fighting against a CUT3 Sti51xx i.o a CUT2 CPU in the maggie.

---

slugworth

04-25-2009, 08:28 AM

Using the original unedited mb400.cfg and mb400_mem.cfg from the burner.rar from the first pages of this thread I get that async error.
I also get that error with both cfg files set for 16meg,so I don't know if I can "downgrade" the settings on mine to 16meg.

YLG80

04-25-2009, 01:19 PM

Are your comfortably sitting ? Do you recognize the capture below ?

setup Clock Generator for ST_5105
CKG_CLOCK_SELECT_CFG = 68
--> Mfactor = 5 Nfactor = 37 Pfactor = 0
PLL Request --> PLL Frequency = 400 . 0 MHz
PLL_A Initialised --> PLL Frequency = 399 . 600 MHz
--> Mfactor = 7 Nfactor = 86 Pfactor = 0
PLL Request --> PLL Frequency = 664 . 0 MHz
PLL_B Initialised --> PLL Frequency = 663 . 428 MHz
Setup PLL dividers for ST_5105 chip
CPU = 200 MHz Phase = 0
LMI = 166 MHz Phase = 0
BLITTER = 100 MHz Phase = 0
SYSTEM = 100 MHz Phase = 0
FDMA = 200 MHz Phase = 0
VIDEO = 50 MHz Phase = 0
FLASH = 100 MHz Phase = 0
Setup PLL Frequency Synthesizers for ST_5105 chip
CKG_PIX_CLK_SETUP0 = 0x00000ebf
CKG_PLLA_CONFIG0 = 0x00002404
CKG_PLLA_CONFIG1 = 0x0000c838
CKG_PLLB_CONFIG0 = 0x00005506
CKG_PLLB_CONFIG1 = 0x0000c838
ST_5105 Clock Generator Successfully setup

==Informations of SDRAM==
SDRAM DEVICE TYPE = DDR SDRAM
SDRAM DEVICE FREQUENCY = 166 MHz
SDRAM DEVICE SIZE = 128 MBits (16 MBytes)
SDRAM DEVICE REFRESH PERIOD = 15.625 microseconds (4K/64ms)
SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.5
==========================

LMI_STR = 0x3d6d4b45
set memory size
LMI_COC_UPPER= 0x000c6750
LMI_COC_LOWER= 0x00002000
LMI Config Done

Configuring registers for FMI ...
All Banks[0-3] have been configured
the ident from EPLD is 0x00000020
reseting flash

FMI configure done
Setup Complete

STBOOT_Init()=Ok

```
STPIO_Init(0-4)=Ok
ST_GetClockSpeed() = 200000000
```

```
STUART_Init()=STUART_ERROR_PIO - error calling PIO command (0x00160005)
ROMTOOL V1.0:
Chain : DCU: Osprey2.0.0

ROMTOOL> Enter Flash Base address (default: STx5105--0x40000000):
Flash Base address = [0x40000000]

ROMTOOL>Operation:
1) Erase
2) Program
3) Verify

4) Read
5) Exit
Enter a number : 4

ROMTOOL> Enter Flash to filename : DTS1601_WTMEC.bin

ROMTOOL> Enter Read start address : 40000000
read_start_address = 0x2625a00

ROMTOOL> Enter Read Size (byte) : 200000
read_size = 0x30d40
```

If you remember the first package that I've supplied a few month ago... (Wtmec). .
I've decided to unzip it from your slug dir.

It boots like a charm up to the menu on the same PC.

I did not remember that I had modified the cfg files for 16Mb RAM and 2Mb Flash/
[correction : of course because it was originally written for a 16Mb...]
It runs right as it was a few months ago.

But it's not properly configured for my flash rom.It hangs even on reading the flash.

It takes a while when the memset trap command is executed up to the flashprg menu
(about 20sec with nothing happening in the ucTAP window).
I think it is uploading and "digesting" the trap.

I will try to transpose the config into a compilable flashprog.

Some hope at least for a 16Mb SDRAM config (for effesse ...)

YLG80

04-25-2009, 01:45 PM

Here is the capture file - stopped at flash reading selected from the menu.

---

effesse

04-28-2009, 07:26 AM

(*) TAP.BAT:

```
LOAD 808 to c0815bc0
LOAD 1000 to c0815ee8
LOAD 1000 to c08162d0
LOAD 1000 to c08166b8
LOAD 244 to c0816aa0
memset 00 1113020 to c0816b94

memset trap 0
memset 00 720 to 80000080
memset trap 0
LOAD 1000 to c0926750
LOAD 88 to c0926b38
```

LOAD 608 to c0926b90
LOAD 272 to c0926df0

memset 00 704 to c0926f00
memset trap 0
memset 00 36 to 80000350
memset trap 0
LOAD 84 to 80000374
LOAD 56 to 800003c8
LOAD 112 to 80000400
LOAD 188 to 80000470
LOAD 12 to c09271c0
LOAD 12 to c09271cc
LOAD 24 to c092e9e4

Poke 3 words.
peek 1 words from 30000004
peeked ffffffff
Poke 8 words.
peek 1 words from 30000100
peeked ffffffff
Poke 1 words.
peek 1 words from 30000004
peeked ffffffff
peek 1 words from 30000004
peeked ffffffff
Poke 2 words.
LOAD 152 to c05ff848
Poke 1 words.
LOAD 268 to c05ff924
Poke 1 words.
LOAD 68 to c05ff8e0
Poke 8 words.

(*) NOGUI.BAT:

C:\5105_flash_burner>nogui sti5105box

C:\5105_flash_burner>gmake run TARGET=sti5105box
st20run -l . -i mb400.cfg -t sti5105box wtmecburner.lku

MB400/STi5105 BOARD CONFIGURATION

setup Clock Generator for ST_5105
CKG_CLOCK_SELECT_CFG = 68

--> Mfactor = 5 Nfactor = 37 Pfactor = 0
PLL Request --> PLL Frequency = 400 . 0 MHz
PLL_A Initialised --> PLL Frequency = 399 . 600 MHz
--> Mfactor = 7 Nfactor = 86 Pfactor = 0
PLL Request --> PLL Frequency = 664 . 0 MHz
PLL_B Initialised --> PLL Frequency = 663 . 428 MHz
Setup PLL dividers for ST_5105 chip
CPU = 200 MHz Phase = 0
LMI = 166 MHz Phase = 0
BLITTER = 100 MHz Phase = 0
SYSTEM = 100 MHz Phase = 0
FDMA = 200 MHz Phase = 0
VIDEO = 50 MHz Phase = 0
FLASH = 100 MHz Phase = 0
Setup PLL Frequency Synthesizers for ST_5105 chip

CKG_PIX_CLK_SETUP0 = 0xffffffff
CKG_PLLA_CONFIG0 = 0xffffffff
CKG_PLLA_CONFIG1 = 0xffffffff
CKG_PLLB_CONFIG0 = 0xffffffff
CKG_PLLB_CONFIG1 = 0xffffffff

ST_5105 Clock Generator Successfully setup

==Informations of SDRAM==
SDRAM DEVICE TYPE = DDR SDRAM
SDRAM DEVICE FREQUENCY = 166 MHz
SDRAM DEVICE SIZE = 128 MBits (16 MBytes)
SDRAM DEVICE REFRESH PERIOD = 15.625 microseconds (4K/64ms)
SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.5
==========

LMI_STR = 0x3d6d4b45
set memory size
LMI_COC_UPPER= 0xffffffff

LMI_COC_LOWER= 0xffffffff
LMI Config Done

Configuring registers for FMI ...
Bank[0-3] not configured fully!!
the ident from EPLD is 0x000000ff
reseting flash
FMI configure done

Setup Complete

Hi all, my problem is the same with jKeys too when I try to connect to the Flash.

I always get mfg/code of the Flash SSt M29W160ET (top boot) = 'FFFFFFFF/FFFFFFFF'. In this picture .JPG of my receiver

http://digilander.libero.it/effesse2002/Telestar_Srs_1_Top.jpg

you can see a little copper round paddle called TP63 between the CPU and the oscillator which is directly connected to the pin #130 (reset) of the Stx5118, while the #131 (flash_rst) goes to pin #12 of the Flash SSt M29W160ET.

The true is I don't know how to use this Test Point #63!

However many thanks again for the opportunity you gave to me to try talking with my Flash M29W160ET, but I think I'm still so far from rewriting atleast the bootloader (1st 64kb of my Flash) and restart my dead receiver with the homebuild serial RS-232<-->TTL interface. :-(

Best regards, good luck to all and have great days, Stefano.

---

slugworth

04-28-2009, 05:26 PM

The ram size must be set to 16meg during the make .lku stage.
Traphandler set to internal.
Dcu_mb400 for the \board folder,the other two in the same folder as the .lku.

YLG80

04-30-2009, 09:11 AM

Thanks for pointing out that the cfg files in the /board directory had also to be changed.:five:
I completely forgot to do that, as these files were available in the flashprog directory.
Now my build goes up to the help screen (when -h option is selected).:cucumber:
Together with the wtmecburner, this is enough for me to analyze the trap setup.

---

slugworth

04-30-2009, 11:37 AM

A good test for wtmecburner would be if it detects the flash correctly then do a flash dump.

That would do no harm to a working receiver.

---

slugworth

04-30-2009, 08:09 PM

lku for receivers with 16meg of ram and 4meg flash.
Should also work with receivers with more ram.

YLG80

05-02-2009, 12:41 PM

Sorry, I have deleted the original message and edited a new one, because the attachement was wrong.

Thanks for that build.
I've tested that config as is. (only with a target change)
Tested on a 16Mb RAM - 2Mb Flash STi5105
It runs perfectly .
The setup is very fast up to the menu.

When using the verify option, it returns a "verified failed" but my flash chip is different than the one setup with that config.

Attachements : the lku launch capture and the ucTap capture.

---

slugworth

05-02-2009, 03:15 PM

2meg flash support would be a tweak of the sdm.c file and flashprg.c file.
You would then choose -b sdm on the commandline instead of -b mb400 or mb385.

---

effesse

05-02-2009, 03:50 PM

Thanks for that build.
I've tested that config as is. (only with a target change)
Tested on a 16Mb RAM - 2Mb Flash STi5105
It runs perfectly .
The setup is very fast up to the menu.

When using the verify option, it returns a "verified failed" but my flash chip is different than the one setup with that config.

Attachements : the lku launch capture and the ucTap capture.

Dear YLG80, I would like to try this build to see if I can go until the menu too, can you post me attachements in PM, please? Thanks in advance, best regards, Stefano.

YLG80

05-03-2009, 01:19 PM

@effesse
Ok, tomorrow I will zip two builds for you.
However I think that you have a AM29LV160 flash chip, so I 'm afraid there will more work to do to handle that chip.

---

effesse

05-04-2009, 09:07 AM

@effesse
Ok, tomorrow I will zip two builds for you.
However I think that you have a AM29LV160 flash chip, so I 'm afraid there will more work to do to handle that chip.

Really I have SST M29W160ET (top boot) 60ns 2MByte/16MBit Flash Memory, I know it because I have desold and reprogram it with a serial Rs232 Dump done by myself and Willem programmer with Tsop48 adapter 1 time once ago, the receiver worked fine for about two months, but now I killed it again trying Jtaging it. :-(

YLG80

05-11-2009, 11:37 AM

@effesse
Would you like to try that version.
It has a menu and has been compiled for a LV160 flash chip.(sdm board)
Just decompress it in the examples directory under the STM tree.

After having launched ucTAP, type gomenu until you have the menu in the second DOS window.

Try first the Verify option (3) against a flash dump that should be in the same directory. (bin file) . If you have a current dump of you STB it should finally return verify OK.

---

v2neo

05-12-2009, 01:24 AM

wonder if we can get this going on XXXXX's using st5517's

Terryl

05-12-2009, 01:59 AM

No, because we don't talk about modifying providers equipment here.

---

effesse

05-12-2009, 04:24 AM

@effesse
Would you like to try that version.
It has a menu and has been compiled for a LV160 flash chip.(sdm board)
Just decompress it in the examples directory under the STM tree.

After having launched ucTAP, type gomenu until you have the menu in the second DOS window.

Try first the Verify option (3) against a flash dump that should be in the same directory. (bin file) . If you have a current dump of you STB it should finally return verify OK.

Unfortunately I get always same output screens:

C:\STM\ST20R2.3.1\examples\GoMenu>uctapsrv.exe -p 0x378 -v 1
ucTAP v 0.2 beta May 31 2006 (c) 2006 TAPDancers
ucTAP is a program to emulate ST20 Micro Connect functions over a basic JTAG in terface.
Operazione completata.

Il servizio specificato Þ stato segnato per l'eliminazione.

Impossibile avviare il servizio. Il servizio Þ disabilitato oppure non Þ associato ad alcuna periferica attiva.

Your CPU is an STi5105 DCU_CTRL 30000004, DCU_DEVICE_ACCESS_MASK 00000200

Starting a server on 9737
Starting a server on 9735
[process_request_9735][9735] data = 128
05 00 00 00 00 00 00 00
packet 05
[process_request_9735][9735] data = 128
01 00 00 00 00 00 00 00
[process_request_9735][9735] data = 128
00 00 00 00 75 63 72 2e
[process_request_9735][9735] data = 16384
71 21 58 72 e0 71 22 50
[process_request_9735][9735] data = 16384
48 76 2d 29 2f 92 25 fa
[process_request_9735][9735] data = 16384
67 49 6e 74 20 20 20 20

closing 9735
Administrator@PC connected on Tue May 12 12:20:21 2009

IDCODE read 759328833
Reset with no boot.
Check IDCODE 2d427041
[get_hosted_poke] val = 000000ff
Ppeek 1 words from 30000000
peeked ffffffff
peek 1 words from 30000040
peeked ffffffff
peek 1 words from 30000000
peeked ffffffff

peek 1 words from 30000040
peeked ffffffff
peek 1 words from 30000000
peeked ffffffff
peek 1 words from 30000040
peeked ffffffff
peek 1 words from 30000000
peeked ffffffff
peek 1 words from 30000040
peeked ffffffff
peek 1 words from 30000100
peeked ffffffff
Poke 50 words.
Poke 43 words.
peek 1 words from 30000100
peeked ffffffff
Poke 50 words.
Poke 43 words.
peek 1 words from 30000004
peeked fffffdff
Poke 8 words.
peek 1 words from 20f00110
peeked ffffffff
Poke 34 words.
peek 1 words from 20f00054
peeked ffffffff
Poke 8 words.
peek 1 words from 20f00000
peeked ffffffff
peek 1 words from 20f00004
peeked ffffffff

peek 1 words from 20f00008
peeked ffffffff
peek 1 words from 20f0000c
peeked ffffffff
peek 1 words from 20f00004
peeked ffffffff
peek 1 words from 20f0000c
peeked ffffffff
Poke 4 words.
peek 1 words from e0000028
peeked ffffffff
Poke 19 words.
peek 1 words from e000002c
peeked ffffffff
peek 1 words from e0000028

peeked ffffffff
Poke 26 words.
peek 1 words from 20200010
peeked ffffffff
peek 1 words from 45000000

peeked ffffffff
peek 1 words from 20402004

peeked ffffffff
Poke 1 words.
peek 1 words from 20402014
peeked ffffffff
Poke 1 words.
LOAD 40 to c0024a30
LOAD 1000 to c0000000
LOAD 1000 to c00003e8
LOAD 1000 to c00007d0
LOAD 1000 to c0000bb8
LOAD 1000 to c0000fa0

LOAD 1000 to c0001388
LOAD 1000 to c0001770
LOAD 1000 to c0001b58
LOAD 1000 to c0001f40
LOAD 1000 to c0002328
LOAD 1000 to c0002710
LOAD 1000 to c0002af8
LOAD 1000 to c0002ee0
LOAD 1000 to c00032c8
LOAD 1000 to c00036b0
LOAD 1000 to c0003a98
LOAD 1000 to c0003e80
LOAD 1000 to c0004268
LOAD 1000 to c0004650
LOAD 1000 to c0004a38
LOAD 1000 to c0004e20
LOAD 1000 to c0005208
LOAD 1000 to c00055f0
LOAD 1000 to c00059d8
LOAD 1000 to c0005dc0
LOAD 1000 to c00061a8
LOAD 1000 to c0006590
LOAD 1000 to c0006978
LOAD 1000 to c0006d60
LOAD 1000 to c0007148
LOAD 1000 to c0007530
LOAD 1000 to c0007918
LOAD 1000 to c0007d00
LOAD 1000 to c00080e8
LOAD 1000 to c00084d0
LOAD 1000 to c00088b8

LOAD 1000 to c0008ca0
LOAD 1000 to c0009088
LOAD 1000 to c0009470
LOAD 1000 to c0009858
LOAD 1000 to c0009c40
LOAD 392 to c000a028
LOAD 636 to c000a1b0
memset 00 107752 to c000a42c
memset trap 0
LOAD 12 to c0024914
LOAD 12 to c0024920
LOAD 260 to c002492c
LOAD 40 to c002aa2c
Poke 3 words.
peek 1 words from 30000004

peeked ffffffff
Poke 8 words.
peek 1 words from 30000100
peeked ffffffff
Poke 1 words.

peek 1 words from 30000004
peeked fffffdff

peek 1 words from 30000004
peeked fffffdff
Poke 2 words.
LOAD 152 to c05ff848
Poke 1 words.
LOAD 268 to c05ff924
Poke 1 words.
LOAD 68 to c05ff8e0
Poke 8 words.

C:\STM\ST20R2.3.1\examples\GoMenu>gomenu

MB400/STi5105 BOARD CONFIGURATION

setup Clock Generator for ST_5105
CKG_CLOCK_SELECT_CFG = 68
--> Mfactor = 5 Nfactor = 37 Pfactor = 0
PLL Request --> PLL Frequency = 400 . 0 MHz
PLL_A Initialised --> PLL Frequency = 399 . 600 MHz
--> Mfactor = 7 Nfactor = 86 Pfactor = 0
PLL Request --> PLL Frequency = 664 . 0 MHz
PLL_B Initialised --> PLL Frequency = 663 . 428 MHz
Setup PLL dividers for ST_5105 chip
CPU = 200 MHz Phase = 0
LMI = 166 MHz Phase = 0
BLITTER = 100 MHz Phase = 0
SYSTEM = 100 MHz Phase = 0
FDMA = 200 MHz Phase = 0
VIDEO = 50 MHz Phase = 0
FLASH = 100 MHz Phase = 0
Setup PLL Frequency Synthesizers for ST_5105 chip
CKG_PIX_CLK_SETUP0 = 0xffffffff
CKG_PLLA_CONFIG0 = 0xffffffff
CKG_PLLA_CONFIG1 = 0xffffffff
CKG_PLLB_CONFIG0 = 0xffffffff
CKG_PLLB_CONFIG1 = 0xffffffff
ST_5105 Clock Generator Successfully setup

==Informations of SDRAM==
SDRAM DEVICE TYPE = DDR SDRAM
SDRAM DEVICE FREQUENCY = 166 MHz
SDRAM DEVICE SIZE = 128 MBits (16 MBytes)
SDRAM DEVICE REFRESH PERIOD = 15.625 microseconds (4K/64ms)
SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.5
==========

LMI_STR = 0x3d6d4b45
set memory size
LMI_COC_UPPER= 0xffffffff
LMI_COC_LOWER= 0xffffffff
LMI Config Done

Configuring registers for FMI ...
Bank[0-3] not configured fully!!
the ident from EPLD is 0x000000ff

resetting flash
FMI configure done

Setup Complete

Setup Complete

Thank you again for all and have great days, best regards, Stefano.

---

slugworth

05-12-2009, 08:09 AM

LMI_STR = 0x3d6d4b45

that isn't in any of my .cfg files and doesn't appear to be correct.

YLG80

05-12-2009, 11:04 AM

@Effesse
Your capture shows a SDRAM problem.
I've checked the datasheet of your SDRAM (A2V28S40TCP) which is a 133 Mhz max speed.
So it appears that you are overclocking your chip.
Would you like to open the mb400.cfg file with a text editor and change the first lines as follows :

```
SDRAM_DEVICE_TYPE = "DDR_RAM"
## SDRAM_DEVICE_TYPE = "SDR_RAM"
##
##
## SDRAM_SIZE_CONFIG = "BUILT_FOR_256MBIT"
SDRAM_FREQ_CONFIG = "166"
SDRAM_SIZE_CONFIG = "BUILT_FOR_128MBIT"
## SDRAM_FREQ_CONFIG = "133"
```

to

```
SDRAM_DEVICE_TYPE = "DDR_RAM"
## SDRAM_DEVICE_TYPE = "SDR_RAM"
##
##
## SDRAM_SIZE_CONFIG = "BUILT_FOR_256MBIT"
## SDRAM_FREQ_CONFIG = "166"
SDRAM_SIZE_CONFIG = "BUILT_FOR_128MBIT"
SDRAM_FREQ_CONFIG = "133"
```

Uncomment (remove the ##) the line with SDRAM_FREQ_CONFIG = ""133"

Comment (add ##) the line with SDRAM_FREQ_CONFIG = ""166"

Try to rerun the flash utility.
You should not have any 0xFFFFFFFF shown in your capture.

YLG80

05-12-2009, 11:21 AM

That value is peeked from a register and poked to the LMI_STR.
It should be something like 0x3d6d4b45 for a 166Mhz SDRAM
and 0x35485235 for a 133 Mhz SDRAM

@ effesse,
If it does not work with the 133Mhz RAM speed you should give us more info on your CPU (the complete ID number) .

It is perhaps also overclocked @ 200Mhz.

YLG80

05-12-2009, 12:12 PM

Here what's written in the short DS :

Here what's written in the short DS :

The STi5118 features the ST20 CPU that is standard across the OMEGA range but boosts

the clock speed to 200 MHz which, when combined with the 2D graphics engine gives the new device outstanding graphics performances, for mosaic composition, fast channel change time and richer graphic content. Targeting low-cost, single tuner STB applications,
the STi5118 is hardware and software binary compatible with the STi5105. This allows manufacturers to use a unified platform for all mass market set-top boxes.
The integration of the VCXO and audio DACs ensures that manufacturers using the STi5118 will cut their total bill of materials and significantly simplify board design and assembly.

So a 200 Mhz clock is likely OK for your CPU.

---

effesse

05-12-2009, 03:01 PM

Here what's written in the short DS :

So a 200 Mhz clock is likely OK for your CPU.

Tnx again for all replies, I just post output screens after mb400.cfg modifications:

```
C:\STM\ST20R2.3.1\examples\GoMenu>tap

C:\STM\ST20R2.3.1\examples\GoMenu>uctapsrv.exe -p 0x378 -v 1
ucTAP v 0.2 beta May 31 2006 (c) 2006 TAPDancers
ucTAP is a program to emulate ST20 Micro Connect functions over a basic JTAG i
terface.
Il servizio specificato Þ stato segnato per l'eliminazione.

Il servizio specificato Þ stato segnato per l'eliminazione.

Impossibile avviare il servizio. Il servizio Þ disabilitato oppure non Þ associ
to ad alcuna periferica attiva.

Your CPU is an STi5105 DCU_CTRL 30000004, DCU_DEVICE_ACCESS_MASK 00000200

Starting a server on 9737
Starting a server on 9735
[process_request_9735][9735] data = 128
05 00 00 00 00 00 00 00
packet 05
[process_request_9735][9735] data = 128
01 00 00 00 00 00 00 00
[process_request_9735][9735] data = 128
00 00 00 00 75 63 72 2e
[process_request_9735][9735] data = 16384
71 21 58 72 e0 71 22 50
[process_request_9735][9735] data = 16384
48 76 2d 29 2f 92 25 fa
[process_request_9735][9735] data = 16384
67 49 6e 74 20 20 20 20
closing 9735
Administrator@PC connected on Tue May 12 22:59:04 2009

IDCODE read 759328833
Reset with no boot.

Check IDCODE 2d427041
[get_hosted_poke] val = 000000ff
Ppeek 1 words from 30000000
peeked ffffffff
peek 1 words from 30000040
```

peeked ffffffff
peek 1 words from 30000000

peeked ffffffff
peek 1 words from 30000040
peeked ffffffff
peek 1 words from 30000000
peeked ffffffff
peek 1 words from 30000040
peeked ffffffff
peek 1 words from 30000000
peeked ffffffff
peek 1 words from 30000040
peeked ffffffff

peek 1 words from 30000100
peeked ffffffff
Poke 50 words.
Poke 43 words.
peek 1 words from 30000100
peeked ffffffff
Poke 50 words.
Poke 43 words.
peek 1 words from 30000004
peeked fffffdff
Poke 8 words.
peek 1 words from 20f00110
peeked ffffffff
Poke 34 words.
peek 1 words from 20f00054
peeked ffffffff
Poke 8 words.
peek 1 words from 20f00000
peeked ffffffff
peek 1 words from 20f00004
peeked ffffffff
peek 1 words from 20f00008
peeked ffffffff
peek 1 words from 20f0000c
peeked ffffffff
peek 1 words from 20f00004
peeked ffffffff
peek 1 words from 20f0000c
peeked ffffffff
Poke 4 words.
peek 1 words from e0000028

peeked ffffffff
Poke 19 words.
peek 1 words from e000002c
peeked ffffffff
peek 1 words from e0000028
peeked ffffffff
Poke 26 words.
peek 1 words from 20200010
peeked ffffffff
peek 1 words from 45000000
peeked ffffffff
peek 1 words from 20402004
peeked ffffffff
Poke 1 words.
peek 1 words from 20402014

peeked ffffffff
Poke 1 words.
LOAD 40 to c0024a30
LOAD 1000 to c0000000
LOAD 1000 to c00003e8

LOAD 1000 to c00005e8
LOAD 1000 to c00007d0
LOAD 1000 to c0000bb8

LOAD 1000 to c0000fa0
LOAD 1000 to c0001388
LOAD 1000 to c0001770
LOAD 1000 to c0001b58
LOAD 1000 to c0001f40
LOAD 1000 to c0002328
LOAD 1000 to c0002710
LOAD 1000 to c0002af8
LOAD 1000 to c0002ee0
LOAD 1000 to c00032c8
LOAD 1000 to c00036b0

LOAD 1000 to c0003a98
LOAD 1000 to c0003e80
LOAD 1000 to c0004268
LOAD 1000 to c0004650
LOAD 1000 to c0004a38
LOAD 1000 to c0004e20
LOAD 1000 to c0005208
LOAD 1000 to c00055f0
LOAD 1000 to c00059d8
LOAD 1000 to c0005dc0
LOAD 1000 to c00061a8
LOAD 1000 to c0006590
LOAD 1000 to c0006978
LOAD 1000 to c0006d60
LOAD 1000 to c0007148
LOAD 1000 to c0007530
LOAD 1000 to c0007918
LOAD 1000 to c0007d00
LOAD 1000 to c00080e8
LOAD 1000 to c00084d0
LOAD 1000 to c00088b8
LOAD 1000 to c0008ca0
LOAD 1000 to c0009088
LOAD 1000 to c0009470
LOAD 1000 to c0009858
LOAD 1000 to c0009c40
LOAD 392 to c000a028
LOAD 636 to c000a1b0
memset 00 107752 to c000a42c
memset trap 0
LOAD 12 to c0024914

LOAD 12 to c0024920
LOAD 260 to c002492c
LOAD 40 to c002aa2c
Poke 3 words.
peek 1 words from 30000004
peeked fffffdff
Poke 8 words.
peek 1 words from 30000100
peeked ffffffff
Poke 1 words.
peek 1 words from 30000004
peeked fffffdff
peek 1 words from 30000004
peeked fffffdff
Poke 2 words.

LOAD 152 to c05ff848
Poke 1 words.
LOAD 268 to c05ff924
Poke 1 words.

LOAD 68 to c05ff8e0
Poke 8 words.

C:\STM\ST20R2.3.1\examples\GoMenu>gomenu

MB400/STi5105 BOARD CONFIGURATION

setup Clock Generator for ST_5105
CKG_CLOCK_SELECT_CFG = 68
--> Mfactor = 5 Nfactor = 37 Pfactor = 0
PLL Request --> PLL Frequency = 400 . 0 MHz
PLL_A Initialised --> PLL Frequency = 399 . 600 MHz

--> Mfactor = 7 Nfactor = 69 Pfactor = 0
PLL Request --> PLL Frequency = 533 . 0 MHz
PLL_B Initialised --> PLL Frequency = 532 . 285 MHz
Setup PLL dividers for ST_5105 chip
CPU = 200 MHz Phase = 0
LMI = 133 MHz Phase = 0
BLITTER = 100 MHz Phase = 0
SYSTEM = 100 MHz Phase = 0
FDMA = 200 MHz Phase = 0
VIDEO = 50 MHz Phase = 0
FLASH = 100 MHz Phase = 0
Setup PLL Frequency Synthesizers for ST_5105 chip
CKG_PIX_CLK_SETUP0 = 0xffffffff
CKG_PLLA_CONFIG0 = 0xffffffff
CKG_PLLA_CONFIG1 = 0xffffffff
CKG_PLLB_CONFIG0 = 0xffffffff
CKG_PLLB_CONFIG1 = 0xffffffff
ST_5105 Clock Generator Successfully setup

==Informations of SDRAM==
SDRAM DEVICE TYPE = DDR SDRAM
SDRAM DEVICE FREQUENCY = 133 MHz
SDRAM DEVICE SIZE = 128 MBits (16 MBytes)
SDRAM DEVICE REFRESH PERIOD = 15.625 microseconds (4K/64ms)
SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.0
==========

LMI_STR = 0x35485235
set memory size
LMI_COC_UPPER= 0xffffffff
LMI_COC_LOWER= 0xffffffff

LMI Config Done

Configuring registers for FMI ...
Bank[0-3] not configured fully!!
the ident from EPLD is 0x000000ff
reseting flash
FMI configure done

Setup Complete

The LMI_STR now seem to be right but I get always too many FFFFFFFF in the other fields. :-(

YLG80 05-13-2009, 12:47 AM

It looks like the JTAG connection is not working properly.
Most of the peeks are wrong.
I've got your board picture. I don't see a JATG connector

I've got your board picture. I don't see a JATG connector.
Where did you make the connections ?

---

effesse

05-13-2009, 03:57 AM

It looks like the JTAG connection is not working properly.
Most of the peeks are wrong.
I've got your board picture. I don't see a JATG connector.
Where did you make the connections ?

I use CN61 the 10 pin header connector as Jtag and with jKeys too I'm able to get the right CPU ID and correct flash dump changing address for reading to 0x40000000 lenght 0x200000, but get always 'FFFFFFFF/FFFFFFFF' when I try to go to programming the flash.

http://digilander.libero.it/effesse2002/TsSrs1_Jtag.jpg

http://digilander.libero.it/effesse2002/Telestar_Srs_1_Top.jpg

http://digilander.libero.it/effesse2002/jKeys.jpg

http://digilander.libero.it/effesse2002/jKeys2.jpg

I follow with my multimeter the wires from CN61 to CPU and watching to Vestel DVBS Receiver with 5105 CPU datasheet it seems to be identical to Redline 14500 receiver even thought it uses the 5518 CPU!

Thank you again for all help, best regards, Stefano.

---

slugworth

05-13-2009, 06:30 AM

I use CN61 the 10 pin header connector as Jtag and with jKeys too I'm able to get the right CPU ID and correct flash dump changing address for reading to 0x40000000 lenght 0x200000, but get always 'FFFFFFFF/FFFFFFFF' when I try to go to programming the flash.

Like was stated many times before you can't program DCU3 with jkeys,so you get that error.
You are missing the sti5118 id in the .def file so the id comes up as unknown.

---

slugworth

05-13-2009, 06:33 AM

The clock registers are all based on the sys_services_base address,
which may be different for the sti5118

clock registers definition

SYS_SERVICES_BASE_ADDRESS = 0x20F00000

CKG_PLLA_CONFIG0 = (SYS_SERVICES_BASE_ADDRESS + 0x000)
CKG_PLLA_CONFIG1 = (SYS_SERVICES_BASE_ADDRESS + 0x004)
CKG_PLLB_CONFIG0 = (SYS_SERVICES_BASE_ADDRESS + 0x008)
CKG_PLLB_CONFIG1 = (SYS_SERVICES_BASE_ADDRESS + 0x00C)
CKG_FSA_SETUP = (SYS_SERVICES_BASE_ADDRESS + 0x010)
CKG_PCM_CLK_SETUP0 = (SYS_SERVICES_BASE_ADDRESS + 0x020)
CKG_PCM_CLK_SETUP1 = (SYS_SERVICES_BASE_ADDRESS + 0x024)
CKG_SPDIF_CLK_SETUP0 = (SYS_SERVICES_BASE_ADDRESS + 0x030)
CKG_SPDIF_CLK_SETUP1 = (SYS_SERVICES_BASE_ADDRESS + 0x034)
CKG_SC_CLK_SETUP0 = (SYS_SERVICES_BASE_ADDRESS + 0x040)

CKG_SC_CLK_SETUP1 = (SYS_SERVICES_BASE_ADDRESS + 0x044)
CKG_PIX_CLK_SETUP0 = (SYS_SERVICES_BASE_ADDRESS + 0x054)
CKG_PIX_CLK_SETUP1 = (SYS_SERVICES_BASE_ADDRESS + 0x058)
CKG_FDMA_CLK_SETUP0 = (SYS_SERVICES_BASE_ADDRESS + 0x070)

CKG_FDMA_CLK_SETUP1 = (SYS_SERVICES_BASE_ADDRESS + 0x074)
CKG_AUX_CLK_SETUP0 = (SYS_SERVICES_BASE_ADDRESS + 0x070)
CKG_AUX_CLK_SETUP1 = (SYS_SERVICES_BASE_ADDRESS + 0x074)

CKG_MODE_CONTROL = (SYS_SERVICES_BASE_ADDRESS + 0x110)
CKG_CLOCK_SELECT_CFG = (SYS_SERVICES_BASE_ADDRESS + 0x180)
CKG_REGISTER_LOCK_CFG = (SYS_SERVICES_BASE_ADDRESS + 0x300)
CKG_OBSERVATION_CFG_AUX = (SYS_SERVICES_BASE_ADDRESS + 0x188)

YLG80

05-13-2009, 10:00 AM

Thanks for that additional information
It looks like your CPU has still the control over your flash chip.
It cannot enter in DCU3 mode.
You can get the CPU ID but that's all you can do.

Have you tried to Power your STB OFF then ON and immediately after power it
ON start uCTAP. Afterwards you start flashprg.
Perhaps you will gain the control over the flash chip.
(You can do that also with JKEYS if your interface is wired for JKEYS and not for ucTAP.)

Are you able to see (via the STB display) if the JTAG program is able to reset the box.
If you no longer have anything displayed you cannot verify the correct RESET, so can you probe (scope) the jtag connector pin 10 to see if it's really pulled down to level 0.
If your parallel interface is not able to pull that signal LOW you cannot enter in DCU mode.
Sometimes it locks HIGH because the driven transistor base current is too high.

Could you remind me to which pin is connected the TP test point located between the CPU and the XTAL. Is that pin to pin 131 ?

YLG80

05-13-2009, 11:46 AM

I've sent a correct jkeys.def file to effesse.
In fact his Sti5118 CPU has the same CPU id as the Sti5105.(2D427041)
So he has almost the same configuration as the DTS1601. (my STB)
(Same flash type, same SDRAM size, same CPU ID)

---

effesse

05-13-2009, 05:40 PM

I've sent a correct jkeys.def file to effesse.
In fact his Sti5118 CPU has the same CPU id as the Sti5105.(2D427041)
So he has almost the same configuration as the DTS1601. (my STB)
(Same flash type, same SDRAM size, same CPU ID)

Tnx YLG80, your .DEF file worked for reading the Flash, but I'm already able to read it with jKeys. For the TP63, it goes to the pin #130 of the Stx5118alb, called RESET and how you can see in this attachment .PDF file maybe I have to implement a Reset circuit on my Buffered Jtag Interface!

Thank you very much again, best regards, Stefano.

YLG80

05-14-2009, 12:18 AM

OK.
So if you can read your flash with JKEYS, you need now to rewire correctly the connection for ucTAP.
I believe that you need to use the pin 130 signal/TP63 which is the nRST JTAG reset signal needed by ucTAP.
You should connect that signal following Slugworth connection schematic shown in a previous post.

(You can see in the attach pdf that pin 130 is driven fron pin 17 of the JTAG-CON through a pair of transistors and is identified as JTAG_RESET)

---

effesse

05-14-2009, 03:55 AM

OK.
So if you can read your flash with JKEYS, you need now to rewire correctly the connection for ucTAP.
I believe that you need to use the pin 130 signal/TP63 which is the nRST JTAG reset signal needed by ucTAP.
You should connect that signal following Slugworth connection schematic shown in a previous post.

(You can see in the attach pdf that pin 130 is driven fron pin 17 of the JTAG-CON through a pair of transistors and is identified as JTAG_RESET)

So I think your receiver implements this little Reset circuit on the mainboard and your Jtag interface is able to give the JTAG_RESET signal to your Jtag header? Kind regards, Stefano.

YLG80

05-14-2009, 04:47 AM

Yes.
My board has a 20 pin JTAG connector with that signal.
When I launch ucTAP I can see on the display that the STB is running in DCU mode.
The STB displays the current time, and the clock is frozen during a DCU3 session.
uCTAp needs that signal, although I've been able to run a correct ucTAP session on a STI5518 by power cycling the STB before starting ucTAP.
(This was done keeping the same JTAG connector wiring as for JKEYS.)
But this is not good and you can have various results and problems.

You need to pickup Slugworth's connector configuration (see his post) and adapt it to your 10 pin connector plus TP63.

I use a buffered JTAG and I had to modify the buffer wiring, i.e cutting tracks to ground and adddin wires in order to reactivate and use a free gate for the extra connection needed on the LPT side.

I've posted a diagram in this thread. But if you have connectors you could perhaps simply build the Slugworth simple unbuffered connector. He had no proble with that config, providing that you connect your PC and STB while they are unpowered.

YLG80

05-14-2009, 05:03 AM

As far as I can see I have the same double transistors JTAG RST circuit on my board.
That circuit is connected to pin 17 of the JTAG_Connector.

When you connect something to TP63 DO NOT MAKE A DIRECT CONNECTION TO THE LPT PORT !
Use a "via" 100 ohm resistor.

---

effesse

05-15-2009, 08:17 AM

As far as I can see I have the same double transistors JTAG RST circuit on my board.
That circuit is connected to pin 17 of the JTAG_Connector.

When you connect something to TP63 DO NOT MAKE A DIRECT CONNECTION TO THE LPT PORT !
Use a "via" 100 ohm resistor.

I don't understand if I can use those schemes I attached in Thumbnail
or if I have to modify my buffered Jtag or my simple Jtag with 5 resistors (100ohm).

What's the name of your JTAG Pin #17, is it nRST? At which pin of your LPT port is connected to?

I have only TCK, TDO, TMS, TRST and GND on the CN61 10 pin header connector of the receiver and the TRST goes to pin #5 of the LPT port, the TP63 then will correspond to nRST signal?

However the 2 thumbnails are not equivalent in signal names, I have to investigate more deeply...

Many thanks again, best regards, Stefano.

---

slugworth

05-15-2009, 09:04 AM

I believe you use pin 2 instead of pin 8
vestel sti5105 10pin jtag connector.

YLG80

05-15-2009, 12:04 PM

@effesse
I would advise you to use the simple interface with via resistors.
(see also Slugworth's post #189 page 19). It's KISS, it's good and Slugworth is happy with that type of interface.

I've taken my X4 magnifier and test nails and I've probed my circuit from pin 130 to ... pin 17 of the STB 20 PIN JTAG Connector which is RST or JPI_RST or TRST.

This is not a direct connection. There are two NPN smd transistors 1AM=mMBT3904 wired as shown in the CTV2991 pdf schematic.
Between pin 130 and the transistor base, there is a 100ohm resistor.

If I'm not wrong you should connect that TP63 pin via a 100 ohm to pin 5 of the LPT connector.
I would connect pin 7 of your 10 pin JTAG connector to pin 12 of the LPT CON.
Then uCTAP would be happy !

On my modified JTAG buffered interface:
I've desoldered the 100 ohm resistor end connected normally to pin 19 of the 20 PIN JTAG CON (nRST).
Using a wire, I've connected that resistor end to pin 17 of the JTAG CON (JPI_RST)

I've rewired JTAG CON PIN 19 (left open after desoldering the resistor end) to a free buffer in the HC244, i.e pin 16 via an additional 100 ohm resistor.

Pin 16 is a buffer output.
The corresponding buffer input pin is pin 4 (you will need to cut the short to ground in order to use that buffer input.)

Pin 4 of the HC244 is connected via a 100ohm to pin 12 of the LPT1 port.
From pin 4 I've also added a pull-up resistor of 1K.

So finally JTAG CON pin 19 is connected to LPT PIN 12 via a buffer.

---

effesse

05-15-2009, 04:35 PM

Tnx to all u 2 (Slug and YLG) , after rewiring correctly my simple Jtag, I got something new:

C:\STM\ST20R2.3.1\examples\Slugworth>uctapsrv.exe -p 0x378 -v 1
ucTAP v 0.2 beta May 31 2006 (c) 2006 TAPDancers
ucTAP is a program to emulate ST20 Micro Connect functions over a basic JTAG in terface.
Il servizio specificato Þ stato segnato per l'eliminazione.

Il servizio specificato Þ stato segnato per l'eliminazione.

Impossibile avviare il servizio. Il servizio Þ disabilitato oppure non Þ associa

to ad alcuna periferica attiva.

Your CPU is an STi5105 DCU_CTRL 30003004, DCU_DEVICE_ACCESS_MASK 00000200

Starting a server on 9737
Starting a server on 9735
[process_request_9735][9735] data = 128
05 00 00 00 00 00 00 00
packet 05
[process_request_9735][9735] data = 128
01 00 00 00 00 00 00 00
[process_request_9735][9735] data = 128
00 00 00 00 75 63 72 2e
[process_request_9735][9735] data = 16384

71 21 58 72 e0 71 22 50
[process_request_9735][9735] data = 16384
48 76 2d 29 2f 92 25 fa
[process_request_9735][9735] data = 16384
67 49 6e 74 20 20 20 20
closing 9735
Administrator@PC connected on Sat May 16 00:31:54 2009

IDCODE read 759328833
Reset with no boot.
Check IDCODE 2d427041
peek 1 words from 30000000
peeked 20022121
peek 1 words from 30000040
peeked 01212121
peek 1 words from 30000000
peeked 20022121
peek 1 words from 30000040
peeked 01212121
peek 1 words from 30000000
peeked 20022121
peek 1 words from 30000040
peeked 01212121
peek 1 words from 30000100
peeked 0000a240
Poke 12 words.
peek 1 words from 30000100

peeked 0000a240
Poke 12 words.
peek 1 words from 30000004
peeked 0000fc10
Poke 8 words.
peek 1 words from 20f00110
peeked 00000002
Poke 34 words.
peek 1 words from 20f00054
peeked 00000ebf
Poke 8 words.
peek 1 words from 20f00000
peeked 00002404
peek 1 words from 20f00004
peeked 0000c838

peek 1 words from 20f00008
peeked 00004406
peek 1 words from 20f0000c
peeked 0000c838

peek 1 words from 20f00004
peeked 0000c838
peek 1 words from 20f0000c

peeked 0000c838
Poke 4 words.
peek 1 words from e0000028
peeked 00002000
Poke 19 words.
peek 1 words from e000002c
peeked 000c6750
peek 1 words from e0000028
peeked 00002000
Poke 26 words.
peek 1 words from 20200010
peeked 0000000f
peek 1 words from 45000000
peeked 24202020
peek 1 words from 20402004
peeked 00000000
Poke 1 words.
peek 1 words from 20402014
peeked 0e000000
Poke 1 words.
LOAD 40 to c0006ae8
LOAD 1000 to c0000000
LOAD 1000 to c00003e8
LOAD 1000 to c00007d0
LOAD 1000 to c0000bb8
LOAD 1000 to c0000fa0
LOAD 1000 to c0001388
LOAD 1000 to c0001770
LOAD 1000 to c0001b58
LOAD 1000 to c0001f40
LOAD 1000 to c0002328
LOAD 1000 to c0002710
LOAD 1000 to c0002af8
LOAD 1000 to c0002ee0
LOAD 1000 to c00032c8
LOAD 1000 to c00036b0
LOAD 1000 to c0003a98
LOAD 1000 to c0003e80
LOAD 1000 to c0004268
LOAD 56 to c0004650
LOAD 328 to c0004688
memset 00 8700 to c00047d0
memset trap 0
LOAD 12 to c00069cc
LOAD 12 to c00069d8
LOAD 260 to c00069e4
LOAD 25 to c000caf0
Poke 3 words.
peek 1 words from 30000004
peeked 0000fc10
Poke 9 words.
peek 1 words from 30000100
peeked 0000a240
Poke 2 words.
peek 1 words from 30000004
peeked 0000fc10
Poke 3 words.

my_write() failed; 6 -1 err=10053RECV ERROR: err=10053
closing 9737

C:\STM\ST20R2.3.1\examples\Slugworth>ho

```
C:\STM\ST20R2.3.1\examples\Slugworth>st20run -i mb400.cfg -t mag flashprg_c1c.l
u



-----------------------------------------------------
MB400/STi5105 BOARD CONFIGURATION

setup Clock Generator for ST_5105
CKG_CLOCK_SELECT_CFG = 68
--> Mfactor = 5 Nfactor = 37 Pfactor = 0
PLL Request --> PLL Frequency = 400 . 0 MHz
PLL_A Initialised --> PLL Frequency = 399 . 600 MHz
--> Mfactor = 7 Nfactor = 69 Pfactor = 0
PLL Request --> PLL Frequency = 533 . 0 MHz

PLL_B Initialised --> PLL Frequency = 532 . 285 MHz
Setup PLL dividers for ST_5105 chip
CPU = 200 MHz Phase = 0
LMI = 133 MHz Phase = 0
BLITTER = 100 MHz Phase = 0
SYSTEM = 100 MHz Phase = 0
FDMA = 200 MHz Phase = 0
VIDEO = 50 MHz Phase = 0
FLASH = 100 MHz Phase = 0
Setup PLL Frequency Synthesizers for ST_5105 chip
CKG_PIX_CLK_SETUP0 = 0x00000ebf
CKG_PLLA_CONFIG0 = 0x00002404
CKG_PLLA_CONFIG1 = 0x0000c838
CKG_PLLB_CONFIG0 = 0x00004406
CKG_PLLB_CONFIG1 = 0x0000c838
ST_5105 Clock Generator Successfully setup

==Informations of SDRAM==
SDRAM DEVICE TYPE = DDR SDRAM
SDRAM DEVICE FREQUENCY = 133 MHz
SDRAM DEVICE SIZE = 128 MBits (16 MBytes)
SDRAM DEVICE REFRESH PERIOD = 15.625 microseconds (4K/64ms)
SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.0
============================

LMI_STR = 0x35485235
set memory size
LMI_COC_UPPER= 0x000c6750
LMI_COC_LOWER= 0x00002000
LMI Config Done

Configuring registers for FMI ...
All Banks[0-3] have been configured
the ident from EPLD is 0x00000020
reseting flash
FMI configure done


Setup Complete
-----------------------------------------------------

error-address space not stopped
```

I got this error, but I don't know if all other value fields are correct. Maybe I'm near to the end???

Thank you very much again, kind regards, Stefano.

---

slugworth

05-15-2009, 04:43 PM

Sometimes you have to kick it in the ass to get it to work.
I sometimes get that error and just run the .bat file again.
You can still jkeys dump with the new wiring,so I leave it that way.

---

effesse

05-15-2009, 05:25 PM

Yeah!
I can still read with jKeys the entire 2MB flash dump, but did you put an eye
to my last output (specially to ho.bat)? Did all values seem correct for you?

I think it's very important for me to continue this neverending job, tnx.

Kind regards, have great night, Stefano.

---

slugworth

05-15-2009, 05:45 PM

The ho looked good,you should get further to the choices screen soon.

YLG80

05-16-2009, 12:34 AM

Yes your config looks now good.
I've changed my settings to 133Mhz and I've exactly the same debug message output.
You are near the end of the beginning.
As Slugworth mentioned, I've sometimes to restart ucTAP.
Usually the first time I start ucTAP, it gives a write error during the setup.
The second time it's OK.

YLG80

05-16-2009, 02:54 AM

I've the same error when the dcu_mb400.cfg is incorrect in the board directory.
I'm still wondering why flashprg has to fetch data from that board directory at runtime as the config files are available in the flashprog directory.

Could you try this:
1- Backup dcu_mb400.cfg and dcu_mb400_conf.cfg in your toolset ../board directory

2- Decompress the file attached to this post in that directory.
It will replace the above files with the same configured for 2Mb Flash and 16Mb SDRAM

---

effesse

05-16-2009, 04:43 AM

I've the same error when the dcu_mb400.cfg is incorrect in the board directory.
I'm still wondering why flashprg has to fetch data from that board directory at runtime as the config files are available in the flashprog directory.

Could you try this:
1- Backup dcu_mb400.cfg and dcu_mb400_conf.cfg in your toolset ../board directory

2- Decompress the file attached to this post in that directory.
It will replace the above files with the same configured for 2Mb Flash and 16Mb SDRAM

Thanks, I will try soon. In this meanwhile I would like to see a Picture .JPG of your LPT port connector, to see if I made all connection that way.

Thank you very much again, kind regards, Stefano.

---

slugworth

05-16-2009, 07:47 AM

I've the same error when the dcu_mb400.cfg is incorrect in the board directory.

All that should have been fixed on page 39 if he was following directions.
http://www.ttatalk.com/showthread.php?t=164232&page=39

---

effesse

05-16-2009, 09:29 AM

All that should have been fixed on page 39 if he was following directions.
http://www.ftatalk.com/showthread.php?t=164232&page=39

I fellow the directions but however the problem for me isn't still solved!

:frusty:

YLG80

05-17-2009, 04:48 AM

Would you like to test your config using the attached build.
It's a simple "Hello World" test that should only display :

Hello World from O s p r e y

This is a simple Sti51xx board DCU3 test.
If you read me, I'm OK !

You decompress the build in your toolset examples directory.

Run ucTAP
Open a DOS box and cd to the test51xx directory created under examples.

Type go.

If there is an error or no "hello world" message, do the following from the test51xx directory :
nmake clean <this will remove the executable and debug file>
nmake < this will recompile the executable>

This will normally recompile the hello.c program.
If there are errors, please report them.

---

effesse

05-17-2009, 07:20 AM

Would you like to test your config using the attached build.
It's a simple "Hello World" test that should only display :

Hello World from Osprey
This is a simple Sti51xx board DCU3 test.
If you read me, I'm OK !

You decompress the build in your toolset examples directory.

Run ucTAP
Open a DOS box and cd to the test51xx directory created under examples.

Type go.

If there is an error or no "hello world" message, do the following from the test51xx directory :
nmake clean <this will remove the executable and debug file>
nmake < this will recompile the executable>

This will normally recompile the hello.c program.
If there are errors, please report them.

In my PC there's no nmake utility installed! What I do install to have nmake working fine? Tnx a lot for all, Stefano.

YLG80

05-17-2009, 07:50 AM

Put the attached files in the toolset bin directory.

(compressed file)

YLG80

05-17-2009, 09:43 AM

I've found why there was always an error_space when running an application compiled with the original config files in the board directory.

With these config files, the target command needs to specify the proc together with arguments. W/o arguments the board is not completely configured.

I attach the simple Hello World example.
There is NO mb400 config file in the directory, so that the compiler uses the original files in the boards directory.

The target is specified in the ST20.rc file as follows :

```
parse ("directory "+(getenv ST20ROOT)+"/board\n")
parse ("include boardprocs.cfg\n")
target sti5105box tap "jei 127.0.0.1" "STi5105MB400 (1) (1)"
```

Note the quotes and the two parms.

The simple Hello World compiles and runs OK on my 128Mbit stb even if I specifiy a BUILT FOR 256Mbit in the very first lines dcu_mb400_conf.cfg.
Snapshot:

```
-----------------------------------------
PROCEDURE (stddefs: ST20C1MemoryInit)
-----------------------------------------
Build for : BUILT_FOR_256MBIT
-----------------------------------------
PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xc0fffc00
size : 0x00000400
-----------------------------------------
Hello World from O s p r e y
This is a simple Sti51xx board DCU3 test.
If you read me, I'm OK !
```

---

effesse

05-17-2009, 11:24 AM

I've found why there was always an error_space when running an application compiled with the original dcu_mb400.cfg and dcu_mb400_conf.cfg files.

With these config files, the target command needs to specify the proc together with arguments. W/o arguments the board is not completely configured.

I attach the simple Hello World example.
There is NO mb400 config file in the directory, so that the compiler uses the original files in the boards directory.

The target is specified in the ST20.rc file as follows :

```
parse ("directory "+(getenv ST20ROOT)+"/board\n")
parse ("include boardprocs.cfg\n")
target sti5105box tap "jei 127.0.0.1" "STi5105MB400 (1) (1)"
```

Note the quotes and the two parms.

The simple Hello World compiles and runs OK on my 128Mbit stb even if I specifiy a BUILT FOR 256Mbit in the very first lines dcu_mb400_conf.cfg.
Snapshot:

```
-----------------------------------------
PROCEDURE (stddefs: ST20C1MemoryInit)
```

```
-----------------------------------------
Build for : BUILT_FOR_256MBIT
-----------------------------------------
PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xc0fffc00
size : 0x00000400
-----------------------------------------
Hello World from Osprey
This is a simple Sti51xx board DCU3 test.
If you read me, I'm OK !
```

Thank you very much for your crossed finger, but here we are my output screens after did all you said to me, I deleted all the files in the test51xx directory, unzip the files in the test51xx-2.zip archive and operate as you said:

```
C:\ucTAP>tap

C:\ucTAP>uctapsrv.exe -p 0x0378 -v 1
ucTAP v 0.2 beta May 31 2006 (c) 2006 TAPDancers
ucTAP is a program to emulate ST20 Micro Connect functions over a basic JTAG in
terface.
Il servizio specificato Þ stato segnato per l'eliminazione.

Il servizio specificato Þ stato segnato per l'eliminazione.
Impossibile avviare il servizio. Il servizio Þ disabilitato oppure non Þ associa
to ad alcuna periferica attiva.

Your CPU is an STi5105 DCU_CTRL 30003004, DCU_DEVICE_ACCESS_MASK 00000200

Starting a server on 9737
Starting a server on 9735
[process_request_9735][9735] data = 128
05 00 00 00 00 00 00 00
packet 05
[process_request_9735][9735] data = 128
01 00 00 00 00 00 00 00
[process_request_9735][9735] data = 128
00 00 00 00 75 63 72 2e
[process_request_9735][9735] data = 16384
71 21 58 72 e0 71 22 50
[process_request_9735][9735] data = 1136
48 76 2d 29 2f 92 25 fa
[process_request_9735][9735] data = 16384
21 51 72 e1 71 22 23 59
[process_request_9735][9735] data = 1136
2f f1 72 71 23 28 31 2f
[process_request_9735][9735] data = 16384
f0 71 63 2f 2d 98 c2 c0
[process_request_9735][9735] data = 1136
23 28 9d 75 f4 a2 64 0c
[process_request_9735][9735] data = 16384
2d 21 95 72 37 ef 43 72
[process_request_9735][9735] data = 1136
76 28 25 2c 5c d6 21 01
[process_request_9735][9735] data = 13624
f9 a9 7a 59 13 22 25 29
[process_request_9735][9735] data = 16384
30 00 00 00 00 00 16 00
[process_request_9735][9735] data = 1136
```

00 44 01 10 02 00 00 00
[process_request_9735][9735] data = 16384
04 00 00 00 10 00 00 00

[process_request_9735][9735] data = 1136
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 16384
00 03 00 00 00 00 14 fe
[process_request_9735][9735] data = 1136
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 16384
00 00 00 00 60 00 00 00
[process_request_9735][9735] data = 1136
03 00 00 0c 00 00 31 00
[process_request_9735][9735] data = 16384
11 0a 00 40 09 00 00 15
[process_request_9735][9735] data = 1136
00 00 29 ff 80 01 00 00
[process_request_9735][9735] data = 16384
04 00 4c 90 09 06 00 26
[process_request_9735][9735] data = 1136
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 16384
00 01 20 00 00 00 00 02
[process_request_9735][9735] data = 1136
00 04 80 00 10 00 02 00
[process_request_9735][9735] data = 16384
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 1136
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 4688
00 00 00 00 00 00 00 00
[process_request_9735][9735] data = 3416
00 00 00 00 08 00 00 00
[process_request_9735][9735] data = 28
c7 1f 01 00 bc 3b 01 00
[process_request_9735][9735] data = 16
33 22 01 00 2b 20 01 00
[process_request_9735][9735] data = 10
74 6d 6a 65 69 2e 64 62
[process_request_9735][9735] data = 128
01 00 00 00 75 63 72 2e
my_write() failed; 532 -1 err=10054closing 9735
Administrator@PC connected on Sun May 17 19:17:24 2009

IDCODE read 759328833
Reset with no boot.
Check IDCODE 2d427041
peek 1 words from 30000000
peeked 20022121
peek 1 words from 30000040
peeked 01212121
peek 1 words from 30000000
peeked 20022121
peek 1 words from 30000040
peeked 01212121
peek 1 words from 30000000
peeked 20022121
peek 1 words from 30000040
peeked 01212121
Poke 8 words.
peek 1 words from 20f00110
peeked 00000002
peek 1 words from 20f00000
peeked 00002404

peek 1 words from 20f00008
peeked 00004406
Poke 31 words.

peek 1 words from 20f00054
peeked 00000ebf
Poke 8 words.
peek 1 words from 20f00004
peeked 0000c838
peek 1 words from 20f0000c
peeked 0000c838
Poke 4 words.
peek 1 words from e0000028
peeked 00002000
Poke 19 words.
peek 1 words from e000002c
peeked 000c6750
peek 1 words from e0000028
peeked 00002000
Poke 28 words.
peek 1 words from 30000000
peeked 20022121
peek 1 words from 30000040
peeked 01212121
peek 1 words from 30000100
peeked 0000a240
Poke 12 words.
peek 1 words from 30000100
peeked 0000a240
Poke 12 words.
peek 1 words from 30000004
peeked 0000fc10
LOAD 40 to c000033c
LOAD 784 to c0000000
LOAD 16 to c0000310
memset 00 24 to c0000320
memset trap 0
LOAD 4 to c0000338
LOAD 22 to c0006348
Poke 3 words.
peek 1 words from 30000004
peeked 0000fc10
Poke 9 words.
peek 1 words from 30000100
peeked 0000a240
Poke 2 words.
peek 1 words from 30000004
peeked 0000fc10
Poke 3 words.
my_write() failed; 6 -1 err=10053RECV ERROR: err=10053
closing 9737

C:\STM\ST20R2.3.1\examples\Test51xx>dir
Il volume nell'unità C non ha etichetta.
Numero di serie del volume: 2772-08EC

Directory di C:\STM\ST20R2.3.1\examples\Test51xx

17/05/2009 18.56 <DIR> .
17/05/2009 18.56 <DIR> ..
17/05/2009 12.42 65 go.bat
17/05/2009 12.22 240 hello.c
20/11/2000 13.38 94 makefile
01/10/2001 12.46 96 rm.bat
17/05/2009 16.47 147 st20.rc

```
20/11/2000 13.38 50 targets.mkf
17/05/2009 13.15 374 build.mkf
17/05/2009 19.16 1.614 hello.tc1
```

```
17/05/2009 19.16 13.147 hello_c1c.dbg
17/05/2009 19.16 1.126 hello_c1c.lku
10 File 16.953 byte
2 Directory 6.329.303.040 byte disponibili

C:\STM\ST20R2.3.1\examples\Test51xx>nmake clean

Microsoft (R) Program Maintenance Utility Version 1.50
Copyright (c) Microsoft Corp 1988-94. All rights reserved.


Microsoft (R) Program Maintenance Utility Version 1.50
Copyright (c) Microsoft Corp 1988-94. All rights reserved.

C:\STM\ST20R2.3.1\examples\Test51xx\..\mkf\rm.bat hello_c1c.lku hello_
c.dbg hello.tc1

C:\STM\ST20R2.3.1\examples\Test51xx>nmake

Microsoft (R) Program Maintenance Utility Version 1.50
Copyright (c) Microsoft Corp 1988-94. All rights reserved.


Microsoft (R) Program Maintenance Utility Version 1.50
Copyright (c) Microsoft Corp 1988-94. All rights reserved.

st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o hello.tc1 hel
.c
st20cc -p STi5105MB400 -O0 -LC:\STM\ST20R2.3.1\examples\Test51xx\..\m
/../lib -o hello_c1c.lku hello.tc1

C:\STM\ST20R2.3.1\examples\Test51xx>go
error-address space not stopped

C:\STM\ST20R2.3.1\examples\Test51xx>
```

Only the nmake utility seem to work correctly, at least now I know how to compile and make new .LKUs files!

Thanks again a lot, best regards, Stefano.

---

effesse

05-17-2009, 04:00 PM

Here we are the 2 files you asked to me. Tnx a lot again for all, Stefano.

YLG80

05-18-2009, 12:51 PM

No problem to run the progs and compile them on my machine with your config files.

---

slugworth

05-18-2009, 06:34 PM

Maybe the traphandler has to be external for that processor,which leads back to the other problem we had pages ago with 16meg receivers.(address not in range)

---

effesse

05-18-2009, 11:36 PM

No problem to run the progs and compile them on my machine with your config files.

I installed TS R2.3.1 Patch2, recompiled hello.c but the result remains identical!
:banghead:

Does it matter if my receiver is in dead state because of a wrong BootLoader installed in its Flash?? The hello.c prog have to run as usuals in this case?? Thank you very much again, best regards, Stefano.

YLG80

05-19-2009, 07:22 AM

I may be wrong, but I think that the traphandler is installed in EXTERNAL.
It doesn't matter that the receiver is death.
Slugworth's receiver was also death and he was able to restore the bootloader.

Your Sti5118 has the same CPU ID as the Sti5105, so I think it should have the same behavior. (If I remember MSSFWP mentioned that the STi5118 could be just a second choice based on the same wafer)

You ucTAP capture looks good and just stop when loading the lku.

I'm going to PM you another test.

---

effesse

05-19-2009, 09:55 AM

I may be wrong, but I think that the traphandler is installed in EXTERNAL.
It doesn't matter that the receiver is death.
Slugworth's receiver was also death and he was able to restore the bootloader.

Your Sti5118 has the same CPU ID as the Sti5105, so I think it should have the same behavior. (If I remember MSSFWP mentioned that the STi5118 could be just a second choice based on the same wafer)

You ucTAP capture looks good and just stop when loading the lku.

I'm going to PM you another test.

C:\STM\ST20R2.3.1\examples\test51xx-2>nmake

Microsoft (R) Program Maintenance Utility Version 1.50
Copyright (c) Microsoft Corp 1988-94. All rights reserved.

Microsoft (R) Program Maintenance Utility Version 1.50
Copyright (c) Microsoft Corp 1988-94. All rights reserved.

```
st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o hello.tc1 hello
.c
st20cc -p STi5105MB400 -O0 -LC:\STM\ST20R2.3.1\examples\test51xx-2\..\m
kf/../lib -o hello_c1c.lku hello.tc1
-----------------------------------------------
sti5105chip.cfg : STi5105chip procedure
-----------------------------------------------

-----------------------------------------------
sti5105chip.cfg config vars trace for debug
-----------------------------------------------

memory DCU 0x30000000 size 0x1000
memory DCache 0x30001000 size 0x1000
memory ICache 0x30001000 size 0x1000
memory PWM 0x30003000 size 0x1000
memory INT_CONTROLLER 0x30004000 size 0x1000
memory EMIindex 0x20200000 size 0x1000
memory LMIindex 0xe0000000 16*M
memory INTLEVELCTRL 0x20800000 size 0x1000
memory LPC 0x20808000 size 0x1000
memory IRB 0x20818000 size 0x1000
```

---
sti5105chip.cfg registers setup trace for debug

DCU3_registers 0x30000000
CacheControl3Registers 0x30001000 0
CacheControl3Registers 0x30002000 1
PWMControllerRegisters 0x30003000
InterruptController2Registers 0x30004000 16 0
ExceptionVectorTableRegisters 0x80000040 16
EMI4Registers 0x20200000 0 1 STi5105
InterruptLevelController3Registers 0x20800000 72
LowPowerControllerRegisters 0x20808000 1

PIOControllerRegisters 0x20820000 0
PIOControllerRegisters 0x20821000 1
PIOControllerRegisters 0x20822000 2
PIOControllerRegisters 0x20823000 3
AsynchronousSerialControllerRegisters 0x20830000 0 1 0
AsynchronousSerialControllerRegisters 0x20831000 1 1 0
SynchronousSerialController2Registers 0x20840000 0
SynchronousSerialController2Registers 0x20841000 1

PROCEDURE (stddefs: ST20C1MemoryInit

PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xfffffc00
size : 0x00000400
---
sti5105chip.cfg : STi5105chip procedure

---
sti5105chip.cfg config vars trace for debug
---

memory DCU 0x30000000 size 0x1000
memory DCache 0x30001000 size 0x1000
memory ICache 0x30001000 size 0x1000
memory PWM 0x30003000 size 0x1000
memory INT_CONTROLLER 0x30004000 size 0x1000
memory EMIindex 0x20200000 size 0x1000
memory LMIindex 0xe0000000 16*M
memory INTLEVELCTRL 0x20800000 size 0x1000
memory LPC 0x20808000 size 0x1000
memory IRB 0x20818000 size 0x1000
---
sti5105chip.cfg registers setup trace for debug

DCU3_registers 0x30000000
CacheControl3Registers 0x30001000 0
CacheControl3Registers 0x30002000 1
PWMControllerRegisters 0x30003000
InterruptController2Registers 0x30004000 16 0
ExceptionVectorTableRegisters 0x80000040 16
EMI4Registers 0x20200000 0 1 STi5105
InterruptLevelController3Registers 0x20800000 72
LowPowerControllerRegisters 0x20808000 1

```
PIOControllerRegisters 0x20820000 0
PIOControllerRegisters 0x20821000 1
PIOControllerRegisters 0x20822000 2
```

```
PIOControllerRegisters 0x20823000 3
AsynchronousSerialControllerRegisters 0x20830000 0 1 0
AsynchronousSerialControllerRegisters 0x20831000 1 1 0
SynchronousSerialController2Registers 0x20840000 0
SynchronousSerialController2Registers 0x20841000 1
-----------------------------------------
PROCEDURE (stddefs: ST20C1MemoryInit
-----------------------------------------
-----------------------------------------
PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xc1fffc00
size : 0x00000400
-----------------------------------------

C:\STM\ST20R2.3.1\examples\test51xx-2>go
-----------------------------------------------
sti5105chip.cfg : STi5105chip procedure
-----------------------------------------------

-----------------------------------------------
sti5105chip.cfg config vars trace for debug
-----------------------------------------------

memory DCU 0x30000000 size 0x1000
memory DCache 0x30001000 size 0x1000
memory ICache 0x30001000 size 0x1000
memory PWM 0x30003000 size 0x1000
memory INT_CONTROLLER 0x30004000 size 0x1000
memory EMIindex 0x20200000 size 0x1000
memory LMIindex 0xe0000000 16*M
memory INTLEVELCTRL 0x20800000 size 0x1000
memory LPC 0x20808000 size 0x1000
memory IRB 0x20818000 size 0x1000
-----------------------------------------------
sti5105chip.cfg registers setup trace for debug
-----------------------------------------------

DCU3_registers 0x30000000
CacheControl3Registers 0x30001000 0
CacheControl3Registers 0x30002000 1
PWMControllerRegisters 0x30003000
InterruptController2Registers 0x30004000 16 0
ExceptionVectorTableRegisters 0x80000040 16
EMI4Registers 0x20200000 0 1 STi5105
InterruptLevelController3Registers 0x20800000 72
LowPowerControllerRegisters 0x20808000 1
-----------------------------------------------

PIOControllerRegisters 0x20820000 0
PIOControllerRegisters 0x20821000 1
PIOControllerRegisters 0x20822000 2
PIOControllerRegisters 0x20823000 3
AsynchronousSerialControllerRegisters 0x20830000 0 1 0
AsynchronousSerialControllerRegisters 0x20831000 1 1 0
SynchronousSerialController2Registers 0x20840000 0
SynchronousSerialController2Registers 0x20841000 1
-----------------------------------------

PROCEDURE (stddefs: ST20C1MemoryInit
-----------------------------------------
-----------------------------------------
PROCEDURE (stddefs: PlaceDebugTrapHandler
```

address : 0xc1fffc00
size : 0x00000400
----

error-address space not stopped

C:\STM\ST20R2.3.1\examples\test51xx-2>

YLG80

05-19-2009, 11:05 AM

Would you like to try the following :

In flashmenu because it uses the local config files within the current directory

You open the mb400.cfg and change the first lines as follows:

```
##
SDRAM_DEVICE_TYPE = "DDR_RAM"
## SDRAM_DEVICE_TYPE = "SDR_RAM"
```

to

```
##
## SDRAM_DEVICE_TYPE = "DDR_RAM"
SDRAM_DEVICE_TYPE = "SDR_RAM"
```

Try flashmenu again. (Or recompile it than try it)

I believe that the reason is : your Sti5118 does not support DDR RAM. (according to the brief DS it supports SDRAM (133 Mhz) .
I've checked your SDRAM chip : it is not a DDR.
Could be the reason because progs on your STB need to use ST_5105_IniLMI_SDR_SDRAM and not ST_5105_IniLMI_DDR_SDRAM

I'm curious to see !

---

jvvh5897

05-19-2009, 11:09 AM

Hum...if this were an sti5518 and I had the wrong boot in the box and it was sending the controller to an addr that was not really in memory space, I would expect to have to trigger the boot-from-link to get the jtag to function correctly. Is there a way you can hold the box in reset state till you are ready to kick off the link control--remember that all the printing you are doing to screen slows your computer down by a large factor so the less you send to screen the better (saving to disk takes less of the processors resources).

---

slugworth

05-19-2009, 11:22 AM

The lku files I made lately have the traphandler set to internal;8000 0400
so they could be used on receivers with different ram size than 32meg.
It may need the trick of starting the .lku a split second after you powerup the receiver.

---

effesse

05-19-2009, 11:47 AM

Here we are the output screens:

C:\STM\ST20R2.3.1\examples\flashmenu>nmake

Microsoft (R) Program Maintenance Utility Version 1.50
Copyright (c) Microsoft Corp 1988-94. All rights reserved.

Microsoft (R) Program Maintenance Utility Version 1.50
Copyright (c) Microsoft Corp 1988-94. All rights reserved.

st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o flashprg.tc1 fl
ashprg.c
Warning-st20icc-flashprg.c(823)- Static 'srecord_getnextline' declared but not u
sed
st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb159.tc1 mb159
.c
st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb193.tc1 mb193
.c
st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o sdm.tc1 sdm.c
st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb275.tc1 mb275
.c
st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb282.tc1 mb282
.c
st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb292.tc1 mb292
.c
st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb314.tc1 mb314
.c
Warning-st20icc-mb314.c(538)- Lower precision in wider context: '|'
st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o eval0396.tc1 ev
al0396.c
st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb361.tc1 mb361
.c
st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb382.tc1 mb382
.c
st20cc -c1 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb385.tc1 mb385
.c
st20cc -p ST20DC1MB292A -O0 -I../lib -o flashprg_dc1.ld flashprg.tc1 mb159.tc1 mb193.tc1 sdm.tc1 mb275.tc1
mb282.tc1 mb292.tc1 mb314.tc1 eval0396.tc1 mb361.tc1 mb382.tc1 mb385.tc1

PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xfffffc00
size : 0x00000400

PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0x80001c00
size : 0x00000400
----
st20cc -c2 -c -g -O0 -I../lib -fdisable-device -g -O0 -o flashprg.tc2 fl
ashprg.c
Warning-st20icc-flashprg.c(823)- Static 'srecord_getnextline' declared but not u
sed
st20cc -c2 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb159.tc2 mb159
.c
st20cc -c2 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb193.tc2 mb193
.c
st20cc -c2 -c -g -O0 -I../lib -fdisable-device -g -O0 -o sdm.tc2 sdm.c
st20cc -c2 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb275.tc2 mb275
.c
st20cc -c2 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb282.tc2 mb282
.c
st20cc -c2 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb292.tc2 mb292
.c
st20cc -c2 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb314.tc2 mb314
.c
Warning-st20icc-mb314.c(538)- Lower precision in wider context: '|'
st20cc -c2 -c -g -O0 -I../lib -fdisable-device -g -O0 -o eval0396.tc2 ev
al0396.c
st20cc -c2 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb361.tc2 mb361
.c

```
.c
st20cc -c2 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb382.tc2 mb382
.c
st20cc -c2 -c -g -O0 -I../lib -fdisable-device -g -O0 -o mb385.tc2 mb385
```

```
.c
st20cc -p c2_bank0 -O0 -LC:\STM\ST20R2.3.1\examples\flashmenu\..\mkf/..
/lib -o flashprg_c2b0.lku flashprg.tc2 mb159.tc2 mb193.tc2 sdm.tc2 mb275.tc2 mb2
82.tc2 mb292.tc2 mb314.tc2 eval0396.tc2 mb361.tc2 mb382.tc2 mb385.tc2
----------------------------------------
PROCEDURE (stddefs: c2_bank0
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: c2hw_bank0
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: c2hw_common
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: ST20C2MemoryInit
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xfffffc00
size : 0x00000400
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: c2hw_common
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: c2_bank0
========================================
PROCEDURE (stddefs: c2hw_bank0
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: c2hw_common
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: ST20C2MemoryInit
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0x401ffc00
size : 0x00000400
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: c2hw_common
----------------------------------------
st20cc -p c2_bank1 -O0 -LC:\STM\ST20R2.3.1\examples\flashmenu\..\mkf/..
/lib -o flashprg_c2b1.lku flashprg.tc2 mb159.tc2 mb193.tc2 sdm.tc2 mb275.tc2 mb
282.tc2 mb292.tc2 mb314.tc2 eval0396.tc2 mb361.tc2 mb382.tc2 mb385.tc2
----------------------------------------
PROCEDURE (stddefs: c2_bank1
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: c2hw_bank1
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: c2hw_common
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: ST20C2MemoryInit
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: PlaceDebugTrapHandler
```

PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xfffffc00
size : 0x00000400

PROCEDURE (stddefs: c2hw_common

PROCEDURE (stddefs: c2_bank1

PROCEDURE (stddefs: c2hw_bank1

PROCEDURE (stddefs: c2hw_common

PROCEDURE (stddefs: ST20C2MemoryInit

PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0x501ffc00
size : 0x00000400

PROCEDURE (stddefs: c2hw_common
----
st20cc -p STi5100MB390 -O0 -LC:\STM\ST20R2.3.1\examples\flashmenu\..\mk
f/../lib -o flashprg_5100.lku flashprg.tc2 mb159.tc2 mb193.tc2 sdm.tc2 mb275.tc
2 mb282.tc2 mb292.tc2 mb314.tc2 eval0396.tc2 mb361.tc2 mb382.tc2 mb385.tc2
PROCEDURE (stddefs: ST20C2MemoryInit

PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xfffffc00
size : 0x00000400

PROCEDURE (stddefs: ST20C2MemoryInit

PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xc1fffc00
size : 0x00000400
----
st20cc -p STm5700MB385 -O0 -LC:\STM\ST20R2.3.1\examples\flashmenu\..\mk
f/../lib -o flashprg_c1c.lku flashprg.tc1 mb159.tc1 mb193.tc1 sdm.tc1 mb275.tc1
mb282.tc1 mb292.tc1 mb314.tc1 eval0396.tc1 mb361.tc1 mb382.tc1 mb385.tc1

PROCEDURE (stddefs: ST20C1MemoryInit

PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xfffffc00
size : 0x00000400

PROCEDURE (stddefs: ST20C1MemoryInit

PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xc1fffc00
size : 0x00000400

---

effesse

05-19-2009, 11:47 AM

```
C:\STM\ST20R2.3.1\examples\flashmenu>gomenu

--------------------------------------------------
MB400/STi5105 BOARD CONFIGURATION

------------------------------------------------
sti5105chip.cfg : STi5105chip procedure
------------------------------------------------

------------------------------------------------
sti5105chip.cfg config vars trace for debug
------------------------------------------------

memory DCU 0x30000000 size 0x1000
memory DCache 0x30001000 size 0x1000
memory ICache 0x30001000 size 0x1000
memory PWM 0x30003000 size 0x1000
memory INT_CONTROLLER 0x30004000 size 0x1000
memory EMIindex 0x20200000 size 0x1000
memory LMIindex 0xe0000000 16*M
memory INTLEVELCTRL 0x20800000 size 0x1000
memory LPC 0x20808000 size 0x1000
memory IRB 0x20818000 size 0x1000
------------------------------------------------
sti5105chip.cfg registers setup trace for debug
------------------------------------------------

DCU3_registers 0x30000000
CacheControl3Registers 0x30001000 0
CacheControl3Registers 0x30002000 1
PWMControllerRegisters 0x30003000
InterruptController2Registers 0x30004000 16 0
ExceptionVectorTableRegisters 0x80000040 16
EMI4Registers 0x20200000 0 1 STi5105
InterruptLevelController3Registers 0x20800000 72
LowPowerControllerRegisters 0x20808000 1
------------------------------------------------

PIOControllerRegisters 0x20820000 0
PIOControllerRegisters 0x20821000 1
PIOControllerRegisters 0x20822000 2
PIOControllerRegisters 0x20823000 3
AsynchronousSerialControllerRegisters 0x20830000 0 1 0
AsynchronousSerialControllerRegisters 0x20831000 1 1 0
SynchronousSerialController2Registers 0x20840000 0
SynchronousSerialController2Registers 0x20841000 1
----------------------------------------
PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xc05ff800
size : 0x00000400
----------------------------------------
----------------------------------------
PROCEDURE (stddefs: ST20C1MemoryInit
----------------------------------------
setup Clock Generator for ST_5105
CKG_CLOCK_SELECT_CFG = 68
--> Mfactor = 5 Nfactor = 37 Pfactor = 0

PLL Request --> PLL Frequency = 400 . 0 MHz
PLL_A Initialised --> PLL Frequency = 399 . 600 MHz
--> Mfactor = 7 Nfactor = 69 Pfactor = 0
```

PLL Request --> PLL Frequency = 533 . 0 MHz
PLL_B Initialised --> PLL Frequency = 532 . 285 MHz
Setup PLL dividers for ST_5105 chip
CPU = 200 MHz Phase = 0

LMI = 133 MHz Phase = 0
BLITTER = 100 MHz Phase = 0
SYSTEM = 100 MHz Phase = 0
FDMA = 200 MHz Phase = 0
VIDEO = 50 MHz Phase = 0
FLASH = 100 MHz Phase = 0
Setup PLL Frequency Synthesizers for ST_5105 chip
CKG_PIX_CLK_SETUP0 = 0x00000ebf
CKG_PLLA_CONFIG0 = 0x00002404
CKG_PLLA_CONFIG1 = 0x0000c838
CKG_PLLB_CONFIG0 = 0x00004406
CKG_PLLB_CONFIG1 = 0x0000c838
ST_5105 Clock Generator Successfully setup

==Informations of SDRAM==
SDRAM DEVICE TYPE = SDR SDRAM
SDRAM DEVICE FREQUENCY = 133 MHz
SDRAM DEVICE SIZE = 128 MBits (16 MBytes)
SDRAM DEVICE REFRESH PERIOD = 15.625 microseconds (4K/64ms)
SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.0
==========

LMI_STR = 0x35485235
set memory size
LMI Config Done

Configuring registers for FMI...
All Banks[0-3]have been configured
the ident from EPLD is 0x00000020
reseting flash
FMI configure done

Setup Complete
---

error-address space not stopped
C:\STM\ST20R2.3.1\examples\flashmenu>

Tnx again for all, kind regards, Stefano.

YLG80

05-19-2009, 11:48 AM

Could you remind me which build was loading the TH in INTERNAL.(181menu ?)
I've tried to build the simple Hello World for INTERNAL, but it does not run.
I think it's because it loads the TH (1Kb) in INTERNAL (0x80000400) which does not leave any space free for the "Hello world" proggie (1Kb also).
So the only solution was to load the lku in External.
However for another flashprog not using the TS I would need only the special Pionero-like TH (1Kb) without any other prog. It would fit in the INTERNAL.

OK SOLVED

---

effesse

05-20-2009, 07:27 AM

Could you remind me which build was loading the TH in INTERNAL.(181menu ?)
I've tried to build the simple Hello World for INTERNAL, but it does not run.

I think it's because it loads the TH (1Kb) in INTERNAL (0x80000400) which does not leave an space free for the "Hello world" proggie (1Kb also).
So the only solution was to load the lku in External.

However for another flashprog not using the TS I would need only the special Pionero-like TH (1Kb) without any other prog. It would fit in the INTERNAL.

Is it this file that you need? Look inside this .ZIP archive to find what do you need for

help me more again. Even thought I don't understand very well your prev post. :-(

Best regards, Stefano.

---

effesse

05-24-2009, 03:24 AM

Can you confirm to me (probe or meter) that Pin 131 is connected directly to Pin called FLASH_RESET (I think Pin 12) of the Flash of the STi5105 STB as the Output for the correctly driven Pin 130?

That is, if I set correctly Pin 130 (TP63 in my STB) then the CPU will drive correctly the Pin 131 to put the Flash in 'Delete/Write Enable' Mode? Can you confirm that too?

TNX again for all, Stefano.

---

effesse

05-24-2009, 05:29 PM

:cucumber:

```
-------------------------------------------------
MB400/STi5105 BOARD CONFIGURATION

-----------------------------------------------
sti5105chip.cfg : STi5105chip procedure
-----------------------------------------------

-----------------------------------------------
sti5105chip.cfg config vars trace for debug
-----------------------------------------------

memory DCU 0x30000000 size 0x1000
memory DCache 0x30001000 size 0x1000
memory ICache 0x30001000 size 0x1000
memory PWM 0x30003000 size 0x1000
memory INT_CONTROLLER 0x30004000 size 0x1000
memory EMIindex 0x20200000 size 0x1000
memory LMIindex 0xe0000000 16*M
memory INTLEVELCTRL 0x20800000 size 0x1000
memory LPC 0x20808000 size 0x1000
memory IRB 0x20818000 size 0x1000
-----------------------------------------------
sti5105chip.cfg registers setup trace for debug
-----------------------------------------------

DCU3_registers 0x30000000
CacheControl3Registers 0x30001000 0
CacheControl3Registers 0x30002000 1
PWMControllerRegisters 0x30003000
InterruptController2Registers 0x30004000 16 0
ExceptionVectorTableRegisters 0x80000040 16
EMI4Registers 0x20200000 0 1 STi5105
InterruptLevelController3Registers 0x20800000 72
LowPowerControllerRegisters 0x20808000 1
-----------------------------------------------

PIOControllerRegisters 0x20820000 0
PIOControllerRegisters 0x20821000 1
PIOControllerRegisters 0x20822000 2
PIOControllerRegisters 0x20823000 3
AsynchronousSerialControllerRegisters 0x20830000 0 1 0
```

```
AsynchronousSerialControllerRegisters 0x20831000 1 1 0
SynchronousSerialController2Registers 0x20840000 0
SynchronousSerialController2Registers 0x20841000 1
-----------------------------------------
```

```
PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xc05ff800
size : 0x00000400
-----------------------------------------
-----------------------------------------
PROCEDURE (stddefs: ST20C1MemoryInit
-----------------------------------------
setup Clock Generator for ST_5105
CKG_CLOCK_SELECT_CFG = 68
--> Mfactor = 5 Nfactor = 37 Pfactor = 0
PLL Request --> PLL Frequency = 400 . 0 MHz
PLL_A Initialised --> PLL Frequency = 399 . 600 MHz
--> Mfactor = 7 Nfactor = 69 Pfactor = 0
PLL Request --> PLL Frequency = 533 . 0 MHz
PLL_B Initialised --> PLL Frequency = 532 . 285 MHz
Setup PLL dividers for ST_5105 chip
CPU = 200 MHz Phase = 0
LMI = 133 MHz Phase = 0
BLITTER = 100 MHz Phase = 0
SYSTEM = 100 MHz Phase = 0
FDMA = 200 MHz Phase = 0
VIDEO = 50 MHz Phase = 0
FLASH = 100 MHz Phase = 0
Setup PLL Frequency Synthesizers for ST_5105 chip
CKG_PIX_CLK_SETUP0 = 0x00000ebf
CKG_PLLA_CONFIG0 = 0x00002404
CKG_PLLA_CONFIG1 = 0x0000c838
CKG_PLLB_CONFIG0 = 0x00004406
CKG_PLLB_CONFIG1 = 0x0000c838
ST_5105 Clock Generator Successfully setup

==Informations of SDRAM==
SDRAM DEVICE TYPE = SDR SDRAM
SDRAM DEVICE FREQUENCY = 133 MHz
SDRAM DEVICE SIZE = 128 MBits (16 MBytes)
SDRAM DEVICE REFRESH PERIOD = 15.625 microseconds (4K/64ms)
SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.0
==========================

LMI_STR = 0x35485235
set memory size
LMI Config Done

Configuring registers for FMI ...
All Banks[0-3] have been configured
the ident from EPLD is 0x00000020
reseting flash
FMI configure done


Setup Complete
-----------------------------------------------------

Hello World from Osprey
This is a simple Sti51xx board DCU3 test.
If you read me, I'm OK !


TT HH AA NN KK YY OO UU VV EE RR YY MM UU CC HH !!
```

to all users but in particular way to the great YLG80 !!!

---

spartacussat

05-25-2009, 04:05 AM

can someone summarize the whole discussion? At the end, how can we erase/write sti5119 and 4MB flash.

---

effesse

05-27-2009, 04:34 PM

can someone summarize the whole discussion? At the end, how can we erase/write sti5119 and 4MB flash.

Really I still trying to understand how I will be able to Erase my 2MB flash eprom, maybe a day I will got it and I will write some rows about this extraterrestrial procedure. Tnx a lot for reading, kind regards,Stefano.

---

effesse

05-27-2009, 05:08 PM

I get always this kind of errors:

Setup Complete

JTAG Flash program Release Date: 20090511 : Auth: YLG

Flash Programmer - version dedicated to the STi51xx for TS R2.3.1.
Credits to Slugworth, YLG80, MSSFWP and other folks around the world
---

Supported Flash devices:-

Intel 28F004-TL (MB159, STi5500)
SGS-Thomson 28F411-TL (MB193, ST20TP3)
AMD 29LV160BB (SDM, STi5505)
SGS-Thomson 28F411-TL (MB231, STi5510)
ST M29W800AT (MB275, STi5508/18)
ST M29W800AT (MB282, STi5512)
ST M29W800AT (MB292, ST20DC1)
Intel 28F640J3A (MB314, STi5514)
ST M58LW064B (MB314, STi5514)
ST M58LW032A (MB361, STi5516)
ST M58LW064D (MB382, STi5516/17)
ST M29KW032E (MB385, STm5700)
ST M58LW064D (MB390, STi5100)
ST M58LW064D (MB391, STi7710)
ST M58LW064D (MB395, STi5100)
ST M58LW064D (MB400, STi5105)
ST M58LW064D (MB424, STi5100)
ST M28W320CT (EVAL0396, STV0396)

Please select the Operation:
1) Erase
2) Program
3) Verify
4) Exit
Enter a number : 1

Flash Programmer v1.0 (2009) - Special version STi5105 for TS R2.3.1.
Credits to Slugworth, YLG80, MSSFWP and other folks around the world
---

board = SDM

mode = ERROR_MODE
report = FULL
asynchronous event error-address not in memory range address 0xc1ffdbf8 not in k

nown range

---

effesse

05-28-2009, 04:58 PM

```
-------------------------------------------------
MB400/STi5105 BOARD CONFIGURATION

-----------------------------------------------
sti5105chip.cfg : STi5105chip procedure
-----------------------------------------------

-----------------------------------------------
sti5105chip.cfg config vars trace for debug
-----------------------------------------------

memory DCU 0x30000000 size 0x1000
memory DCache 0x30001000 size 0x1000
memory ICache 0x30001000 size 0x1000
memory PWM 0x30003000 size 0x1000
memory INT_CONTROLLER 0x30004000 size 0x1000
memory EMIindex 0x20200000 size 0x1000
memory LMIindex 0xe0000000 16*M
memory INTLEVELCTRL 0x20800000 size 0x1000
memory LPC 0x20808000 size 0x1000
memory IRB 0x20818000 size 0x1000
-----------------------------------------------
sti5105chip.cfg registers setup trace for debug
-----------------------------------------------

DCU3_registers 0x30000000

CacheControl3Registers 0x30001000 0
CacheControl3Registers 0x30002000 1
PWMControllerRegisters 0x30003000
InterruptController2Registers 0x30004000 16 0
ExceptionVectorTableRegisters 0x80000040 16
EMI4Registers 0x20200000 0 1 STi5105
InterruptLevelController3Registers 0x20800000 72
LowPowerControllerRegisters 0x20808000 1
-----------------------------------------------

PIOControllerRegisters 0x20820000 0
PIOControllerRegisters 0x20821000 1
PIOControllerRegisters 0x20822000 2
PIOControllerRegisters 0x20823000 3
AsynchronousSerialControllerRegisters 0x20830000 0 1 0
AsynchronousSerialControllerRegisters 0x20831000 1 1 0
SynchronousSerialController2Registers 0x20840000 0
SynchronousSerialController2Registers 0x20841000 1
-----------------------------------------
PROCEDURE (stddefs: PlaceDebugTrapHandler
address : 0xc05ff800
size : 0x00000400
-----------------------------------------
-----------------------------------------
PROCEDURE (stddefs: ST20C1MemoryInit
-----------------------------------------
setup Clock Generator for ST_5105
CKG_CLOCK_SELECT_CFG = 68
--> Mfactor = 5 Nfactor = 37 Pfactor = 0
PLL Request --> PLL Frequency = 400 . 0 MHz
PLL_A Initialised --> PLL Frequency = 399 . 600 MHz

--> Mfactor = 7 Nfactor = 69 Pfactor = 0
PLL Request --> PLL Frequency = 533 . 0 MHz
PLL_B Initialised --> PLL Frequency = 532 . 285 MHz
```

Setup PLL dividers for ST_5105 chip
CPU = 200 MHz Phase = 0
LMI = 133 MHz Phase = 0
BLITTER = 100 MHz Phase = 0

SYSTEM = 100 MHz Phase = 0
FDMA = 200 MHz Phase = 0
VIDEO = 50 MHz Phase = 0
FLASH = 100 MHz Phase = 0
Setup PLL Frequency Synthesizers for ST_5105 chip
CKG_PIX_CLK_SETUP0 = 0x00000ebf
CKG_PLLA_CONFIG0 = 0x00002404
CKG_PLLA_CONFIG1 = 0x0000c838
CKG_PLLB_CONFIG0 = 0x00004406
CKG_PLLB_CONFIG1 = 0x0000c838
ST_5105 Clock Generator Successfully setup

==Informations of SDRAM==
SDRAM DEVICE TYPE = SDR SDRAM
SDRAM DEVICE FREQUENCY = 133 MHz
SDRAM DEVICE SIZE = 128 MBits (16 MBytes)
SDRAM DEVICE REFRESH PERIOD = 15.625 microseconds (4K/64ms)
SDRAM DEVICE REFRESH CAS LATENCY(CL) = 2.0
==========

LMI_STR = 0x35485235
set memory size
LMI Config Done

Configuring registers for FMI ...
All Banks[0-3] have been configured

the ident from EPLD is 0x000000ff --> be4 erasing 1st time was '0x00000020' !!!!!
reseting flash
FMI configure done

Setup Complete

Flash Programmer v1.3 - MB159, MB193, SDM, MB231, MB275, MB282, MB292, MB314, MB
361, MB382, MB385, MB390, MB391, MB395, MB400, MB424 and EVAL0396.
(c) STMicroelectronics Ltd 1998-2004.

board = SDM
operation = PROGRAM
report = VERBOSE
file = wholeflash.bin

Erasing Flash
Checking FlashROM Identifiers ...
Manufacturer ID 0020 --> I modded the '0001' AMD to '0020' SSt
Device ID 22c4 --> it's alright !
Flash Identifiers are recognised OK.
Performing Chip Erase...
Checking FLASH memory is fully erased...
ROM is fully erased with no errors.

Data file wholeflash.bin opened
Writing 4096 bytes to address 0x40000000
Failed at 40000001 (wrote 2020, read 2000) --> how can it be possible ???

Program/verify failed

I try reading with jKeys and I can dump all 2MB of 'FF' so I think it's really blank, it isn't

?

But I'm still unable to program the flash and I don't know why. Maybe the BottomBoot of the AMD 29LV160BB is hurting the TopBoot of my M29W160ET ?

I would like to know what and how I have to modify again to Program and Verify ATLEAST the 1st 64Kb area of my Flash (the bootloader). MANY THANK TO ALL, AND IN PARTICULAR WAY TO YLG80 AND SLUGWORTH FOR ALL THE SUPPORT GIVEN TO ME.

Kind regards, Stefano.

---

jvvh5897

05-29-2009, 01:36 PM

Get a datasheet for the flash and see where the write protect pin is--many boxes have the pin connected to a processor pin so that the processor can control it--maybe through a couple of transistors because there could be a need for +12V to enable writing. If you can't control the write protect pin with software you will have to mod the transistor control--likely just have to jumper the base of a transistor to a specific state. In the US the FCC required flashes to be hard to re-write because of piracy issues, so the newer the box, the more likely that some write protect is in effect.

---

slugworth

05-29-2009, 02:05 PM

timing problem.
I had the same problem with the original sst39vf3201 programming until I added the delay to the .c file.
The 2meg flash programming is based on a different .c file (sdm.c) that didn't have a delay added.

---

slugworth

05-29-2009, 06:00 PM

The sdm.c is actually for the old sti5505,it was just convenient because it was set up for a 2meg flash already.That is why you see the piocontroller stuff.
You may have to edit the mb385.c or other mbxxx.c file instead to support your flash chip.

Alex_Trask

05-30-2009, 04:22 PM

Greatings to all of you.

I was not here with you for a simple reason, a motorbike accident, but now I'm fine.

And now my question:
using the test51xx files i get this:

C:\STM\ST20R2.3.1\examples\ods>go
executing connect procedure STi5105MB400 (1) (1)
error-Reference to an uninitialised variable "ST_5105_InitFMI"
error-address space not stopped

C:\STM\ST20R2.3.1\examples\ods>

Thanks

---

slugworth

05-30-2009, 07:52 PM

Reference to an uninitialised variable "ST_5105_InitFMI"
Means st_5105_initfmi isn't in your .cfg file

---

effesse

05-31-2009, 10:25 AM

Damn it!
I would like to add a connection between the Pin 130 of the Stx 5118 ALB (nRST / TP64) and the Pin 5 of the LPT port on the Simple Jtag Cable for uCtap working much well but...

aaaarrrrgggggghhhhh (p0rc@ z0zz@!)

A little bit of solder broken down and now I connected for mystake four Pins of the CPU together :-(

Any tricks/tips/suggestions about removing the unwanted leads (ways and tools) between the very importants four Pins? :-((

Thanks in advance, kind regards, Stefano.

:cry: :cry: :cry:

This is the original picture taken with 600dpi color scanner:

http://digilander.libero.it/effesse2002/Telestar_Srs_1_Top.jpg

---

ukronic

05-31-2009, 11:56 AM

A little bit of solder broken down and now I connected for mystake four Pins of the CPU together :-(

Any tricks/tips/suggestions about removing the unwanted leads (ways and tools) between the very importants four Pins? :-((

There are tools (Wicked or Braided copper) willl help remove unwanted solder, there is also a vaccuum type of tool that pulls off solder (so long as the solder is in a melted state).

I also find a Dentist pick to be quite handy for soldering within small areas aswell.

Alex_Trask

05-31-2009, 12:12 PM

Reference to an uninitialised variable "ST_5105_InitFMI"
Means st_5105_initfmi isn't in your .cfg file

Fixed.

Now I get this:

C:\STM\ST20R2.3.1\examples\ods>go
error-address space not stopped

C:\STM\ST20R2.3.1\examples\ods>

I'm trying the test files with uctap
Thanks slugworth

---

slugworth

05-31-2009, 01:32 PM

read post #432 to present.
I think effe was having the same problem.

---

slugworth

05-31-2009, 01:50 PM

I believe at this time people that have receivers other than st5105/32m/ 4meg should start their own threads on their type receiver.
It is getting too difficult to keep track of what people have and the stage they are at.There are people with sti5105/16meg but with 2meg flash of different vendors and people with st511x 16meg with 2meg flash different vendors.

---

effesse

05-31-2009, 02:37 PM

Fixed.

Now I get this:

C:\STM\ST20R2.3.1\examples\ods>go
error-address space not stopped

C:\STM\ST20R2.3.1\examples\ods>

I'm trying the test files with uctap
Thanks slugworth

You have to use the TRST signal of Jtag Header (often Pin 19) connected to Pin 12 of the LPT port via a 100ohm res. and the nRST signal (often Pin 17) -in my case a little round copper pad called TP64- to the Pin 5 of the LPT port via a 100ohm res. too (better will do a NPN inverter) as uCtap can work as better than ever. So the nRST signal will put Low for a very short time (as triggering) by the Pin 5 of the LPT to bring the CPU in a good state to execute the .LKU you just compile. If this way don't work as well just try using Pin 6 instead of 5. Read previous post to see a confirmatin to that.

Good luck for all and kind regards, Stefano.

YLG80

06-01-2009, 04:38 AM

It's not very difficult to remove solder bridges between pin's : use desoldering wick. Do not overheat the chip.
http://en.wikipedia.org/wiki/Solder_wick

---

jvvh5897

06-01-2009, 11:40 AM

I find lots of use for a magnifying glass when working on boxes. I also have a probe that I built out of an old pen body with a straight pin glued in one end--I use it to clean between pins on flash and processor and when connected to multimeter to do resistance/continuity tests (not brave enough to use it directly to pins for voltage tests--even a straight pin is too big to trust that you will not bridge two close-spaced pins).

---

effesse

06-01-2009, 03:02 PM

...now I'm trying to clean solder between only two close-spaced pins. :cry:
I'm thinking to use a piece of cutter blade (used for paper) on top of a used ballpen or a piece of little blade of unshavers. :Cry:

Tnx to all for any kind of support, Stefano.

Flat

06-02-2009, 02:54 AM

Why do I have so little time now to play around? This must be one of the most interesting threads I have read and reread in a long time.

If this is working well, it will be surely one of the best ways to start a source project.

Just a quick heads-up and big thank you to all who contributed. Can't wait to start playing with this.

Regards,
Envious Flat

---

remal1

06-29-2009, 04:13 AM

Hello to all!

First of all, thanks for your great work and sorry for my english :)

I know this is a FTAtalk (the unit is not FTA), but if someone can help me, i really appreciate it.
My unit has STi5100KUC and all i got is the idcode (with every tool).
The question is: for reading out the registers do i need to fully set up the st20 toolset?

Because even the first (30003004)readout has failed :(
I'm not 100% sure but it seems to me the memory access via jtag is disabled.
Or is this just a problem with traphandler?
Is there some info about this kind of protection?
Thanks in advance.
ReMal

---

slugworth

06-29-2009, 06:59 AM

The 5100 is a DCU3 device,but core2 instead of core1 like the sti5105.
I would try a variation of the pace3100 jtag,which was also core2 DCU3 but a different processor.

---

bobabooie

06-30-2009, 01:30 PM

The Ipro2000+ use the STi5100 as well and everything needed to dump and write can be made up out of the last st toolset...

---

niebieski20

08-19-2009, 10:09 PM

Hi
Is it possible to connect ucTAPsrv with Sti7100? I mean what do i need to change in cfg file to proper read/write flash?? Regards

---

slugworth

08-19-2009, 11:44 PM

Uctap is just an interface from parallel port to the jtag program you are using.
The 7100 is a st40 based processor I believe and there is no jtag program for it.

---

niebieski20

08-20-2009, 01:08 AM

Yes but i think that those program is more than just interface, it emulate st micro connect1 so if orginaly st20r2.3.1 works fine with it, then st40r4.. must work too. St40r have option to connect to board via st micro conect 2 or 1. Or im wrong??

---

slugworth

08-20-2009, 08:27 AM

I don't have an st40 based receiver, so I never played with the st40 toolkits.
They seem totally different than the st20 toolkits anyway, java based?

---

niebieski20

08-20-2009, 10:31 AM

If You ask about OS as i known stlinux, os21...
When I connect sti7100 base receiver with sh4xrun i have something like:
SHDEBUG [ERROR] :: Unable to initialise SDI interface
Unable to connect to remote target ....
but in window of ucTAPsrv there is some movement, so it connect with receiver.

---

minipc

09-23-2009, 05:19 AM

Thanks to YLG80, here it is:

h**p://www.hostclip.com/dl/a9d525a33ff36a9cd0b3ae299a8e737a

apologies for raising the dead, but would this download be available anywhere else? the "corrected" link above produces an "unable to connect ... to the server host" error, and i can't find any references to it in subsequent posts.

edit: file was supposed to be 13M of sti5101 sources grabbed from pudn, aka the great wall of intellectual property exchange;)

slugworth

09-24-2009, 05:44 PM

A service manual for an lcd tv that shows the pinouts to the sti5100.
In this case the jtag pins are test points.

YLG80

09-26-2009, 12:56 AM

hostclip seems to be down.
Here is another link for the DTV 5105 file:
http://www.megaupload.com/?d=YICCK6NW

---

mkanet

10-04-2009, 12:05 AM

I have a receiver with a STi5107KYA CPU a standard JTAG port (20 pin header). Here's the actual picture of it:

http://i67.photobucket.com/albums/h283/mkanet/20pinheader2.jpg

Unfortunately, I can't use a standard st20 receiver ribbon cable since the pinouts need to be DCU3 compatible.

I know this sounds pathetic, but I wouldn't ask unless I was really despirate. But, can someone please sell me a custom Buffered JTAG Programmer with the correct 20 pin connector on it which is compatible with my receiver?

Does anyone know of a vendor that will just sell me the Buffered JTAG Programmer with this already done for me?

Thanks in advance guys! I know someone here will come through!

YLG80

10-04-2009, 01:28 PM

Hello,
The JTAG pinout has nothing to do with DCU3 which is a protocol to handle the interrupts and the transactions between a CPU and the external world via the JTAG interface.
The only thing that is different, is the wiring of the interface when using the ST Toolset.

This is because, up to now, the only way to repair a firmware via jtag with a DCU3 CPU is to use the original ST software with the ucTAP ST20 connect emulator. The software needs to be recompiled for each specific configuration (DRAM, flash size ...) which is not really for the beginner....

So your 20 pin jtag pinout is likely standard. The standard JTAG interface is not too difficult to change for ucTAP.
You should better use the simple direct connect interface. (See Slugworth posts on that interface.)

---

slugworth

10-31-2009, 07:49 PM

this entire thread should be archived so it isn't lost forever.
